

A MERGED RESEARCH REPORT

Vector and Tensor Fields over the Surreal Numbers

Admissible calculus in every dimension,
three-dimensional vector calculus, and Minkowski fields

September 22, 2026

Abstract

Surreal coefficients extend finite-dimensional tensor algebra without changing its algebraic identities, but they do not determine a calculus on their own. This report, merged from three manuscripts that build the same layered theory, separates three settings: finite algebra over a set-sized real-closed surreal subfield; smooth real manifolds whose fields are Hahn series of ordinary smooth tensor fields; and genuine surreal-coordinate neighbourhoods reached by support-controlled Taylor prolongation and scale charts.

Part I works in every finite dimension and every nondegenerate signature: prolongation with intrinsic derivatives, tensors, forms, admissible metrics, Levi-Civita connections, curvature, Hodge operators, coefficientwise integration, Stokes and Poincaré theorems, de Rham base change, Hodge decomposition for a real metric and for a regular Riemannian Hahn metric, standard-part reduction, near-real flows, exact inversion of positive-order operator perturbations, and nonlinear lifting from an invertible real linearization. Part II specializes to three dimensions: dot and cross products, leading-order rank, vector-calculus identities, integral theorems, a monopole flux that forbids a global vector potential, an explicit Helmholtz decomposition, infinitesimally forced stationary incompressible flow on the three-torus, and Maxwell, stress and elasticity equations. Part III treats Minkowski space: Lorentz transformations of arbitrary scale, a split reduction of the bounded Lorentz group with an exponential description of its kernel, a causal-stability criterion, the Lorentzian Hodge operator, Maxwell and scalar fields, stress-energy with exact doubling of the leading field scale, Cauchy and retarded lifting, and a transfinite cubic-wave deformation.

Every infinite operation carries an explicit support hypothesis. The obstructions, among them pointwise-invertible fields without admissible inverses, infinite-frequency oscillations, infinite boosts and infinite times, are proved rather than assumed away. Nothing here is a physics claim, a regularization of singularities, or a transfer of all real analytic existence theorems.

Built from three manuscripts prepared in relation to the `VladimirReshetnikov/Surreal` project,

all pinned to commit `d22a5b3`, and merged after placement commit `5fe7f8d`.

An AI-assisted research draft with proofs and finite symbolic checks; not refereed;

not a claim of a completed Lean formalization.

Contents

I	The theory in every finite dimension	1
1	Scope and the central choices	1
2	Scalars, size, and admissible infinite sums	5
3	Finite-dimensional vectors and tensors	7
4	The sheaf of smooth Hahn fields	8
5	Intrinsic derivatives and Taylor–Hahn prolongation	12
6	Genuinely surreal-coordinate neighborhoods	15
7	Vector fields, tensor fields, and exterior calculus	16
8	Admissible metrics and their Levi–Civita connections	17
9	Curvature and geometric identities	19
10	Volume, Hodge duality, and vector calculus in every dimension	20
11	Integration, Stokes, and local potentials	21
12	Global cohomology and a precise Hodge theorem	24
13	Standard-part geometry and scale stability	27
14	Examples at finite, infinitesimal, and infinite scales	28
15	Flows and parallel transport by positive-support recursion	30
16	Nonlinear field equations and a support-controlled lifting theorem	31
17	Variational calculus and gauge-valued fields	33
18	Scalar derivations and scale connections	35
19	Boundaries of the coherent calculus	36
20	Computation, proof boundaries, and implementation	38
II	Three-dimensional fields	41
21	Scope of Part II	41
22	Vectors and tensors in three dimensions	41
23	Vector calculus in three dimensions	43
24	Two metric examples	45
25	Integral theorems in three dimensions	45

26	Exact inversion on the three-torus	47
27	The scale connection in three dimensions	49
28	Continuum applications	49
III Minkowski space		52
29	Scope and conventions of Part III	52
30	Algebraic Minkowski geometry over a surreal field	54
31	Tensor algebra, exterior algebra, and Hodge duality	56
32	Bounded Lorentz transformations and real shadows	57
33	Fields on real Minkowski spacetime	60
34	Electromagnetic fields over surreal coefficients	61
35	Stress–energy, positivity, and leading scales	64
36	Causal linear equations and retarded solution operators	66
37	Actions, scalar fields, fluids, and conserved charges	69
38	A nonlinear wave theorem by transfinite coefficient recursion	71
39	Worldlines, local scales, and derivative choices	73
40	Covariance, singular sources and interpretation	74
IV Synthesis		75
41	What the theory establishes, and what it leaves as choices	75
A	Notation and hypothesis ledger	77
B	Lorentzian sign conventions	78
C	Sources, provenance and repository audit	79
D	A proof-dependency summary	83

Part I

The theory in every finite dimension

1 Scope and the central choices

An n -dimensional vector with surreal components is an element of $\mathbf{No}^n$, understood class-theoretically. A vector *field* requires more: its base space, allowed functions, derivative, coordinate transformations, and any integration operation must be specified. The two natural questions are

$$X : U \subseteq \mathbb{R}^n \longrightarrow \mathbf{No}^n, \quad X : W \subseteq \mathbf{No}^n \longrightarrow \mathbf{No}^n.$$

They are different questions. This report treats both, without identifying the usual topology of the real domain with a topology inherited from all surreals.

Throughout, $n \geq 1$ is an ordinary finite integer. Indices run from 1 to n , and repeated upper/lower indices are summed. Tensor rank is finite as well. The default metric is Euclidean; arbitrary nondegenerate signatures are introduced explicitly. No step of Part I depends on three dimensions unless it is labeled as such. Parts II and III specialize to three Euclidean dimensions and to four-dimensional Minkowski space.

1.1 Three complementary models

Model	Base and coefficients	What it supplies
Finite algebra	K^n , with $K \subset \mathbf{No}$ a set-sized real closed field	Tensors, metrics at a point, contractions, determinants, orthogonal algebra; no infinite analysis is implicit.
Smooth Hahn geometry	An ordinary smooth n -manifold M , with locally Hahn-supported smooth coefficients	Coordinate-independent differential calculus, real-chain integration, global topology and scale-by-scale geometric equations.
Intrinsic coordinate geometry	Open subsets of K^n , with rational or support-certified Taylor–Hahn functions	Actual derivatives with respect to surreal coordinates, finite halos, arbitrary-center scale charts and algebraic integration.

The main bridge is [theorem 5.2](#): a smooth Hahn field has a canonical Taylor–Hahn realization on a nearstandard open subset of K^n . The second model therefore is not merely symbolic bookkeeping. Its differential identities become genuine intrinsic ordered-field identities on that subset. It is nevertheless only one specified class of intrinsic functions.

1.2 Three manuscripts, one report

This report is built from three manuscripts, written independently on September 22, 2026 against the same repository pin ([section C](#)). Each constructs the same layered theory: finite algebra over a set-sized real-closed surreal subfield, Hahn series of ordinary smooth tensor fields, and Taylor prolongation or scale charts to genuine surreal-coordinate neighbourhoods. Their numbers are local to this directory.

Source 01 *Vector and Tensor Fields over the Surreal Numbers: admissible calculus, differential geometry, and multiscale field equations*, an n -dimensional framework. It is the base: its text is Part I, and its statements are the most general ones (every finite dimension, every nondegenerate signature, intrinsic derivatives of the prolongation). Its files carry the prefix `01-vector-tensor-fields-n-`.

Source 02 *Surreal-Valued Fields on Minkowski Space: Lorentz geometry, tensor calculus, and causal Hahn dynamics*. Its Lorentzian material is Part III. Its dimension-free remarks, the cohomology of a fixed coherent complex, the algebraic Poincaré lemma, the infinite-frequency and infinite-time obstructions and its distributional coefficients are placed in Part I. Its files carry the prefix `02-minkowski-`; its delivered source audit is `02-minkowski-SOURCE_AUDIT.md`.

Source 03 *Three-Dimensional Surreal Vector and Tensor Fields: support-controlled differential geometry, integral theorems, and multiscale continuum equations*. Its three-dimensional material is Part II. Its dimension-free results (convolution extension of real multilinear maps, exact inversion of positive-order perturbations, cohomology of a Hahn-extended cochain complex, the Hodge theorem for a regular Riemannian Hahn metric, quadratic lifting, the scale-connection calculus and the finite-partition obstruction) are placed in Part I. Its files carry the prefix `03-three-dimensional-fields-`.

When two or three sources prove the same statement it is printed once, in source 01's formulation when 01 has one, and the other sources are named there; [section C.2](#) lists these places. Genuinely different proofs are kept, for example the tree expansion for quadratic lifting ([theorem 16.2](#)). Before a result from the three-dimensional or the Lorentzian source was attached to a general statement, its proof was re-read for uses of dimension three, of the Euclidean metric, or of the Lorentzian signature. Each such attachment is marked where it is made. Every label carries the prefix `vtf:`; material from source 02 carries `vtf:mink:` and material from source 03 `vtf:3d:`, wherever it is placed.

Part III is mathematics of fields on Minkowski space. It is not an assessment in the sense of the collection's report *Surreal Scalars in Theoretical Physics* (`docs/physics/surreal-scalars-and-spacetime`), and it respects that report's claim-status firewall: no statement in this report is a physics claim. Words such as energy, radiation, observer, charge and fluid name mathematical objects in the fixed conventions of [section B](#). None of them asserts a measurable prediction, a physical regime, or the regularization of a singular field.

1.3 Relation to the repository and the literature

At the pin of all three manuscripts ([section C](#)) the repository's documentation distinguished, as it still does, finite algebra, set-sized Hahn workspaces, common-domain coefficient rings, strong summability, fine differentiation, and the Berarducci–Mantova scalar derivation [1, 3, 2]. We retain those distinctions. In particular, the one-complex-variable coherent calculus is a precedent for our evaluation construction; the present treatment extends its *method* to several real variables, smooth coefficients, tensor sheaves, and geometric equations.

Conway normal forms and real closedness are background results [10, 11, 12]; the support lemma is classical Hahn–Neumann algebra [13, 14]. Differential calculus over non-real topological fields already has a broad theory [20]. The tensor identities themselves are classical differential geometry [22, 23, 24]. We give direct arguments for the support-sensitive extensions rather than claiming that ordinary completeness or compactness transfers to K .

Our contribution is a unified construction with explicit sufficient hypotheses and proofs of its compatibility results. No claim of priority for every result, resolution of a named open problem, or exhaustive novelty search is made. Repository observations are pinned in [section C](#); no code in that repository is modified. Sources 02 and 03 make the same disclaimers ([section C.5](#)).

1.4 Neighbouring reports and the Lean development

Several results of this report restate, in the real tensor setting, material the collection already contains, and several connect to newer reports.

- `docs/surcomplex/contours-and-stokes` constructs the Hahn de Rham complex with complex coefficients on any smooth manifold, with coefficientwise Stokes (`contours:thm:hahnstokes`), a support-preserving Poincaré lemma (`contours:prop:poincare`), the cohomology of the globally supported Hahn complex (`contours:thm:cohomology`), polynomial-chain Stokes (`contours:thm:polstokes`), Stokes for Hahn-deformed chains (`contours:thm:deformedstokes`) and the finite-partition obstruction (`contours:prop:mesh`). [Theorems 11.1](#), [11.3](#) and [12.1](#), the polynomial Stokes theorem of [section 11](#) and [theorem 5.1](#) are real-coefficient versions of those statements and are not claimed as new. Source 03 said so at its pin; source 01 cited that report only through the documentation catalogue.
- `docs/surcomplex/analysis` develops coherent Hahn sections over complex domains; the evaluation construction of [section 5](#) extends its method to several real variables and smooth coefficients.
- `docs/foundations-and-computation/foundations` records the size conventions and the discreteness of small subsets and small-index nets in the full fine topology (`found:thm:discrete`), which [section 5](#) uses.
- `docs/physics/surreal-scalars-and-spacetime` is the collection’s assessment of surreal scalars in physics. Its bounded-frame valuation invariance (`phys:prop:frame`) is the statement of [section 3](#) and of [theorem 32.2](#). Its standard-part reduction over a Hahn coefficient algebra for a Lorentzian g_0 (`phys:thm:reduction`) is the four-dimensional Lorentzian case of [theorem 13.1](#), together with its Einstein-tensor consequence. Its conclusion that scalar extension does not regularize a singular field is shared by [sections 14.2](#), [25.2](#) and [40](#).
- `docs/surquaternions/surquaternions` treats Hamilton quaternions over real-closed fields, including the spin description of $\mathrm{SO}(3, F)$; the cross product of Part II is the vector part of the product of two pure quaternions. Its no-oscillation theorem (`squat:thm:nooscillation`) concerns the scalar Berarducci–Mantova derivation. The infinite-frequency obstruction [theorem 19.1](#) concerns coefficientwise derivatives. The two statements are different.
- `docs/surreal/euclidean-three-space`, placed in the same batch as this report, develops the coordinate geometry of $\mathbf{No}^3$, the group $\mathrm{SO}(3, \mathbf{No})$, Rodrigues rotations and finite-angle trigonometry. The finite algebra of K^3 in [section 22](#) overlaps its first layer and leaves angles to it.
- `docs/surcomplex/trigonometry` treats the unit circle and its angle group; the duality rotations of [section 34](#) use only the rational parametrization of the circle.
- `docs/surcomplex/spectral-theory` and `docs/surcomplex/infinite-dimensional-hahn-spectral-theory` treat spectral theory over real-closed and Hahn fields; Part II uses only the pointwise symmetric spectral theorem.

- `docs/surreal/hahn-valued-measures-and-probability` develops coefficientwise Hahn-valued measures. The positivity statement of [section 11](#) concerns smooth common-support densities only.
- `docs/surcomplex/differential-equations` studies differential equations for the scalar derivation; its phase obstruction is not the coefficientwise obstruction of [section 19](#).

The repository’s Lean Hahn layer proves the full Neumann support lemma, including finiteness across all word lengths (`Surreal/HahnSeries/Neumann.lean`, `NeumannWords.lean`), positive-order evaluation and composition in finitely many variables (`MvEvaluation.lean`, `MvComposition.lean`), binomial roots near one and standard part on the nonnegative-order subring. These are the scalar-level inputs of [section 2](#) and of the support half of [theorem 5.2](#). No field-level statement of this report has a Lean counterpart, and `docs/FORMALIZATION.md` maps no `vtf`: label.

1.5 Conventions fixed for the whole report

The three sources use overlapping symbols with different meanings. The choices below are made once; the source text has been renamed to match, and [section A](#) repeats them.

- *Monomials and scales.* $t^\gamma = \omega^{-\gamma}$ as in [\(2.1\)](#); when $1 \in \Gamma$, $t = t^1 = \omega^{-1}$. Source 02 wrote this monomial as ε^γ . Here ε always denotes a chosen positive infinitesimal t^η with $\eta > 0$, which is source 01’s usage. Ordinary time is τ , never a monomial.
- *Derivatives.* D_i in Part I and ∂_i, ∂_μ in Parts II and III are the coefficientwise derivatives of [proposition 4.1](#). They annihilate every constant of K and agree with intrinsic derivatives on halos ([theorem 5.2](#)). The intrinsic derivative of a realized function is written ∂_{y^i} . The Berarducci–Mantova scalar derivation is written ∂_{BM} and the character derivations of [section 18](#) are written $\mathcal{D}_\chi$. Neither is a spatial derivative.
- *Alternating symbols and the Minkowski metric.* Alternating symbols are ϵ_{ijk} and $\epsilon_{\mu\nu\rho\sigma}$; source 03 wrote ε_{ijk} . In Part III, $\eta_{\mu\nu}$ and $\eta(u, w)$ denote the Minkowski metric, its quadratic form is $q_\eta(u) = \eta(u, u)$, and no exponent is called η there; elsewhere η is an exponent. The letter q alone is the number of negative signs of a signature, as in [\(10.2\)](#). The group $\text{SO}^+(1, 3; K)$ is written $\mathcal{G}$.
- *Valuations.* v is the valuation of a scalar. v_V is the least component valuation of a finite tensor in a stated frame (source 02’s ν). v_H is the least exponent of a nonzero coefficient function of a field (source 02’s ord_Γ and source 03’s $v(F)$ for fields). The number $v_H(F)$ bounds the valuations of the values $F(x)$ from below but need not equal them. The same symbol v_H denotes the least exponent of an element of a Hahn vector space $E((t^\Gamma))$ ([section 4.2](#)).
- *Tolerances.* In [\(5.1\)](#) and the proof of [theorem 5.2](#), ϵ without indices is a positive element of K ; with indices, ϵ is always an alternating symbol.
- *Classes of fields and metrics.* A *coherent* field on U (source 02’s term) is an element of $\mathcal{A}_\Gamma(U)$, with one well-ordered support on all of U . An *admissible* metric has determinant a unit of the coefficient algebra ([section 8](#)). A *regular* metric is $g_0 + h$ with g_0 real and nondegenerate of any signature and h of positive support ([8.1](#)). A *regular Riemannian* metric (source 03’s “regular Hahn metric”) is a regular metric with g_0 positive definite. “Positive support” always means support in $\Gamma_{>0}$; “positive” for a scalar, a form or a density refers to the order of K .

- *Laplacians and codifferentials.* $\Delta_g = \operatorname{div}_g \operatorname{grad}_g$ and the Cartesian $\Delta = \sum_i \partial_i^2$ have the positive-second-derivative sign; $\Delta_H = d\delta + \delta d$ is the Hodge Laplacian, with $\Delta_H = -\Delta_g$ on functions; $\square = \partial_\tau^2 - \Delta$. The symbol δ with no index is a codifferential; δ_{ij} is Kronecker's symbol and δ_0 a Dirac distribution.
- *Renamings in the added material.* Source 02's exponent set Δ and exponent δ (in the retarded-perturbation, cubic-wave and infinite-frequency statements) are S_Q, S_r and θ ; its angular current M is $\mathcal{M}$ and its fluid expansion θ is ϑ . Source 03's Maxwell permittivity and permeability are κ_E, κ_B , its Lamé constants are λ_L, μ_L , and its metric density $J = \sqrt{\det g}$ is μ_g , as in source 01.
- *Curvature.* The three sources' conventions agree: $R(D_i, D_j)D_k = R^\ell_{kij}D_\ell$ and $\operatorname{Ric}_{kj} = R^i_{kij}$ (9.1); source 03's formulas were checked to be the same after renaming indices. Christoffel symbols Γ^k_{ij} always carry indices and are not the exponent group Γ .
- *Actions.* Source 01's action density $\frac{1}{2}|d\phi|_g^2 + V(\phi)$ and source 02's Lagrangian $\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi)$ use opposite signs for the potential; see [theorem 17.1](#).

2 Scalars, size, and admissible infinite sums

2.1 A fixed real closed Hahn workspace

Write a surreal normal form in increasing-exponent notation as

$$a = \sum_{\gamma \in S} a_\gamma t^\gamma, \quad a_\gamma \in \mathbb{R}, \quad t^\gamma := \omega^{-\gamma}, \quad (2.1)$$

where $S \subset \mathbf{No}$ is a well-ordered *set*. Here $\omega^{-\gamma}$ denotes Conway's monomial map, not an unannounced identification with the Gonshor exponential $\exp(-\gamma \log \omega)$. The relevant law is $t^{\gamma+\eta} = t^\gamma t^\eta$.

Fix a nonzero, set-sized, divisible ordered additive subgroup $\Gamma \subset \mathbf{No}$, and put

$$K = K_\Gamma = \mathbb{R}((t^\Gamma)). \quad (2.2)$$

The normal-form embedding identifies this field with a subfield of $\mathbf{No}$. The Hahn-field algebraic-closure theorem applied to $\mathbb{C}((t^\Gamma)) = K[i]$ implies that K is real closed [12]. Thus positive square roots, finite orthogonal algebra and signature are available in K .

Proposition 2.1 (Localization of set-sized scalar data). *Every set-sized collection of surreal numbers is contained in a workspace of the form (2.2).*

Proof. Take the union of the normal-form supports of the given numbers. It is a set. Its rational additive span in $\mathbf{No}$, enlarged by a nonzero exponent if necessary, is a set-sized divisible ordered subgroup Γ . Every original normal form is a Hahn series over Γ . $\square$

This proposition localizes scalar data, not arbitrary regularity claims about functions. It does not assert that every set-valued field has smooth Hahn coefficients. Nor does it assert that a selected workspace is closed under every additional scalar derivation.

Remark 2.2 (Abstract exponent groups and the class level). An abstract divisible ordered group may serve as Γ once a strictly order-preserving additive embedding into $\mathbf{No}$ is fixed; its image replaces Γ in (2.1) (sources 02 and 03). No part of the construction needs a positive exponent to be cofinal in Γ , and larger groups accommodate several Archimedean classes of scales. At the level of

the full class, $\mathbf{No}^n$ and its finite tensor carriers may be treated as classes. Every finite algebraic computation takes place in a set-sized workspace and inherits the identities proved below. A field on the proper class $\mathbf{No}^n$ would need a separately specified class-function notion; it is not an object of the real-domain theory, and no proper-class sum is formed (source 02).

2.2 Valuation, order, and standard part

For $a \neq 0$ define $v(a) = \min \text{supp}(a)$, and set $v(0) = +\infty$. Then

$$v(ab) = v(a) + v(b), \quad v(a + b) \geq \min(v(a), v(b)).$$

The order of K is the sign of the first nonzero real coefficient. Define

$$\mathcal{O}_K = \{a : v(a) \geq 0\}, \quad \mathfrak{m}_K = \{a : v(a) > 0\}.$$

These are, respectively, the finite elements and the infinitesimals relative to $\mathbb{R}$. Every $a \in \mathcal{O}_K$ has a unique decomposition

$$a = \text{st}(a) + h, \quad \text{st}(a) \in \mathbb{R}, \quad h \in \mathfrak{m}_K,$$

and coefficient extraction at exponent zero gives a ring homomorphism $\text{st} : \mathcal{O}_K \rightarrow \mathbb{R}$ with kernel $\mathfrak{m}_K$. Infinite elements have no standard part under this definition.

Coefficient extraction at exponent zero is not multiplicative on all of K : $[t^0](\varepsilon\varepsilon^{-1}) = 1$, although both separate extractions vanish (source 02).

Definition 2.3 (Strong summability). A family $(a_j)_{j \in J}$ of Hahn series is strongly summable when the union of its supports is well ordered and every exponent occurs in the support of only finitely many a_j . Its sum is obtained by finite coefficientwise addition. The same definition applies to coefficients in any real algebra, or in a real vector space when only addition is used.

Lemma 2.4 (Classical support lemma). *If S, T are well-ordered subsets of an ordered abelian group, then $S + T$ is well ordered and a fixed group element has only finitely many representations $s + t$, with $s \in S, t \in T$. If $T \subset \Gamma_{>0}$ is well ordered, the additive monoid*

$$\langle T \rangle = \{0\} \cup \{\tau_1 + \cdots + \tau_r : r \geq 1, \tau_i \in T\}$$

is well ordered. Every exponent has only finitely many word representations from T , counting all lengths. Consequently, $S + \langle T \rangle$ is well ordered and all the corresponding coefficient contributions are finite.

This is the support theorem imported from Hahn–Neumann theory [13, 14]. Its last assertion follows by first applying the finite-pair statement to S and $\langle T \rangle$, then the finite-word statement to the second summand. We will name these supports whenever a Taylor substitution, inverse, or nonlinear recursion is used.

Corollary 2.5 (Geometric inversion). *If H is a scalar or finite square matrix of Hahn series and the union of its entry supports is strictly positive, then*

$$(I - H)^{-1} = \sum_{r \geq 0} H^r.$$

The sum is strong, including all matrix-index choices. The result is an exact Hahn identity, not a claim about convergence of ordinary partial sums.

Proof. There are finitely many matrix entries. Apply lemma 2.4 to their support union. At any fixed exponent, only finitely many lengths and index words contribute. The finite identity $(I - H) \sum_{r=0}^N H^r = I - H^{N+1}$ therefore gives the asserted identity at each exponent once all words contributing there have been included. $\square$

For $h \in \mathfrak{m}_K$ the families

$$(1 + h)^{-1} = \sum_{m \geq 0} (-h)^m, \quad e^h = \sum_{m \geq 0} \frac{h^m}{m!}, \quad \log(1 + h) = \sum_{m \geq 1} \frac{(-1)^{m+1}}{m} h^m \quad (2.3)$$

are strong sums by the same argument, and finite matrices with infinitesimal entries obey the same support rules (source 02). Formal identities between these series, such as $\exp \log(1 + h) = 1 + h$, hold exactly, because only finitely many words contribute at each exponent.

2.3 Strong sums are not ordinary limits

For example, $\sum_{m \geq 1} t^{m/(m+1)}$ is a Hahn series over $\mathbb{Q}$. Its tails have valuations below 1, so its finite partial sums do not converge to that series in the valuation topology. Likewise, if $0 < \eta \ll \lambda$ in Γ , meaning $m\eta < \lambda$ for every positive integer m , then $\sum_{m \geq 0} t^{m\eta}$ is strongly summable but its partial sums are not cofinal in valuation precision. A support argument, not the phrase “the terms become small,” is the justification for our sums. Sources 02 and 03 give the same examples.

3 Finite-dimensional vectors and tensors

3.1 Tensor spaces and transformation laws

Let $V = K^n$, with basis e_i and dual basis e^i . A tensor of type (r, s) is

$$T \in T_s^r(V) := V^{\otimes r} \otimes_K (V^*)^{\otimes s}, \quad \dim_K T_s^r(V) = n^{r+s}.$$

Equivalently, it is a multilinear map on r covector inputs and s vector inputs. Its components are

$$T = T^{i_1 \dots i_r}_{j_1 \dots j_s} e_{i_1} \otimes \dots \otimes e_{i_r} \otimes e^{j_1} \otimes \dots \otimes e^{j_s}.$$

Contraction, tensor product, symmetrization, antisymmetrization and permutation of slots use only finite sums and products. They are therefore valid over K and, for any given finite data, over the full surreal field.

If $e'_a = e_i P^i_a$, with $P \in \text{GL}_n(K)$, then

$$T'^{a_1 \dots a_r}_{b_1 \dots b_s} = (P^{-1})^{a_1}_{i_1} \dots (P^{-1})^{a_r}_{i_r} P^{j_1}_{b_1} \dots P^{j_s}_{b_s} T^{i_1 \dots i_r}_{j_1 \dots j_s}. \quad (3.1)$$

Coordinate transformations $y = \Phi(x)$ instead give vector components $X^a_y = (\partial y^a / \partial x^i) X^i_x$ and covector components $\alpha_{y,a} = (\partial x^i / \partial y^a) \alpha_{x,i}$. Distinguishing a basis change from a coordinate map prevents an inverse-Jacobian ambiguity.

3.2 Metrics and norms at a point

A metric at a point is a symmetric nondegenerate bilinear form $g : V \times V \rightarrow K$. Sylvester's law of inertia holds over a real closed field: in a suitable basis,

$$g(u, u) = u_1^2 + \cdots + u_p^2 - u_{p+1}^2 - \cdots - u_n^2, \quad p + q = n.$$

This follows by symmetric Gaussian elimination and rescaling nonzero diagonal entries by positive square roots; the numbers of positive and negative terms are intrinsic by the ordinary finite-dimensional intersection argument.

For positive-definite g , put $\|u\|_g = \sqrt{g(u, u)} \in K_{\geq 0}$. For $v \neq 0$, positivity of $g(u - av, u - av)$ at $a = g(u, v)/g(v, v)$ proves

$$g(u, v)^2 \leq g(u, u)g(v, v).$$

Consequently the triangle inequality and Gram–Schmidt orthonormalization hold. These are ordered-field inequalities. The norm is K -valued, not a real-valued Banach norm. No completeness theorem is being asserted.

The metric identifies vectors and covectors by $X_i^\flat = g_{ij}X^j$ and $\alpha^{\sharp i} = g^{ij}\alpha_j$. A surreal infinitesimal is not nilpotent: for nonzero ε , every finite power ε^m is nonzero. Dropping higher powers therefore changes an exact tensor into a truncation and must be indicated.

3.3 Which component scales are invariant?

The coordinate minimum $v_V(X) = \min_i v(X^i)$ depends on a chosen lattice $\mathcal{O}_K^n \subset K^n$. If $P \in \mathrm{GL}_n(\mathcal{O}_K)$, where the inverse also has entries in $\mathcal{O}_K$, both P and P^{-1} cannot lower this minimum. Thus

$$v_V(PX) = v_V(X).$$

The same argument applies to the minimum component valuation of any fixed finite tensor type. Under arbitrary $\mathrm{GL}_n(K)$ changes this is false: $e'_1 = t^\eta e_1$ changes the first component of e_1 to $t^{-\eta}$. Invariant scale claims should refer either to a specified integral lattice or to scalar contractions such as $g(X, X)$, $\mathrm{tr}(T^2)$ and curvature invariants.

When X has entries in $\mathcal{O}_K$ as well, coefficient extraction commutes with such a change of frame: $\mathrm{st}(PX) = (\mathrm{st} P) \mathrm{st}(X)$ for $P \in \mathrm{GL}_n(\mathcal{O}_K)$, because st is a ring homomorphism on $\mathcal{O}_K$. Source 02 proves both statements for the bounded Lorentz group (theorem 32.2) and shows with an infinite boost that neither extends to all of $\mathrm{GL}_n(K)$ (theorem 32.3); the invariance is also the collection's `phys:prop:frame`. Source 03 records the case of real frames and adds leading-order information in three dimensions (theorem 22.3).

4 The sheaf of smooth Hahn fields

4.1 A common-domain algebra

For an ordinary open $U \subseteq \mathbb{R}^n$ define

$$\mathcal{A}_\Gamma(U) = C^\infty(U)((t^\Gamma)) = \left\{ \sum_{\gamma \in S} f_\gamma t^\gamma : S \subset \Gamma \text{ well ordered, } f_\gamma \in C^\infty(U) \right\}. \quad (4.1)$$

Support means the exponents with coefficient function not identically zero. Evaluation at any real $x \in U$ is a well-defined map into K . The representation as a function $U \rightarrow K$ is injective: equality at every x forces equality of each real coefficient function.

Proposition 4.1 (Differential algebra). *The algebra (4.1) is a commutative K -algebra. The operators*

$$D_i \left(\sum_{\gamma} f_{\gamma} t^{\gamma} \right) = \sum_{\gamma} (\partial_i f_{\gamma}) t^{\gamma}$$

are commuting K -linear derivations. They preserve strong summability and never enlarge support. Pullback along an ordinary smooth map is a strongly linear algebra homomorphism and satisfies the ordinary chain rule.

Proof. Products have support in $S + T$. By lemma 2.4, each coefficient is a finite sum of products of real smooth functions. Differentiating that finite sum gives Leibniz's rule. Mixed derivatives commute coefficientwise, and restriction, differentiation and real pullback can only delete exponents. All the other assertions reduce to finite coefficient operations and the real smooth chain rule. $\square$

This definition does *not* mean that $U \rightarrow K$ is continuous when U has its usual topology and K its intrinsic order topology. It is the chosen smooth coefficient structure. The intrinsic realization is supplied later.

4.2 Hahn vector spaces and positive-order perturbations

The same coefficientwise constructions apply to any real vector space. This subsection, from source 03, records the two facts used repeatedly below.

For a real vector space E , define

$$E((t^{\Gamma})) = \left\{ \sum_{\gamma \in S} t^{\gamma} e_{\gamma} : e_{\gamma} \in E, S \text{ well ordered} \right\}.$$

This is a K -vector space. A real bilinear map $B : E \times F \rightarrow G$ extends by convolution:

$$B \left(\sum t^{\alpha} e_{\alpha}, \sum t^{\beta} f_{\beta} \right) = \sum_{\gamma} t^{\gamma} \sum_{\alpha+\beta=\gamma} B(e_{\alpha}, f_{\beta}). \quad (4.2)$$

The inner sum is finite by lemma 2.4.

For a real smooth manifold M and real vector bundle $E \rightarrow M$, source 03 writes

$$\mathcal{A}_{\Gamma}(M) = C^{\infty}(M)((t^{\Gamma})), \quad (4.3)$$

$$\mathfrak{X}_{\Gamma}(M) = C^{\infty}(M, TM)((t^{\Gamma})), \quad (4.4)$$

$$\mathcal{T}_{s,\Gamma}^r(M) = C^{\infty}(M, T_s^r M)((t^{\Gamma})), \quad (4.5)$$

$$\Omega_{\Gamma}^k(M) = \Omega^k(M)((t^{\Gamma})). \quad (4.6)$$

These are global-support spaces; on a compact M , $\Omega_{\Gamma}^k(M)$ is the space $\Omega_H^k(M)$ of section 12. Evaluation at a real point is coefficientwise and gives a tensor over K . Evaluation on all real points is injective: a coefficient field vanishing everywhere is zero.

Theorem 4.2 (Algebraic closure of the field calculus; source 03). *Addition, tensor product, contraction, permutation, symmetric and alternating products, and every fixed real-linear differential operator preserve these spaces. A real-linear operator does not enlarge support. A bilinear operation has support contained in the sum of the input supports. All finite polynomial identities between such operations extend from real fields to Hahn-smooth fields.*

Proof. Linear operations act on each coefficient. Bilinear operations use (4.2). For an identity, fix an exponent. Its coefficient is a finite sum of the corresponding real multilinear identities, after polarization when necessary. Thus it is zero. This proof is algebraic; continuity in a real function-space norm is not needed for the extension itself. $\square$

Theorem 4.3 (Support-controlled inverse; source 03). *Let E, F be real vector spaces and $L_0 : E \rightarrow F$ a real-linear isomorphism. Let*

$$P = \sum_{\sigma \in S} t^\sigma P_\sigma, \quad S \subseteq \Gamma_{>0} \text{ well ordered}, \quad P_\sigma : E \rightarrow F \text{ real linear.}$$

Then $L = L_0 + P : E((t^\Gamma)) \rightarrow F((t^\Gamma))$ is an isomorphism. Its inverse is the strongly defined operator expression

$$L^{-1} = \sum_{n \geq 0} (-L_0^{-1}P)^n L_0^{-1}. \quad (4.7)$$

For data f with support T , the solution has support in $T + \langle S \rangle$. If $P \neq 0$ and $s = \min S$, then

$$v_H(L^{-1}f - L_0^{-1}f) \geq v_H(f) + s.$$

The bound allows the left side to be $+\infty$.

Proof. Expanding the n th term produces compositions of n real operators with exponents in words from S . By lemma 2.4, only finitely many such terms contribute to each output exponent, for all n together. Thus (4.7) defines an actual map on Hahn vectors, and finite geometric-series cancellation proves both inverse identities coefficientwise. Alternatively, uniqueness follows by comparing the least exponent of a nonzero u : the leading coefficient of $L_0 u$ is nonzero, and every coefficient of Pu has strictly larger exponent. The support and valuation bounds follow directly from the displayed expansion. $\square$

There is no smallness condition in an ordinary operator norm. The coefficients P_σ may be differential operators unbounded in some Hilbert norm. What is required is that each one be defined on all of the specified real space E with values in F , and that L_0^{-1} exist on all of F . Domain and boundary conditions cannot be discarded.

Remark 4.4 (More than a scalar extension). Usually $E \otimes_{\mathbb{R}} K$ is a strict subspace of $E((t^\Gamma))$: its elements involve only finitely many real coefficient vectors. For example $F(x) = \sum_{m \geq 0} t^m x^m \in C^\infty(\mathbb{R})((t^\mathbb{Q}))$ has infinitely many linearly independent coefficient functions, so it is not a finite sum $\sum_{j=1}^k a_j f_j(x)$ with $a_j \in K$. This completion in the scale direction is essential; it is not a completion in the spatial variable (sources 02 and 03).

Warning 4.5 (The field algebra is not a field; source 02). $C^\infty(U)$ has zero divisors, for example two nonzero bump functions with disjoint supports, and so does $\mathcal{A}_\Gamma(U)$. The least exponent $v_H(f) = \min\{\gamma : f_\gamma \neq 0\}$ satisfies $v_H(fg) \geq v_H(f) + v_H(g)$, not equality in general. Statements about the valuation of numbers in K must not be copied verbatim to coefficient functions. Since D_i acts on coefficients,

$$v_H(D_i f) \geq v_H(f). \quad (4.8)$$

Membership in the subring $\mathcal{A}_{\geq 0}(U)$ of fields with nonnegative support is equivalent to finiteness of the value at every real point: if a negative-exponent coefficient function is nonzero at some point, the value there has a negative leading exponent. Likewise, infinitesimal values at every real point characterize the positive-support ideal $\mathcal{A}_{> 0}(U)$. No uniform bound on the sizes of the coefficient functions or their derivatives accompanies this (source 02). For instance, $\sum_{m \geq 0} t^m \sin(mx^1)$ belongs to $\mathcal{A}_{\mathbb{Q}}(\mathbb{R}^n)$ and has derivatives of every order; the growing real factors m^k in its k th derivative are harmless to the Hahn support. This does not assert convergence after replacing t by a positive real number, and it does not put an infinite frequency $\sin(t^{-1}x^1)$ in the algebra (section 19).

4.3 Why local support is necessary

The presheaf $U \mapsto \mathcal{A}_{\Gamma}(U)$ need not be a sheaf for arbitrary open covers. On disjoint intervals U_m one can prescribe the constant section $t^{1/m}$. Each local section is admissible, but the union of the exponents $\{1/m : m \geq 1\}$ is not well ordered. The glued function has no single representation (4.1) on $\bigcup_m U_m$. Sources 02 and 03 give the same obstruction with the sections t^{-m} , whose exponents are unbounded below.

Define $\mathcal{A}_{\Gamma}^{\text{loc}}$ to consist of functions that admit such a representation near every real base point. This is a sheaf. Uniqueness of coefficient extraction makes gluing unambiguous. All local algebra and calculus statements apply to it. On a smooth manifold, use real coordinate charts and the pullbacks of proposition 4.1.

On a compact real manifold there is no global support defect. A finite subcover of local presentations gives a finite union of well-ordered supports, hence one well-ordered set. Coefficient functions, or coefficient tensors, agree on overlaps and glue to global smooth objects. The analogous statement holds for a section restricted to a neighborhood of a compact set after a finite localization.

Global theorems below use explicitly specified common-domain sections. Global support is not inferred from pointwise well ordering or from local admissibility on a noncompact domain (sources 02 and 03).

A pointwise family that is not Hahn-smooth (source 03). Suppose $\Gamma = \mathbb{R}$ and put $F(x) = t^x$ for real $0 < x < 1$. Every value lies in K , but $F \notin C^{\infty}((0, 1))((t^{\mathbb{R}}))$: the coefficient of t^{γ} would be the indicator of $\{\gamma\}$, which is not smooth, and the supports have no common well ordering. “All components are surreal” is weaker than the condition the calculus needs.

Field valuation and point values (source 03). A leading coefficient field may vanish at some points. Thus $v_H(F)$ controls every value $F(x)$ from below but need not equal its valuation. Estimates below use v_H ; pointwise statements evaluate leading coefficients explicitly.

4.4 Not every nonzero field is a unit

A sufficient local unit condition is

$$f = t^{\lambda}(a + h), \quad a \in C^{\infty}(U), \quad a(x) \neq 0 \text{ on } U, \quad \text{supp}(h) \subset \Gamma_{> 0}. \quad (4.9)$$

Then

$$f^{-1} = t^{-\lambda} a^{-1} \sum_{m \geq 0} (-a^{-1} h)^m \in \mathcal{A}_{\Gamma}(U).$$

If $a > 0$, divisibility of Γ also gives

$$\sqrt{f} = t^{\lambda/2} \sqrt{a} \sum_{m \geq 0} \binom{1/2}{m} (h/a)^m.$$

All these supports are translates of positive generated monoids.

Example 4.6 (A pointwise inverse outside the smooth Hahn algebra; sources 01 and 03). Fix $\varepsilon = t^\eta$, $\eta > 0$, and set $f(x) = x^2 + \varepsilon^2$ on a real interval around zero. Every $f(x)$ is positive and invertible in K . Nevertheless, f^{-1} is not in $\mathcal{A}_\Gamma^{\text{loc}}$ at zero.

Indeed, at $x = 0$ its coefficient at exponent -2η is 1. At every nonzero real x the exact expansion is

$$\frac{1}{x^2 + \varepsilon^2} = \sum_{m \geq 0} (-1)^m x^{-2m-2} \varepsilon^{2m},$$

so that same coefficient is zero. It could not be a continuous real coefficient function on a neighborhood of zero. This example will distinguish pointwise nondegeneracy from an admissible metric inverse.

5 Intrinsic derivatives and Taylor–Hahn prolongation

5.1 The topology that is actually used

Give K its *intrinsic* order topology, whose radii lie in $K_{>0}$. On K^n use the equivalent coordinate maximum norm $\|x\|_\infty = \max_i |x^i|$. A derivative of $F : W \subseteq K^n \rightarrow K^m$ at y is a K -linear map L satisfying

$$\forall \epsilon \in K_{>0} \exists \rho \in K_{>0} \quad 0 < \|k\|_\infty < \rho \implies \|F(y+k) - F(y) - Lk\|_\infty < \epsilon \|k\|_\infty, \quad (5.1)$$

with a ball about y contained in W . The linear map is unique: test the difference of two candidates on small multiples of each coordinate vector. Polynomial and rational functions are differentiable on their denominator nonzero domains by finite algebraic remainder identities.

The topology inherited by a set-sized $K \subset \mathbf{No}$ from the *full* surreal fine topology is different: it is discrete. To see the underlying obstruction, let S be any set of surreals and fix x . There is a positive surreal smaller than every nonzero distance $|s - x|$, $s \in S$, by the small-cut property. A sufficiently small ball isolates x from $S \setminus \{x\}$. Every set-sized subset is thus closed and discrete. In particular, ordinary real-parameter curves into full $\mathbf{No}$ that are continuous in that fine topology must be constant. These size-sensitive facts are also emphasized in the repository [2]. We never give K that inherited topology when using (5.1).

Sources 02 and 03 add three observations about the intrinsic topology. First, a non-Archimedean ordered field $K \supset \mathbb{R}$ has $\mathfrak{m}_K$ as an open and closed additive subgroup, and translates and rescalings of it separate points; so K and K^n are totally disconnected, and an intrinsically continuous map from a connected real interval into K^n is constant. Second, the map $\mathbb{R} \rightarrow K$ sending a real number to the same constant is not continuous from the ordinary topology to the order topology: an infinitesimal ball about 0 meets $\mathbb{R}$ only in 0. The derivative (5.1) is therefore used on open subsets of K^n , never along ordinary real curves, and the smooth structure of $\mathcal{A}_\Gamma$ is specified coefficientwise. Third, a fixed support presentation can be given coefficientwise smooth convergence; no argument below equates strong summation, coefficientwise convergence and valuation-topological convergence.

Proposition 5.1 (Finite-partition obstruction; source 03). *Let $K \supset \mathbb{R}$ be non-Archimedean. Ordinary finite-partition Darboux integration, with error required to be smaller than every positive element of K , fails for $f(x) = x$ on $[0, 1]_K$.*

Proof. For a partition with N subintervals and lengths $\ell_j > 0$, the difference between upper and lower sums is $\sum_j \ell_j^2$. Since $\sum_j \ell_j = 1$, finite Cauchy–Schwarz gives $\sum_j \ell_j^2 \geq 1/N$. Every positive infinitesimal is smaller than $1/N$ for every ordinary finite N . Therefore no such partition achieves infinitesimal error. The argument permits partition points in K , not just in $\mathbb{R}$. $\square$

The same obstruction, with an explicit bound, is `contours:prop:mesh`. Integrals below are either coefficientwise over real chains or algebraic on polynomial chains.

5.2 Prolongation of arbitrary smooth coefficients

For a real open $U \subset \mathbb{R}^n$ put

$$U^h = \{a + h : a \in U, h \in \mathfrak{m}_K^n\} \subset K^n.$$

It is intrinsically open. The decomposition $a = \text{st}(y)$, $h = y - a$ is unique. For $F = \sum_\gamma f_\gamma t^\gamma \in \mathcal{A}_\Gamma(U)$ define

$$(\text{Ev } F)(a + h) = \sum_{\gamma \in \text{supp}(F)} \sum_{\alpha \in \mathbb{N}^n} \frac{\partial^\alpha f_\gamma(a)}{\alpha!} t^\gamma h^\alpha. \quad (5.2)$$

This is a Taylor-*jet* construction. It does not assert convergence of a real Taylor series to a real smooth function on an ordinary neighborhood.

Theorem 5.2 (Multivariable smooth prolongation). *Formula (5.2) is strongly summable and defines an injective K -algebra homomorphism into functions on U^h . It extends evaluation on real points. Its values are intrinsically C^∞ in the sense of iterated (5.1), and*

$$\partial_{y^i}(\text{Ev } F) = \text{Ev}(D_i F). \quad (5.3)$$

For every ordinary smooth map $\Phi : U \rightarrow V$,

$$\text{Ev}(\Phi^* F) = (\text{Ev } F) \circ (\text{Ev } \Phi), \quad (5.4)$$

where $\text{Ev } \Phi$ is componentwise prolongation. These statements localize to $\mathcal{A}_\Gamma^{\text{loc}}$.

Proof. Let $S = \text{supp}(F)$, and let T be the union of the nonzero coordinate supports of h . It is a finite union of well-ordered positive sets. Every summand of (5.2) has support in $S + \langle T \rangle$. A fixed exponent has only finitely many choices of γ , word length, monomials from the coordinates of h , and multiindex: this is lemma 2.4, with the finite coordinate labels attached to each word. Thus the displayed double sum is strong. Evaluation at $h = 0$ gives the asserted restriction and proves injectivity by coefficient extraction on all real points.

For each a , taking the full Taylor jet $J_a : C^\infty(U) \rightarrow \mathbb{R}[[Z_1, \dots, Z_n]]$ is an algebra homomorphism. Leibniz’s formula proves this degree by degree. Substitution of positive-order h into these jets is a homomorphism by the same support lemma. This proves additivity, multiplicativity and compatibility with K constants.

We give the derivative estimate, since it is the important additional step. Fix $y = a + h$. For $k \in \mathfrak{m}_K^n$, formal Taylor translation and strong regrouping yield

$$\text{Ev } F(y + k) = \text{Ev } F(y) + \sum_i \text{Ev}(D_i F)(y) k^i + \sum_{i,j} k^i k^j R_{ij}(h, k). \quad (5.5)$$

The remainders are evaluations of formal real series with Hahn supports in $S + \langle T_h \cup T_k \rangle$. If $F \neq 0$ and $\beta = \min S$, every R_{ij} has valuation at least β , independently of the small k . This follows by removing two of the k factors from terms of k -degree at least two; the remaining factors all have nonnegative valuation. In particular,

$$v \left(\text{Ev } F(y + k) - \text{Ev } F(y) - \sum_i \text{Ev}(D_i F)(y) k^i \right) \geq \beta + 2v(\|k\|_\infty).$$

Given $\epsilon > 0$, choose $\rho = t^\theta$ with $\theta > 0$ and $\beta + \theta > v(\epsilon)$. For $0 < \|k\|_\infty < \rho$, the remainder divided by $\|k\|_\infty$ has valuation strictly greater than $v(\epsilon)$ and hence absolute value less than ϵ . The ball stays in the same halo. This proves (5.3). Applying the same argument to every $D^\alpha F$ proves existence of all iterated derivatives; its first-order difference estimate proves their continuity.

Finally, the ordinary smooth chain rule in every finite jet degree gives the formal identity

$$J_a(f \circ \Phi) = (J_{\Phi(a)} f) \circ (J_a \Phi - \Phi(a)).$$

The inner series have zero constant terms and evaluate to infinitesimals. The common positive-support certificate justifies every regrouping, proving (5.4). Agreement of local representatives proves localization. $\square$

Remark 5.3 (Three proofs of the prolongation). Sources 02 and 03 prove the algebraic half of [theorem 5.2](#) by the same support argument, with the same formal Taylor homomorphism: strong summability on $\text{supp } F + \langle T \rangle$, the homomorphism property, compatibility with formal differentiation, and the chain rule. Source 03 also proves the intrinsic derivative at each halo point, with a slightly different bound: for any fixed $\rho \in \Gamma_{>0}$ the remainder is at most $t^{\beta-\rho} r^2$, where $r = \max_i |k^i|$ is infinitesimal. Source 01's statement is the one printed, since it includes all iterated intrinsic derivatives and the naturality (5.4). Source 02 phrases the prolongation on the set $\{z \in \mathcal{O}_K^n : \text{st } z \in U\}$ of finite points over U , which is U^h .

Remark 5.4 (Smooth is not analytic). If $f(x) = e^{-1/x^2}$ for $x \neq 0$ and $f(0) = 0$, then $\text{Ev } f$ vanishes on the whole halo $\mathfrak{m}_K$ of zero, because its jet there is zero. It still agrees with f at every nonzero real point. This is a well-defined smooth prolongation, not an identity theorem for analytic functions.

Source 03 sharpens the point with the one-sided function $f(x) = e^{-1/x^2}$ for $x > 0$ and $f(x) = 0$ for $x \leq 0$. Its prolongation vanishes at every positive infinitesimal, whereas the surreal exponential of $-1/x^2$ does not. There is no contradiction: $-1/x^2$ is not smooth at 0, and the halo of $(0, \infty)$ contains no point with standard part 0. Taylor prolongation is therefore not a canonical extension of all smooth expressions to all surreal arguments, and it does not address the extension problems of Costin and Ehrlich [19]. For real-analytic coefficients the prolongation agrees with the admissible infinitesimal Taylor expansion. At an unlimited argument neither a real base point nor this prescription exists (source 02).

5.3 Nonuniqueness outside the chosen class

On $\mathbb{R}^h = \mathcal{O}_K$, the functions $y \mapsto y$ and $y \mapsto \text{st}(y)$ agree at every real point. The first has derivative 1; the second is locally constant on each open halo and has derivative 0. Thus restriction to real points

plus intrinsic smoothness does not determine an extension. Theorem 5.2 chooses the extension by all real Taylor jets. The kernel of the derivative is constants on a connected *real base* inside our Hahn class; it is much larger among all intrinsic smooth functions.

5.4 Surreal-valued coordinate changes

Let $\Phi_0 : U \rightarrow V$ be real smooth and let $H \in \mathcal{A}_\Gamma(U)^m$ have strictly positive support. Write $\Phi = \Phi_0 + H$. For $F = \sum_\sigma f_\sigma t^\sigma \in \mathcal{A}_\Gamma(V)$ define

$$\Phi^* F = \sum_{\sigma, \alpha} \frac{(\partial^\alpha f_\sigma) \circ \Phi_0}{\alpha!} t^\sigma H^\alpha. \quad (5.6)$$

The support is contained in $\text{supp}(F) + \langle \text{supp}(H) \rangle$. The proof of theorem 5.2 applies verbatim to show that pullback is an algebra map, obeys the chain rule, and realizes actual composition on halos.

Proposition 5.5 (Lifting coordinate diffeomorphisms). *If Φ_0 is a real diffeomorphism and H has positive support, then $\text{Ev } \Phi : U^h \rightarrow V^h$ is a bijection with an inverse of the same Taylor–Hahn type. The correction to Φ_0^{-1} has support in $\langle \text{supp}(H) \rangle \setminus \{0\}$.*

Proof. Write $\Psi = \Phi_0^{-1} + Q$. At exponent $\gamma > 0$, the equation $\Phi \circ \Psi = \text{id}$ is a linear equation for Q_γ with coefficient matrix $D\Phi_0 \circ \Phi_0^{-1}$. This matrix has a smooth real inverse. All remaining terms depend on strictly smaller positive exponents. The support lemma makes their number finite at each exponent. Transfinite recursion over the generated monoid constructs a unique Q . Construct a left inverse in the same way. A left inverse and a right inverse coincide by associativity of the already certified compositions, proving both the formal and realized statements. $\square$

6 Genuinely surreal-coordinate neighborhoods

6.1 Arbitrary centers and scales

The finite halo is not all of K^n . To work near arbitrary $a \in K^n$, choose nonzero scales $r_1, \dots, r_n \in K$ and coordinates

$$z^i = \frac{x^i - a^i}{r_i}, \quad z \in \mathfrak{m}_K^n.$$

One may evaluate every element of

$$\mathbb{R}[[Z_1, \dots, Z_n]]((t^\Gamma)) \quad (6.1)$$

at such a z : its outer Hahn support is fixed, while the coefficient at each exponent is an arbitrary real formal series. The support proof and the remainder proof of theorem 5.2 still apply. In these coordinates

$$\partial_{x^i} = r_i^{-1} \partial_{z^i}.$$

These scale factors matter: an infinitesimal spatial scale can turn a modest coordinate gradient into an infinite physical-coordinate gradient.

Polynomials and rational functions over K are available independently of this presentation. Near a point where a rational denominator is nonzero, shrink a K -radius so that its relative change has positive valuation, then use geometric inversion. Such rational germs are covered by sufficiently small scale charts even when they fail to be smooth Hahn fields over a large real base, as in section 4.4.

Linear scale charts (source 03). More generally, for $c \in K^n$ and $L \in \mathrm{GL}_n(K)$ put $x = c + Ly$ with $y \in U^h$. This accommodates an infinite centre, an infinitesimal cell, and different scales along non-coordinate directions. For $F(x) = \mathrm{Ev} f(L^{-1}(x - c))$ and a vector field $V(x) = L \mathrm{Ev} v(y)$ the chain rule gives

$$\mathrm{grad}_x F = L^{-T} \mathrm{grad}_y \mathrm{Ev} f, \quad D_x V = L (D_y \mathrm{Ev} v) L^{-1}, \quad \mathrm{div}_x V = \mathrm{div}_y \mathrm{Ev} v,$$

with Euclidean gradients and divergences in both charts. The pulled-back Euclidean metric is $L^T L$ and the volume factor is $|\det L|$. Source 03 states these formulas for $n = 3$; the computation uses no property of dimension three. Gluing two such charts requires a transition map admissible in both chart calculi, exactly as in [section 6.2](#). The three-dimensional cross product transforms with the extra factor in [\(23.15\)](#). Source 02 uses the isotropic case $x = a + t^b X$ in Minkowski space ([section 39.3](#)).

6.2 An important exclusion

An arbitrary element of $K[[Z]]$ is not promised to evaluate on the whole infinitesimal halo. For example,

$$\sum_{m \geq 0} t^{-m^2 \eta} Z^m \quad \text{at} \quad Z = t^\eta$$

has proposed exponents $(-m^2 + m)\eta$, not a well-ordered support. The uniform outer support in [\(6.1\)](#) is a sufficient hypothesis that excludes this example. Other, larger analytic classes are possible, but they need their own support certificates or domains. A formal inverse function theorem alone is not a convergence theorem on a specified domain.

An intrinsic n -dimensional atlas may now be built from such open subsets of K^n , requiring every transition and its inverse to belong to an explicitly admitted calculus. Real diffeomorphism prolongations, their positive-support perturbations, affine scale changes, and rational local coordinate changes provide concrete examples. Tensor and connection formulas below apply in any such atlas once its chain rule and closure hypotheses have been checked. Global integration and cohomology statements will continue to name the real base or algebraic chains to which they apply.

7 Vector fields, tensor fields, and exterior calculus

7.1 Modules of geometric fields

On a real coordinate domain write $\mathcal{A} = \mathcal{A}_\Gamma^{\mathrm{loc}}$ and define

$$\mathfrak{X}_{\mathcal{A}}(U) = \bigoplus_{i=1}^n \mathcal{A}(U) D_i, \quad \Omega_{\mathcal{A}}^1(U) = \bigoplus_{i=1}^n \mathcal{A}(U) dx^i.$$

These glue under the real Jacobian laws. Tensor fields are sections of the corresponding finite tensor powers. For an ordinary bundle $E \rightarrow M$, its Hahn extension has locally the same construction; globally a section has locally well-ordered sums of real sections. We do not identify these geometric fields with every possible algebraic derivation of a huge function ring without a locality or strong-linearity condition.

A vector field acts by $X(F) = X^i D_i F$. Its Lie bracket is

$$[X, Y]^i = X^j D_j Y^i - Y^j D_j X^i. \tag{7.1}$$

Because it is the commutator of derivations, the bracket is antisymmetric and satisfies Jacobi. Coordinate invariance follows from the chain rule and cancellation of the symmetric second derivatives of a coordinate change. It is K -bilinear, but not $\mathcal{A}$ -bilinear: $[X, fY] = f[X, Y] + X(f)Y$.

7.2 Forms, exterior derivative, and Cartan calculus

Define

$$\Omega_{\mathcal{A}}^p = \bigwedge_{\mathcal{A}}^p \Omega_{\mathcal{A}}^1, \quad \text{rank}_{\mathcal{A}} \Omega_{\mathcal{A}}^p = \binom{n}{p}.$$

For increasing multiindices I , set

$$d \left(\sum_I \alpha_I dx^I \right) = \sum_{I,j} D_j \alpha_I dx^j \wedge dx^I.$$

Alternation and commutation of the D_i prove $d^2 = 0$. For homogeneous α of degree p ,

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta.$$

Insertion is $(\iota_X \alpha)(X_2, \dots, X_p) = \alpha(X, X_2, \dots, X_p)$. Define the Lie derivative on forms by Cartan's formula

$$\mathcal{L}_X = d\iota_X + \iota_X d. \quad (7.2)$$

It follows algebraically that $[\mathcal{L}_X, d] = 0$, $[\mathcal{L}_X, \iota_Y] = \iota_{[X,Y]}$ and $[\mathcal{L}_X, \mathcal{L}_Y] = \mathcal{L}_{[X,Y]}$. To verify these identities directly, note that both sides are graded derivations and agree on functions and coordinate one-forms, which generate the exterior algebra. No flow theorem is needed for this definition.

For a tensor T of type (r, s) the component formula is

$$\begin{aligned} (\mathcal{L}_X T)^{i_1 \dots i_r}_{j_1 \dots j_s} &= X^k D_k T^{i_1 \dots i_r}_{j_1 \dots j_s} \\ &\quad - \sum_{a=1}^r (D_k X^{i_a}) T^{i_1 \dots k \dots i_r}_{j_1 \dots j_s} + \sum_{b=1}^s (D_{j_b} X^k) T^{i_1 \dots i_r}_{j_1 \dots k \dots j_s}. \end{aligned} \quad (7.3)$$

Its signs express the contravariant and covariant transformation laws. Theorem 5.2 transports all these identities to halo realizations, and the intrinsic atlas construction transports them to any admitted coordinate domain.

8 Admissible metrics and their Levi–Civita connections

8.1 Pointwise versus differential nondegeneracy

A symmetric matrix $g_{ij} \in \mathcal{A}(U)$ may be nondegenerate at every real point without its inverse belonging to $\mathcal{A}(U)$. For a *metric in this calculus*, we require $\det g$ to be a unit of the chosen local coefficient algebra. When a volume form is used, its square root is required in that algebra as well. This is an admissibility condition, not an extra condition on finite-dimensional metric algebra at one point.

A particularly useful class is

$$g = g_0 + h, \quad g_0 \text{ a real smooth nondegenerate metric, } \text{supp}(h) \subset \Gamma_{>0}. \quad (8.1)$$

Finite matrix inversion gives the exact formula

$$g^{-1} = \sum_{m \geq 0} (-g_0^{-1}h)^m g_0^{-1}. \quad (8.2)$$

Its support is in $\langle \text{supp}(h) \rangle$. The signature is that of g_0 : locally diagonalize the real metric smoothly, use nonzero real principal minors there, and observe that infinitesimal changes cannot alter their signs. The same argument holds at every point of the prolonged halo.

More generally, constant scale factors are allowed. If $B \in \text{GL}_n(K)$ is constant and $g = B^T(g_0 + h)B$, its inverse remains admissible. This includes strongly anisotropic metrics not having a nondegenerate real standard part.

8.2 Construction and uniqueness

An affine connection satisfies

$$\nabla_f X Y = f \nabla_X Y, \quad \nabla_X(fY) = X(f)Y + f \nabla_X Y.$$

Its coefficients are $\nabla_{D_i} D_j = \Gamma^k_{ij} D_k$. The torsion is $T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]$.

Theorem 8.1 (Levi-Civita construction in the admissible algebra). *For every admissible metric there is a unique torsion-free, metric-compatible connection. It is given by*

$$\Gamma^k_{ij} = \frac{1}{2} g^{k\ell} (D_i g_{j\ell} + D_j g_{i\ell} - D_\ell g_{ij}). \quad (8.3)$$

Proof. The inverse metric and its derivatives lie in the chosen algebra, so the formula is defined. It is symmetric in i, j , hence torsion-free. Multiplying by g_{mk} and adding the appropriate permutations gives $D_i g_{jk} = g_{\ell k} \Gamma^\ell_{ij} + g_{j\ell} \Gamma^\ell_{ik}$, which is metric compatibility. Conversely those three permuted equations, together with symmetry in the lower indices, force (8.3). This proves uniqueness. Coordinate independence can either be checked with the chain rule or read from the coordinate-free Koszul identity

$$\begin{aligned} 2g(\nabla_X Y, Z) &= Xg(Y, Z) + Yg(Z, X) - Zg(X, Y) \\ &\quad - g(X, [Y, Z]) + g(Y, [Z, X]) + g(Z, [X, Y]). \end{aligned}$$

Every term is an operation already established in the differential algebra. $\square$

Source 03 proves the same construction for regular Riemannian metrics, together with their pointwise positive definiteness over K at real and halo points: after normalizing $u \neq 0$ by its least component valuation, the leading coefficient of $g(u, u)$ is $g_0(u_0, u_0) > 0$. It also shows that the inverse, volume density, Hodge star, connection and curvature are Hahn-smooth, with nonzero supports in $\langle \text{supp } h \rangle$ and the real objects at exponent zero. These are the Riemannian cases of (8.2), of the signature argument after it, and of [proposition 13.2](#), so they are printed once, here.

On general tensors the connection is determined by the product rule and contraction compatibility:

$$\begin{aligned} (\nabla_k T)^{i_1 \dots i_r}_{j_1 \dots j_s} &= D_k T^{i_1 \dots i_r}_{j_1 \dots j_s} \\ &\quad + \sum_{a=1}^r \Gamma^{i_a}_{k\ell} T^{i_1 \dots \ell \dots i_r}_{j_1 \dots j_s} - \sum_{b=1}^s \Gamma^\ell_{kj_b} T^{i_1 \dots i_r}_{j_1 \dots \ell \dots j_s}. \end{aligned} \quad (8.4)$$

Metric compatibility makes raising and lowering indices commute with ∇ . For a function, $\text{Hess}_g f_{ij} = D_i D_j f - \Gamma^k_{ij} D_k f$; the Hessian is symmetric because the connection has no torsion.

8.3 Parallel transport and geodesic equations

Along an admitted curve $x(s)$, parallel transport and geodesics satisfy

$$\frac{dY^k}{ds} + \Gamma^k_{ij}(x(s))\dot{x}^i Y^j = 0, \quad \ddot{x}^k + \Gamma^k_{ij}(x(s))\dot{x}^i \dot{x}^j = 0.$$

These are well-defined differential equations in an admitted calculus, not yet universal existence assertions. Metric compatibility and the product rule imply

$$\frac{d}{ds}g(Y, Z) = 0, \quad \frac{d}{ds}g(\dot{x}, \dot{x}) = 0$$

for parallel fields and geodesics, respectively. These quantities are constant in the coefficientwise calculus on a connected real parameter interval, and therefore on its Taylor–Hahn prolongation. In a bare intrinsic calculus, zero derivative alone is not a global constancy theorem; the example in [section 5.3](#) explains the distinction. Existence for positive-support perturbations of real trajectories is proved in [section 15](#); exact polynomial solutions may also be available outside that class.

9 Curvature and geometric identities

Fix the convention

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]}Z, \quad R(D_i, D_j)D_k = R^\ell_{kij} D_\ell.$$

Then

$$R^\ell_{kij} = D_i \Gamma^\ell_{jk} - D_j \Gamma^\ell_{ik} + \Gamma^\ell_{im} \Gamma^m_{jk} - \Gamma^\ell_{jm} \Gamma^m_{ik}. \quad (9.1)$$

In particular R is a tensor although Γ is not. Define

$$\text{Ric}_{kj} = R^i_{kij}, \quad \text{Scal} = g^{kj} \text{Ric}_{kj}, \quad G_{ij} = \text{Ric}_{ij} - \frac{1}{2} \text{Scal} g_{ij}.$$

Proposition 9.1 (Bianchi identities and conservation). *For an admissible Levi–Civita connection, the first and second Bianchi identities and the contracted identity hold:*

$$R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0,$$

$$(\nabla_X R)(Y, Z) + (\nabla_Y R)(Z, X) + (\nabla_Z R)(X, Y) = 0, \quad \nabla^i G_{ij} = 0.$$

The curvature has the usual metric symmetries, and Ric is symmetric.

Proof. Insert (9.1) into the cyclic sum. Derivative terms cancel because $\Gamma^\ell_{ij} = \Gamma^\ell_{ji}$, and the product terms cancel in pairs. For the second identity, write the connection matrix $\Gamma = \Gamma_i dx^i$ and curvature matrix $\Omega = d\Gamma + \Gamma \wedge \Gamma$. Direct expansion, using $d^2 = 0$, gives

$$d\Omega + \Gamma \wedge \Omega - \Omega \wedge \Gamma = 0.$$

This is the covariant second Bianchi identity. Differentiating $g(Z, W)$ twice and commuting derivatives gives antisymmetry in the two metric-paired slots; combining it with the first identity gives pair symmetry and symmetry of Ricci. Contracting the second identity twice and using $\nabla g = 0$ yields $\nabla^i \text{Ric}_{ij} = \frac{1}{2} D_j \text{Scal}$, hence the last equation. All operations are finite tensor contractions in the admissible algebra. $\square$

In dimension two G vanishes identically. In dimensions at least three it need not. Flatness means $R = 0$, not that the coefficients happen to be small. A nonzero infinitesimal curvature is still nonzero curvature.

9.1 First-order metric response with an exact remainder

Let $g = g_0 + t^\eta h$, $\eta > 0$, with h a real symmetric tensor. The coefficient at t^η of the connection difference is the tensor

$$C^k_{ij} = \frac{1}{2}g_0^{k\ell}(\nabla_i^0 h_{j\ell} + \nabla_j^0 h_{i\ell} - \nabla_\ell^0 h_{ij}). \quad (9.2)$$

Subtracting the metric-compatibility equations for the two connections and extracting the first coefficient proves this formula. Substitution into (9.1) gives

$$R^\ell_{kij}(g) = R^\ell_{kij}(g_0) + t^\eta(\nabla_i^0 C^\ell_{jk} - \nabla_j^0 C^\ell_{ik}) + E^\ell_{kij}, \quad v_H(E) \geq 2\eta. \quad (9.3)$$

Here v_H is the minimum nonzero *coefficient-function* exponent in a fixed local real frame. The remainder is a genuine Hahn tensor. Formula (9.3) is an exact decomposition, rather than an equality in a ring where $(t^\eta)^2$ is zero.

For $h = \mathcal{L}_V g_0$, naturality under the prolonged diffeomorphisms implies that the first curvature response is $\mathcal{L}_V R(g_0)$. This identifies the usual infinitesimal coordinate freedom and explains why solving geometric field equations requires a treatment of gauge, not just componentwise inversion of a linear operator.

10 Volume, Hodge duality, and vector calculus in every dimension

10.1 Volume and the Hodge star

For an oriented admissible metric of signature (p, q) set

$$\mu_g = \sqrt{|\det g|}, \quad \text{vol}_g = \mu_g dx^1 \wedge \cdots \wedge dx^n.$$

For (8.1), the square root is obtained by the positive binomial expansion after factoring $|\det g_0|$. The induced inner product on forms is the determinant pairing. Define $*$ by

$$\alpha \wedge * \beta = \langle \alpha, \beta \rangle_g \text{vol}_g, \quad * : \Omega^k_{\mathcal{A}} \rightarrow \Omega^{n-k}_{\mathcal{A}}. \quad (10.1)$$

An oriented pseudo-orthonormal basis gives

$$*^2 \alpha = (-1)^{k(n-k)+q} \alpha. \quad (10.2)$$

Indeed the two permutations contribute $(-1)^{k(n-k)}$, and the product of all metric signs is $(-1)^q$. This proves the formula without a positivity assumption.

10.2 Gradient, divergence, and Laplacians

The definitions and their coordinate formulas are

$$\text{grad}_g f = (df)^\sharp = g^{ij} D_j f D_i, \quad (10.3)$$

$$\mathcal{L}_X \text{vol}_g = (\text{div}_g X) \text{vol}_g, \quad \text{div}_g X = \mu_g^{-1} D_i (\mu_g X^i) = \nabla_i X^i, \quad (10.4)$$

$$\Delta_g f = \text{div}_g \text{grad}_g f = \mu_g^{-1} D_i (\mu_g g^{ij} D_j f). \quad (10.5)$$

For (10.4), use Cartan's formula and expand $d\iota_X \text{vol}_g$. Equality with the connection trace follows from $D_i \det g = (\det g) \text{tr}(g^{-1} D_i g)$ and $\Gamma^j_{ji} = D_i \log \mu_g$, with the latter notation meaning the algebraic quotient $\mu_g^{-1} D_i \mu_g$, whether or not a logarithm has been selected in the function ring.

On k -forms define the codifferential by

$$\delta_g = (-1)^{n(k+1)+q+1} * d*, \quad \Delta_H = d\delta_g + \delta_g d. \quad (10.6)$$

The sign follows by integrating $d(\alpha \wedge *\beta)$ on real compact chains, or by the equivalent local algebraic identity. On functions $\Delta_H = -\Delta_g$. This distinguishes the nonnegative real Riemannian Hodge-Laplacian convention from the positive-coordinate-second-derivative convention in (10.5).

10.3 What replaces curl?

The dimension-independent vorticity is the two-form

$$\mathcal{W}_X = d(X^\flat).$$

Its dual $*\mathcal{W}_X$ has degree $n - 2$. It is a scalar in dimension two, a one-form (hence a vector after raising an index) in dimension three, and a two-form in dimension four. In dimension one it vanishes. Thus a vector-valued curl is not a dimension-independent operation.

Similarly, the metric and orientation give an $(n - 1)$ -ary cross product

$$X_1 \times \cdots \times X_{n-1} = (* (X_1^\flat \wedge \cdots \wedge X_{n-1}^\flat))^\sharp.$$

It is the familiar binary cross product only when $n = 3$. Extra exceptional algebraic structures in other dimensions are not supplied by merely changing the scalar field.

The identities $d^2 = 0$ and $\delta_g^2 = 0$ replace the usual “curl of a gradient” and “divergence of a curl” formulas. In a flat constant metric,

$$\delta_g d(X^\flat) = d(\operatorname{div}_g X) - \Delta_g(X^\flat),$$

where the final Laplacian is taken componentwise. For a variable metric the Hodge identity remains valid, while rewriting it using a rough covariant Laplacian introduces the usual Ricci curvature term. The flat formula must not be transplanted unchanged to curved coordinates or a curved metric.

Remark 10.1 (Specializations checked for the merge). For $n = 3$ and Euclidean signature ($q = 0$), (10.6) gives $\delta_g = (-1)^k * d*$ on k -forms, which is source 03’s convention in Part II. For $n = 4$ and signature $(1, 3)$, so $q = 3$, it gives $\delta = *d*$ in every degree, and (10.2) gives $*^2 = (-1)^{k(4-k)+3}$; these are source 02’s formulas in Part III. For $\eta = \operatorname{diag}(1, -1, -1, -1)$ one has $\Delta_\eta = \square$, so $\Delta_H = -\Delta_g$ on functions is the degree-zero case of source 02’s $(d\delta + \delta d)\alpha = -\square\alpha$. The vector-valued curl of Part II is $(*d(X^\flat))^\sharp$, and the four-dimensional substitutes for the cross product are in [section 31](#).

11 Integration, Stokes, and local potentials

11.1 Integration on real chains is coefficientwise

For a smooth real oriented compact p -chain C and a Hahn p -form defined near it, use a finite localization and define

$$\int_C^H \sum_\gamma \alpha_\gamma t^\gamma = \sum_\gamma \left(\int_C \alpha_\gamma \right) t^\gamma. \quad (11.1)$$

The sum is well defined because its support is a subset of a single well-ordered support. Real integration happens *inside each coefficient* first. Formula (11.1) is not defined as a fine limit of surreal-valued Riemann sums. It is independent of finite real chart choices and is K -linear by the finite-pair support lemma.

Theorem 11.1 (Hahn Stokes theorem). *For C a smooth real compact oriented p -chain and $\alpha \in \Omega_{\mathcal{A}}^{p-1}$ defined near it,*

$$\int_C^H d\alpha = \int_{\partial C}^H \alpha.$$

Proof. Every coefficient form satisfies real Stokes. Taking those equalities as the coefficients of one Hahn series proves the identity. No exchange of an uncertified infinite integral with an infinite sum is involved: the integral was defined coefficientwise on a common support. $\square$

For a real domain with smooth compact boundary this yields

$$\int_{\Omega}^H \operatorname{div}_g X \operatorname{vol}_g = \int_{\partial\Omega}^H \iota_X \operatorname{vol}_g,$$

provided the metric, inverse, density and field are admissible near the closure. The flux form is the primary definition; a unit-normal expression requires the corresponding induced metric and square roots.

For noncompact integration domains each coefficient must satisfy the relevant ordinary integrability condition. Fubini, integration by parts and differentiation under the integral are available coefficientwise under their classical hypotheses; a common compact support or Schwartz coefficients are convenient sufficient assumptions, not consequences of coherence (source 02). When the integrand has nonnegative support, standard part commutes with both sides of [theorem 11.1](#) (source 02; see also [theorem 13.1](#)). Coefficientwise Stokes with complex coefficients on any smooth manifold with corners is `contours:thm:hahnstokes`.

Proposition 11.2 (Positivity for smooth common-support integrands). *Let M be a nonempty compact real manifold without boundary, with a positive real smooth density ν . If $f \in \mathcal{A}_1(M)$ is pointwise nonnegative at every real point and is not the zero field, then $\int_M^H f\nu > 0$.*

Proof. Let γ_0 be the least exponent whose smooth coefficient function is not identically zero. Pointwise nonnegativity forces $f_{\gamma_0}(x) \geq 0$ everywhere: wherever it is nonzero it is the leading coefficient of $f(x)$. It is positive on a nonempty real open set. Hence its real integral is positive, and this is the leading coefficient of the Hahn integral. The same argument works for compact domains with interior whenever the nonzero smooth field is detected there. $\square$

This result concerns smooth common-support densities on real domains. It is not a statement about arbitrary Hahn-valued measures or a change from coefficientwise to strong countable additivity.

11.2 A support-preserving Poincaré lemma

Suppose $U \subset \mathbb{R}^n$ is star-shaped about zero. For a real p -form, $p \geq 1$, define the radial homotopy

$$(H\alpha)_x(v_1, \dots, v_{p-1}) = \int_0^1 s^{p-1} \alpha_{sx}(x, v_1, \dots, v_{p-1}) ds.$$

Extend it coefficientwise to Hahn forms, and put $H = 0$ on functions.

Theorem 11.3 (Support-preserving local exactness). *For common-support forms in $\Omega^p(U; \mathbb{R})((t^\Gamma))$, the homotopy never enlarges Hahn support and satisfies*

$$dH + Hd = \operatorname{id} - e_0^*,$$

where $e_0 : U \rightarrow U$ is the constant map. Consequently every closed positive-degree common-support Hahn form on U is exact with a primitive having no additional exponents. This proves local exactness for the sheaf as well. A Hahn function with all $D_i f = 0$ is constant on every connected real domain.

Proof. For the dilation $r_s(x) = sx$, differentiating $r_s^* \alpha$ in s and using Cartan’s identity gives a total derivative whose integral from 0 to 1 is $\alpha - e_0^* \alpha$. This is the displayed real homotopy identity; each coefficient is smooth at $s = 0$ because the factor s^{p-1} is nonnegative-degree. Applying the identity coefficientwise proves it for Hahn forms. For a positive-degree form $e_0^* \alpha = 0$. Finally, a real coefficient function with zero gradient is constant on a connected real open set, and the same conclusion holds at every Hahn exponent. $\square$

In particular, a curl-free field has a scalar potential locally precisely when $d(X^\flat) = 0$; “curl-free” here means the two-form condition in every dimension. A divergence-free field satisfies $d(\iota_X \text{vol}_g) = 0$ and locally has an $(n - 2)$ -form potential when $n \geq 2$. These are differential-form statements, not a special three-dimensional vector-potential ansatz.

There is also a purely algebraic version on K^n for polynomial forms (source 02, stated on K^4 ; the argument uses no property of dimension four). Perform the radial homotopy with the s -integral replaced by the formal rule $s^m \mapsto 1/(m + 1)$. The identity $dH + Hd = \text{id} - e_0^*$ is then a polynomial identity with rational coefficients. This proves the polynomial Poincaré lemma in characteristic zero without an integral over a surreal interval. The explicit three-dimensional scalar and vector potentials are [theorem 25.1](#).

11.3 Integration on surreal-coordinate simplices

There is a second, algebraic integration theory that does not require a real base. For polynomial forms over K on an oriented affine p -simplex with vertices in K^n , pull back to barycentric coordinates $\lambda_0 + \cdots + \lambda_p = 1$ and define its moments by

$$\int_{\Delta^p}^{\text{alg}} \lambda_0^{a_0} \cdots \lambda_p^{a_p} d\lambda_1 \cdots d\lambda_p = \frac{a_0! \cdots a_p!}{(p + a_0 + \cdots + a_p)!}. \quad (11.2)$$

Extend by K -linearity and the oriented affine pullback. Polynomial Stokes holds. One proof applies the polynomial fundamental theorem successively in simplex coordinates. Another observes that, at each fixed degree, both sides are polynomial expressions in the coefficients and vertices with rational coefficients; their equality for all real data is a polynomial identity and hence remains true over K .

A support-certified formal form on an infinitesimal scaled simplex can also be integrated termwise by these moments. The same positive-support proof justifies its sums. This supplies exact small-scale integrals, but neither construction defines a countably additive Lebesgue theory on every subset of K^n . Classical global integral theorems require the chain category stated in their hypotheses.

A smooth real parameter domain may also describe a surreal-valued chain through an admissible Taylor or scale-chart map. Its integral is defined by the pulled-back form, provided that pullback has a common well-ordered support and commutes with d . [Theorem 5.2](#) and the polynomial constructions supply these properties for the stated chart classes, and Stokes then follows on the ordinary parameter domain (source 03; compare `contours:thm:deformedstokes`). A general pointwise surreal map has neither property, and the parameter map is usually not intrinsically continuous. The parametric sphere integrals of [section 14.3](#) and [section 25.2](#) are to be read in this way.

12 Global cohomology and a precise Hodge theorem

Let M be a compact smooth real manifold without boundary. Denote by $\Omega_H^p(M)$ the global sections of the Hahn form sheaf. Compactness of the *real* base gives

$$\Omega_H^p(M) = \Omega^p(M; \mathbb{R})((t^\Gamma)).$$

This is a statement about supports of real coefficient forms, not compactness of the much larger halo in its intrinsic topology.

Theorem 12.1 (Hahn extension of a real cochain complex; sources 02 and 03). *Let $(E^\bullet, d)$ be a cochain complex of real vector spaces. There is a natural isomorphism*

$$H^k(E^\bullet((t^\Gamma)), d) \cong H^k(E^\bullet, d)((t^\Gamma)).$$

If $H^k(E^\bullet, d)$ is finite dimensional, the right side is naturally $H^k(E^\bullet, d) \otimes_{\mathbb{R}} K$.

Proof. A Hahn series is closed exactly when each coefficient is closed. Send it to the series of coefficient cohomology classes. This map is natural and has support contained in the original support. It is surjective: choose a representative for each nonzero coefficient class, retaining the same support. Its kernel consists of series all of whose coefficients are exact. Choose a primitive for each such coefficient; their series again has the same well-ordered support and differentiates to the given series. Thus the kernel is exactly the exact Hahn series. These existence choices are ordinary set-sized choices, not a proper-class selection. In finite dimension, expansion in a finite basis identifies the Hahn coefficient space with the ordinary tensor product. $\square$

In infinite dimension an arbitrary Hahn family of cohomology classes need not lie in a finite-dimensional real span, so writing the right side as an algebraic tensor product with K would lose information (source 02). Applied to the de Rham complex of a real manifold with globally supported coefficients (source 02 on a fixed real domain, source 03 on a manifold), the theorem gives

$$H^p(\Omega^\bullet(M; \mathbb{R})((t^\Gamma)), d) \cong H_{\text{dR}}^p(M; \mathbb{R})((t^\Gamma)).$$

This is the cohomology of the specified coefficientwise complex, not singular cohomology of the halo or of K^n . Betti dimensions are unchanged after scalar extension, while periods can be arbitrary elements of K . The complex-coefficient version is `contours:thm:cohomology`. For compact M the next theorem, proved independently by source 01, adds the graded-algebra structure.

Theorem 12.2 (Compact de Rham base change). *There is a canonical isomorphism of graded K -algebras*

$$H^\bullet(\Omega_H^\bullet(M), d) \cong H_{\text{dR}}^\bullet(M; \mathbb{R}) \otimes_{\mathbb{R}} K. \quad (12.1)$$

In particular, the dimension in degree p is the ordinary real Betti number $b_p(M)$.

Proof. The real cohomology in each degree is finite-dimensional. Choose real closed forms $\theta_1, \dots, \theta_b$ representing a basis. Choose real linear coordinate maps on closed forms and a linear choice of primitive for exact forms; such a choice exists by a vector-space splitting. Let P apply that primitive choice after subtracting the selected cohomology representative. Thus every closed real form has an identity

$$\alpha = \sum_{j=1}^b c_j(\alpha) \theta_j + dP\alpha,$$

where each c_j and P is real linear. Apply these fixed operators to every coefficient of a closed Hahn form. They cannot add exponents. We obtain

$$\alpha = \sum_{j=1}^b a_j \theta_j + d\beta, \quad a_j \in K, \quad \beta \in \Omega_H^{p-1}(M).$$

This proves surjectivity of the canonical map from the right side of (12.1). If a finite K -linear combination of the θ_j is exact, coefficient extraction gives a real exact combination at every exponent. Real independence forces all its coefficients to vanish, proving injectivity. The map itself sends scalar-extended real cohomology classes to the same representative forms, so it is independent of the auxiliary splittings. Products agree because wedge products have finite coefficient convolutions. $\square$

For an oriented compact M choose an ordinary real Riemannian metric g_0 . The classical real Hodge theorem [24] supplies harmonic projection P_0 and Green operator G_0 on smooth forms, with

$$\text{id} = P_0 + d\delta_0 G_0 + \delta_0 dG_0.$$

Every operator in this identity extends coefficientwise without enlarging support. We immediately obtain the following exact, but carefully scoped, Hodge theorem.

Corollary 12.3 (Coefficientwise Hodge decomposition). *For the real metric g_0 ,*

$$\Omega_H^p(M) = d\Omega_H^{p-1}(M) \oplus \delta_0 \Omega_H^{p+1}(M) \oplus (\mathcal{H}_{g_0}^p(M; \mathbb{R}) \otimes_{\mathbb{R}} K).$$

*The summands are orthogonal for the K -valued pairing $\int_M^H \alpha \wedge *_0 \beta$. This pairing is positive definite. Every Hahn cohomology class has a unique harmonic representative.*

Proof. Extend the real operator identities and orthogonality coefficientwise. Uniqueness follows by applying the real projections at each exponent. For positivity, the leading coefficient of the pointwise squared norm of a nonzero form is the real squared norm of its leading nonzero coefficient form; its integral is positive. Finite dimensionality of the harmonic space identifies its Hahn extension with tensoring by K . $\square$

Source 03 proves the same decomposition for a real background metric, and notes that the three projections P_0 , $d\delta_0 G_0$ and $\delta_0 dG_0$ preserve support and that the harmonic space has K -dimension $b_k(M)$.

This does not claim an unrestricted Hodge theorem for every surreal metric, every intrinsic K -manifold, or a completed infinite-dimensional normed space. The fixed real elliptic theory is the input. For a regular Riemannian Hahn metric, source 03 obtains the g -orthogonal decomposition without constructing a new Green operator (theorem 12.4); for metrics without a nondegenerate real leading part no Hodge theorem is proved here.

12.1 Hodge decomposition for a regular Riemannian Hahn metric

Corollary 12.3 used a real metric. A positive-order change of metric can also be handled without postulating a new surreal Hilbert-space theory.

Let $g = g_0 + h$ be a regular Riemannian metric on a compact oriented manifold without boundary. For each degree k , let $R_k = I + Q_k$ be the pointwise bundle operator defined by

$$(\alpha, \beta)_g = (R_k \alpha, \beta)_{g_0}.$$

It is obtained from the inverse metrics and volume densities, is self-adjoint relative to g_0 , and Q_k has positive support. It has an admissible inverse by [theorem 4.3](#). Comparing formal adjoints gives

$$\delta_g = R_{k-1}^{-1} \delta_0 R_k \quad \text{on degree } k. \quad (12.2)$$

Theorem 12.4 (Regular-metric Hahn–Hodge theorem; source 03). *For a compact oriented boundaryless real manifold and a regular Riemannian Hahn metric,*

$$\Omega_\Gamma^k(M) = d\Omega_\Gamma^{k-1}(M) \perp_g \oplus \mathcal{H}_g^k \perp_g \oplus \delta_g \Omega_\Gamma^{k+1}(M),$$

where $\mathcal{H}_g^k = \ker d \cap \ker \delta_g = \ker \Delta_{H,g}$. Every cohomology class has a unique g -harmonic representative, and $\dim_K \mathcal{H}_g^k = b_k(M)$. If S is the positive metric support and T the input support, all three components have support in $T + \langle S \rangle$.

Proof. Let $Z = \ker d$ and $E = \text{im } d$ in the ordinary real degree- k space. Real Hodge theory supplies support-preserving orthogonal projections P_Z^0 and P_E^0 . Closed and exact Hahn forms are respectively $Z((t^\Gamma))$ and $E((t^\Gamma))$ by [theorem 12.1](#) and its proof.

For $W = Z$ or E , the operator

$$P_W^0 R_k|_{W((t^\Gamma))} : W((t^\Gamma)) \rightarrow W((t^\Gamma))$$

is I plus a positive-support operator. It is invertible by [theorem 4.3](#). Hence

$$P_W^g \alpha = \left(P_W^0 R_k|_{W((t^\Gamma))} \right)^{-1} P_W^0 R_k \alpha$$

is well defined. Its defining equation says that $R_k(\alpha - P_W^g \alpha)$ is g_0 -orthogonal to W , equivalently that $\alpha - P_W^g \alpha$ is g -orthogonal to W . Thus it is the desired g -orthogonal projection.

Write

$$e = P_E^g \alpha, \quad h = P_Z^g \alpha - P_E^g \alpha, \quad c = \alpha - P_Z^g \alpha.$$

Because $E \subseteq Z$, $P_E^g P_Z^g = P_E^g$, and these three fields are mutually orthogonal. The field h is closed and orthogonal to exact fields; integration by parts implies $\delta_g h = 0$. The pairing is positive definite by the leading-coefficient argument used in [corollary 12.3](#), now with leading metric g_0 .

To show c is coexact, observe that $R_k c$ is g_0 -orthogonal to all closed forms. Real Hodge theory coefficientwise gives $R_k c = \delta_0 \beta$. By (12.2), $c = \delta_g (R_{k+1}^{-1} \beta)$. Conversely every coexact form is orthogonal to closed forms. This proves the decomposition and its uniqueness. Closed forms therefore split uniquely into exact and harmonic parts, giving the cohomology isomorphism and dimension statement. Finally, $(\Delta_{H,g} \alpha)_g = \|d\alpha\|_g^2 + \|\delta_g \alpha\|_g^2$ proves the asserted description of the Laplacian kernel. Every inverse used above is a positive-support Neumann inverse, so the support bound follows. $\square$

The theorem does not cover arbitrary singular standard-part metrics such as $\text{diag}(1, t^2, t^4)$ without additional analysis. A metric being pointwise positive definite is not a substitute for the uniform expansion hypotheses.

Remark 12.5 (Dimension and signature in [theorem 12.4](#)). Source 03 proves this theorem inside its three-dimensional development. Its statement and proof use only a compact oriented boundaryless real manifold, real Hodge theory for g_0 , [theorem 12.1](#) and [theorem 4.3](#); no property of dimension three is used, so it is printed here in every dimension. Positive definiteness of g_0 is used, for the positivity of the pairing, and remains a hypothesis: regular metrics of indefinite signature are not covered.

For example, on the real n -torus, $\dim_K H_H^p = \binom{n}{p}$. On a circle, $ad\theta$ has period $2\pi a$ and is nonexact for every $a \in K \setminus \{0\}$, even when a is infinitesimal. Scales enrich the coefficients of topological classes; in this model they do not create extra Betti numbers. This cohomology is not asserted to be singular cohomology of the halo with its fine, highly disconnected topology.

13 Standard-part geometry and scale stability

Let $\mathcal{A}_{\geq 0}$ be the subalgebra with no negative exponents, and $\mathcal{A}_{> 0}$ its ideal of positive-support sections. Coefficient extraction extends standard part to

$$\text{st} : \mathcal{A}_{\geq 0} \rightarrow C^\infty(M), \quad \ker \text{st} = \mathcal{A}_{> 0}.$$

The maps D_i commute with it because they act only on coefficient functions.

Theorem 13.1 (Regular reduction of tensor geometry). *Suppose $g \in \mathcal{A}_{\geq 0}$ is symmetric and $g_0 = \text{st}(g)$ is a real smooth nondegenerate metric. For tensor fields with nonnegative support, standard part commutes with tensor products, contractions, exterior products, d , Lie brackets and Lie derivatives. Moreover,*

$$\text{st}(g^{-1}) = g_0^{-1}, \quad \text{st}(\Gamma(g)) = \Gamma(g_0), \quad \text{st}(R(g)) = R(g_0), \quad \text{st}(\text{Ric}(g)) = \text{Ric}(g_0),$$

and similarly for scalar curvature, volume density, Hodge star, gradient and divergence. Coefficientwise real-chain integrals commute with standard part whenever the integrand has nonnegative support.

Proof. All finite polynomial expressions commute with the ring homomorphism st , and it commutes with D_i . Formula (8.2) shows that g^{-1} has no negative exponents and has constant coefficient g_0^{-1} . The binomial density formula has constant coefficient $\sqrt{|\det g_0|}$. Substituting into the explicit formulas for connection, curvature, star and divergence proves every assertion. Integration commutes by its definition. At a halo point $a + h$, positive powers of h disappear under standard part, so these reductions take place at the real base point a . $\square$

Proposition 13.2 (A quantitative support threshold). *If $g = g_0 + h$ and $\text{supp}(h) \subseteq [\eta, \infty)$ for some $\eta > 0$, then the differences from the corresponding real quantities for g^{-1} , Γ , R , Ric , Scal and the volume density all have support in $[\eta, \infty)$, locally in real coordinates.*

Proof. Differentiation does not lower coefficient exponents. Multiplication by real smooth coefficients does not change them. In every difference expansion, including the inverse and binomial expansions, each term contains at least one factor of h or one of its real derivatives. It therefore has exponent at least η . The expansions are strong by the positive-support lemma. $\square$

The hypothesis is a support bound on a field and its chosen smooth coefficient representation, not merely smallness of its values at a few points. It also uses a nondegenerate real leading metric. If $g = \text{diag}(t^{2\eta}, 1, \dots, 1)$, the metric is perfectly nondegenerate over K , but its standard part is degenerate and its inverse contains $t^{-2\eta}$. The theorem intentionally excludes this situation.

Remark 13.3 (Additions from sources 02 and 03). For a regular Riemannian metric, source 03 adds to [theorem 13.1](#) the g -musical maps, the covariant derivative, $\text{st}(\nabla^g T) = \nabla^{g_0} \text{st} T$, and the Hodge projections of [theorem 12.4](#). These reduce to the g_0 -projections because every Neumann inverse in their construction has the real inverse as constant term. Consequently a regular, nonnegative-support

solution of a polynomial field equation reduces to a real solution of its leading equation; this is a consistency statement, not a converse existence theorem. Source 02 proves the scalar part on real spacetime,

$$\text{st}(fg) = \text{st}(f) \text{st}(g), \quad \text{st}(D_i f) = D_i \text{st}(f) \quad (f, g \in \mathcal{A}_{\geq 0}(U)), \quad (13.1)$$

an exact differential-algebra quotient rather than an interchange of a limit and a derivative. It also notes that standard part preserves rational expressions only when their denominators have nonvanishing real body on the chosen domain: ε is finite with an unlimited inverse, and reduction cannot turn $\varepsilon \varepsilon^{-1} = 1$ into a product of real shadows. The collection's `phys:thm:reduction` is the four-dimensional Lorentzian case with the Einstein tensor.

13.1 Information that standard part destroys

Standard part annihilates every positive-order observable (source 03). A monopole flux $4\pi t^\eta q_0$ becomes zero although it obstructs a global vector potential. The area of the vectors $(1, 0, 0)$ and $(1, t^\eta, 0)$ becomes zero although they are exactly independent. Curvature of order t in (24.3) disappears from the ordinary metric.

To retain this information use the leading normalized observable

$$\text{in}(F) = t^{v_H(F)} F_{v_H(F)},$$

or record several successive coefficients. This associated-scale data is covariant under real frame changes. A surreal frame change may shift the hierarchy and must be recorded with the observable.

14 Examples at finite, infinitesimal, and infinite scales

14.1 Anisotropic flat geometry

For constant $\lambda_i \in \Gamma$ let

$$g = \sum_{i=1}^n t^{2\lambda_i} (dx^i)^2.$$

It has

$$g^{-1} = \text{diag}(t^{-2\lambda_i}), \quad \mu_g = t^{\lambda_1 + \dots + \lambda_n}, \quad \Gamma = 0, \quad R = 0,$$

and

$$\text{grad}_g f = \sum_i t^{-2\lambda_i} D_i f D_i, \quad \Delta_g f = \sum_i t^{-2\lambda_i} D_i^2 f.$$

The coordinate change $y^i = t^{\lambda_i} x^i$ makes the metric Euclidean over K . It is generally not an ordinary real coordinate change. Different component scales can be an exact change of units or frame, without creating curvature. Source 03's three-dimensional metric $\text{diag}(1, t^2, t^4)$ (section 24.2) shows in addition that raising an index and taking standard part do not commute when the standard part is degenerate.

14.2 A smooth intrinsic metric with infinite curvature

For $n \geq 2$ and $\varepsilon = t^\eta > 0$ consider, on genuine K coordinates,

$$g_\varepsilon = (dx)^2 + (x^2 + \varepsilon^2)(dy)^2 + \sum_{j=3}^n (dz^j)^2. \quad (14.1)$$

It is positive definite at every point. Its coefficients and inverse are intrinsically smooth rational functions. The only nonzero Christoffel coefficients in the two-dimensional factor are

$$\Gamma^x_{yy} = -x, \quad \Gamma^y_{xy} = \Gamma^y_{yx} = \frac{x}{x^2 + \varepsilon^2}.$$

Our curvature convention gives

$$\mathcal{K}_\varepsilon = -\frac{\varepsilon^2}{(x^2 + \varepsilon^2)^2}, \quad \text{Scal}(g_\varepsilon) = -\frac{2\varepsilon^2}{(x^2 + \varepsilon^2)^2}. \quad (14.2)$$

The extra flat factors do not change the scalar curvature. These formulas follow directly from (9.1); the supplied symbolic program checks the calculation independently.

At $x = 0$, the sectional curvature is $-\varepsilon^{-2}$, an infinite surreal. At every fixed nonzero real x , it has leading term $-\varepsilon^2/x^4$. The inverse coefficient $(x^2 + \varepsilon^2)^{-1}$ has no smooth Hahn expansion on a real neighborhood crossing zero. Thus this is an intrinsic metric, not an admissible metric over that whole real base in the algebra (4.1). There is no contradiction between those conclusions.

The inner coordinate $x = \varepsilon X$ gives the two-dimensional factor

$$\varepsilon^2((dX)^2 + (1 + X^2)(dy)^2), \quad \mathcal{K}_\varepsilon = -\varepsilon^{-2}(1 + X^2)^{-2}.$$

A scale chart resolves the transition and exposes its curvature scale. Replacing a small real parameter by a nonzero infinitesimal removes a zero of this particular metric coefficient, but does not make the curvature finite. Nondegeneracy, regular standard part and bounded invariants are three different requirements.

14.3 Radial fields and scale-independent flux

In Euclidean K^n let $r = (\sum_i (x^i)^2)^{1/2}$ and, away from $r = 0$, set

$$X^i = c x^i r^{-n}, \quad c \in K.$$

The local algebraic derivative of the positive square root gives $D_i r = x^i/r$. Consequently

$$\text{div } X = c \left(n r^{-n} - n r^{-n-2} \sum_i (x^i)^2 \right) = 0.$$

For $n \neq 2$ this is the gradient of $c r^{2-n}/(2-n)$. For $n = 2$ a local logarithmic primitive may be chosen on an admitted positive radius chart; no unrestricted global trigonometric or logarithmic function is needed for the divergence calculation.

On the sphere parametrized by $x = R\theta$, $R \in K_{>0}$ and $\theta \in S^{n-1}(\mathbb{R})$, the pulled-back flux form is exactly c times the real unit-sphere area form: the factor R^{1-n} in the field cancels R^{n-1} in the area. Define this parametric integral by its smooth Hahn coefficients. Its value is

$$c \text{ area}(S^{n-1}),$$

for infinitesimal, finite or infinite R . This is an explicit admissible parametric calculation, not a claim that $\theta \mapsto R\theta$ is a nonconstant fine-continuous map from a usual real sphere into full $\mathbf{No}^n$. The central singularity at $r = 0$ has not been assigned a value. For $n = 3$ this is the monopole of [section 25.2](#), where Stokes on the sphere also rules out a global vector potential; the Coulomb field of [section 40](#) is the same field in electrostatic language.

15 Flows and parallel transport by positive-support recursion

Let X_0 be a real smooth vector field on an open $U \subset \mathbb{R}^n$. Suppose its real flow $\phi_0(s, a)$ is defined smoothly for (s, a) in a fixed real flow domain containing $s = 0$, and stays in U . Consider

$$X = X_0 + \sum_{\gamma \in S} t^\gamma X_\gamma, \quad S \subset \Gamma_{>0} \text{ well ordered},$$

with all coefficient fields smooth on U .

Theorem 15.1 (Near-real Hahn flows). *There is a solution of*

$$\partial_s \phi = (\text{Ev } X)(\phi), \quad \phi(0, a) = a,$$

of the form

$$\phi(s, a) = \phi_0(s, a) + \sum_{\gamma \in \langle S \rangle \setminus \{0\}} t^\gamma u_\gamma(s, a).$$

Each coefficient is real smooth on the same real flow domain, the support is well ordered, and the equation holds exactly in the Taylor–Hahn calculus. The solution is unique among positive-support Hahn corrections with the specified initial condition. The realized flow maps satisfy the local flow composition law wherever all terms are defined.

Proof. Expand each coefficient vector field by its Taylor jet at ϕ_0 . At exponent γ , the unknown u_γ occurs only in the linear term from X_0 . Every other occurrence contains a strictly positive lower exponent, either from $X - X_0$ or another correction. Thus

$$\partial_s u_\gamma = DX_0(\phi_0)u_\gamma + F_\gamma, \quad u_\gamma(0, a) = 0,$$

where F_γ is a finite expression in already constructed coefficients and real derivatives. Finiteness even across all Taylor orders is [lemma 2.4](#). The ordinary inhomogeneous linear variational equation has a unique smooth solution on the given real flow domain, by its fundamental-matrix formula. This defines the coefficients by transfinite recursion over $\langle S \rangle$.

The family is Hahn-supported by construction. Substitution back into the Taylor–Hahn equation is strong and yields equality at every exponent. For uniqueness against a second solution, take the union of the two positive supports and the forcing support and apply the same least-differing-exponent linear equation. The coefficient at that exponent must agree, a contradiction. Finally, composing two flow maps is admissible by [\(5.6\)](#); the usual autonomous equation and matching initial data, followed by this uniqueness, prove the local flow law. The identities can be prolonged in the initial coordinates by [theorem 5.2](#). $\square$

The real flow domain is deliberately part of the hypothesis. This result asserts neither completeness of arbitrary real fields nor existence of an all-real-time flow for a field with infinite leading coefficients. For example $X = t^{-\eta} D_1$ has the exact intrinsic solution $x^1(s) = a^1 + st^{-\eta}$, which leaves the finite halo for nonzero real s . The rescaled time $s = t^\eta \tau$ makes this motion finite.

For $n \geq 2$, the field

$$X = t^\eta (x^1 D_1 - x^2 D_2)$$

has the exact flow

$$x^1(s) = a^1 \sum_{m \geq 0} \frac{(st^\eta)^m}{m!}, \quad x^2(s) = a^2 \sum_{m \geq 0} \frac{(-st^\eta)^m}{m!}.$$

Its Jacobian determinant is 1, by the strong exponential addition identity; the other coordinates are fixed. This illustrates an exact infinitesimal strain with exact volume conservation.

The same recursion proves parallel transport for a connection that is a positive-support perturbation of a real connection along a near-real curve. Geodesics reduce to a first-order system on the position–velocity space, so [theorem 15.1](#) applies near any real geodesic on its specified real existence interval. Metric conservation then holds coefficientwise and on the intrinsic realizations.

16 Nonlinear field equations and a support-controlled lifting theorem

The preceding ODE construction generalizes to a large class of equations for scalar, vector or tensor fields. The point is not to assume that every real PDE is solvable. It is to identify the exact real solvability input and then prove the Hahn extension.

Let E, F be real vector spaces of smooth sections with specified boundary, normalization or gauge conditions. Suppose a finite-degree polynomial differential operator has an expansion at a real field u_0 ,

$$\mathcal{P}(u_0 + w) = \mathcal{P}(u_0) + Lw + \sum_{r=2}^d B_r(w, \dots, w), \quad (16.1)$$

where $L : E \rightarrow F$ is real linear and each $B_r : E^r \rightarrow F$ is real multilinear. The finite differential order is absorbed into these maps.

Theorem 16.1 (Regular nonlinear Hahn lifting). *Assume $L : E \rightarrow F$ is bijective, with a fixed inverse taking the prescribed real data to smooth sections with the prescribed conditions. Given a Hahn forcing*

$$f = \mathcal{P}(u_0) + \sum_{\gamma \in S} f_\gamma t^\gamma, \quad S \subset \Gamma_{>0} \text{ well ordered,}$$

there is a unique solution $u = u_0 + w$, with w of positive Hahn support, to the coefficientwise equation $\mathcal{P}(u) = f$. The correction w has support contained in $\langle S \rangle \setminus \{0\}$.

Proof. At $\gamma \in \langle S \rangle \setminus \{0\}$ define

$$w_\gamma = L^{-1} \left(f_\gamma - \sum_{r=2}^d \sum_{\gamma_1 + \dots + \gamma_r = \gamma} B_r(w_{\gamma_1}, \dots, w_{\gamma_r}) \right), \quad (16.2)$$

where all $\gamma_i > 0$ and missing forcing coefficients are zero. Every $\gamma_i < \gamma$, so the recursion is well founded. Each inner sum is finite by the support lemma. Since L^{-1} maps real coefficient data to real coefficient solutions, it cannot introduce Hahn exponents. This constructs a Hahn family on the stated support; substitution and finite coefficient comparison prove the equation.

For uniqueness, the union of two proposed positive supports is well ordered. At the least exponent where the solutions differ, expand each nonlinear difference as a telescoping sum of multilinear terms. Every such term contains a difference of coefficients at a strictly smaller exponent and therefore vanishes. The difference equation is $Lz = 0$. Bijectivity gives $z = 0$, a contradiction. $\square$

The quadratic case around $u_0 = 0$ has a second proof and a sharper conclusion (source 03).

Theorem 16.2 (Exact infinitesimal quadratic solution; source 03). *Let E, F be real vector spaces, $L_0 : E \rightarrow F$ a real-linear isomorphism, and $B : E \times E \rightarrow F$ a real-bilinear map. If $f \in F((t^\Gamma))$ has positive support, the equation*

$$L_0 u + B(u, u) = f \quad (16.3)$$

has exactly one positive-support solution $u \in E((t^\Gamma))$. Its support lies in $\langle \text{supp } f \rangle \setminus \{0\}$. For $f \neq 0$,

$$v_H(u) = v_H(f), \quad v_H(u - L_0^{-1}f) \geq 2v_H(f).$$

For $f = 0$ the unique positive-support solution is zero.

Proof. Set $a = L_0^{-1}f$ and $C = L_0^{-1}B$. Then $u = a - C(u, u)$. Expand this identity over finite rooted planar full binary trees. A leaf carries a coefficient of a , and each internal vertex applies $-C$ to its two incoming values. A tree with m leaves has exponent equal to the sum of its m positive leaf exponents. For a fixed total exponent the support lemma allows only finitely many leaf words, and for each word only finitely many planar binary trees. Therefore the tree sum is strongly defined and has the stated support. Splitting trees into the single-leaf tree and trees with a root vertex proves $u = a - C(u, u)$ exactly.

For uniqueness, suppose u and v are positive-support solutions and $w = u - v \neq 0$. Then

$$w = -C(u, w) - C(w, v).$$

If both solutions are nonzero, the right side has valuation at least $v_H(w) + \min\{v_H(u), v_H(v)\} > v_H(w)$, impossible. If one is zero, omit its term and obtain the same contradiction. This also treats zero forcing. For nonzero forcing, $C(u, u)$ has valuation at least $2v_H(u) > v_H(u)$, so comparing leading exponents in $u + C(u, u) = a$ gives $v_H(u) = v_H(a) = v_H(f)$. The remainder estimate follows. $\square$

The proof works even when the sequence $n\eta$ is not cofinal in Γ . It constructs coefficients by finite combinatorics, not by assuming that a Picard sequence converges in a topology. The theorem is a support-controlled formal implicit-function principle; it is not a claim that a real nonlinear problem has solutions for arbitrary real forcing.

With $\mathcal{P}(u) = L_0 u + B(u, u)$, $u_0 = 0$ and $d = 2$, existence, uniqueness and the support statement of [theorem 16.2](#) are those of [theorem 16.1](#). The tree expansion is a different construction, and the valuation estimates are additional.

A scalar recurrence (source 03). The scalar model $u + u^2 = t$ illustrates the tree expansion:

$$u = t - t^2 + 2t^3 - 5t^4 + 14t^5 - 42t^6 + \dots = \frac{\sqrt{1+4t} - 1}{2}.$$

Its coefficients count binary trees with alternating signs. The equality is exact in the Hahn field. The program `code/03-three-dimensional-fields-verify_examples.py` checks the first twelve coefficients against the quadratic equation; the infinite existence and uniqueness statement rests on [theorem 16.2](#), not on this finite test.

If L has only a specified right inverse, the same recursion constructs a solution with each correction in that right inverse's image. Uniqueness then holds only with that normalization. A nontrivial cokernel creates compatibility conditions. Gauge kernels and conservation identities in geometric PDEs must be handled before applying the theorem.

16.1 An exact nonlinear example in dimension n

On the ordinary flat torus $\mathbb{T}^n = (\mathbb{R}/2\pi\mathbb{Z})^n$, take

$$\mathcal{P}(u) = (1 - \Delta)u + u^2, \quad f = \varepsilon \cos x^1, \quad \varepsilon = t^\eta.$$

The real operator $L = 1 - \Delta$ is bijective on smooth periodic functions: its Fourier multiplier on mode $k \in \mathbb{Z}^n$ is $1 + |k|^2$, and division by it preserves rapid Fourier decay. The theorem gives an exact Hahn solution $u = \sum_{m \geq 1} u_m \varepsilon^m$, with

$$u_1 = L^{-1} \cos x^1, \quad u_m = -L^{-1} \sum_{a+b=m} u_a u_b \quad (m \geq 2).$$

The first terms are

$$\begin{aligned} u = & \frac{\varepsilon}{2} \cos x^1 - \varepsilon^2 \left(\frac{1}{8} + \frac{1}{40} \cos 2x^1 \right) \\ & + \varepsilon^3 \left(\frac{11}{160} \cos x^1 + \frac{1}{800} \cos 3x^1 \right) + \sum_{m \geq 4} u_m \varepsilon^m. \end{aligned} \quad (16.4)$$

The supplied program computes six coefficients and verifies the residual through that order with exact rational Fourier arithmetic. The all-orders existence and uniqueness are furnished by the theorem, not by that finite check. The solution is exact even in value groups where the sequence $m\eta$ is not cofinal and ordinary partial-sum convergence fails.

16.2 Tensor and constrained systems

For tensor equations the real spaces E, F may be smooth tensor sections and B_r can include contractions and covariant derivatives of a fixed real background. For equations containing g^{-1} , the positive-support inverse expansion gives an infinite Taylor nonlinearity rather than a polynomial. The same coefficient recursion works when those multilinear terms have the Neumann finite-word certificate. This additional certificate must be checked, as must the invertibility of the gauge-fixed real linearization. The theorem does not, by itself, establish global Einstein evolution, Navier–Stokes regularity, or a singularity-removal result.

Two further instances, with their own hypotheses, are proved in Parts II and III: infinitesimally forced stationary incompressible flow on the three-torus ([theorem 26.1](#)) and the positive-order deformation of a real cubic-wave solution ([theorem 38.1](#)), whose real linearization is a Cauchy problem rather than a bijection of section spaces. The same Neumann mechanism as in [theorem 4.3](#) gives the retarded inverse of a positive-order perturbation of a hyperbolic operator ([theorem 36.3](#)).

17 Variational calculus and gauge-valued fields

17.1 A scalar action and its Euler–Lagrange equation

Let a real compact domain have an admissible metric and density, and let V be a polynomial potential with coefficients in K . For an admissible scalar field set

$$\mathcal{S}[\phi] = \int^H \left(\frac{1}{2} g^{ij} D_i \phi D_j \phi + V(\phi) \right) \text{vol}_g.$$

For a compactly supported variation ψ , take the coefficient of the ordinary formal variation parameter s in $\mathcal{S}[\phi + s\psi]$. Leibniz's rule and Hahn Stokes give

$$\delta\mathcal{S}[\phi](\psi) = \int^H (-\Delta_g\phi + V'(\phi))\psi \operatorname{vol}_g.$$

The fundamental lemma remains valid in this setting: if this integral is zero for every real smooth compactly supported ψ , then each coefficient of the admissible residual density has zero integral against every real test function, so each is zero by the real fundamental lemma. Hence stationarity is equivalent to

$$-\Delta_g\phi + V'(\phi) = 0.$$

This is a precise formal-coefficient variational problem; its stationary points and their existence are separate questions. Minimizers are not inferred from order completeness, which K lacks, and no measure on all of K^n is used (source 02).

Define the stress tensor

$$T_{ij} = D_i\phi D_j\phi - g_{ij} \left(\frac{1}{2} g^{ab} D_a\phi D_b\phi + V(\phi) \right).$$

Metric compatibility, Hessian symmetry and the product rule yield

$$\nabla^i T_{ij} = (\Delta_g\phi - V'(\phi)) D_j\phi.$$

Thus the field equation implies covariant stress conservation exactly, including all infinitesimal orders. This identity makes no positivity claim when the metric has indefinite signature.

Remark 17.1 (Sign of the potential). For $\eta = \operatorname{diag}(1, -1, -1, -1)$ one has $\Delta_\eta = \square$, and the Euler–Lagrange equation above reads $-\square\phi + V'(\phi) = 0$. Source 02 (Part III) uses the Lagrangian $\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi)$, so its V is $-V$ here. Its equation $\square\phi + V'(\phi) = J$ and its stress $\partial^\mu\phi\partial^\nu\phi - \eta^{\mu\nu}L_0$ are the equations above after $V \mapsto -V$ and the addition of a source. Neither sign is a positivity statement; the energy positivity of Part III uses its own hypotheses $m^2, \lambda \geq 0$.

17.2 Finite-rank gauge fields

Let A be a matrix-valued admissible one-form acting on a finite-rank K -bundle. Its curvature and covariant exterior derivative are

$$F_A = dA + A \wedge A, \quad d_A\beta = d\beta + A \wedge \beta - (-1)^{\deg\beta} \beta \wedge A.$$

Direct expansion gives $d_A F_A = 0$ and $d_A^2\beta = [F_A, \beta]$ in the graded sense. Under an admissible frame transformation P with admissible inverse,

$$A' = P^{-1}AP + P^{-1}dP, \quad F_{A'} = P^{-1}F_AP.$$

The derivative of $P^{-1}P = I$ proves the transformation law. Positive-support transformations $P = P_0(I + H)$ with real invertible P_0 are a concrete admissible class.

In any dimension, the Yang–Mills-type equation $d_A(*F_A) = 0$ is therefore well defined whenever the metric star and products are admissible. As with the scalar action, finite-rank algebra and variational identities do not supply an automatic global existence theorem. Surcomplex coefficients can be introduced by $K[i]$, with conjugation and Hermitian pairings specified separately; they do not alter the real-coordinate support arguments.

18 Scalar derivations and scale connections

18.1 Spatial and scalar derivatives are different

Spatial differentiation in this report fixes every K constant. In particular,

$$D_i \omega = 0.$$

The Berarducci–Mantova derivation ∂_{BM} on surreal scalars instead has $\partial_{\text{BM}} \omega = 1$ and is strongly additive on summable normal forms [16]. It describes differential-field structure, not the variation of a constant scalar field across space. All three sources insist that the two derivatives must not be substituted for one another. The infinite-argument function and composition calculus of omega-series [17, 18] is also additional structure, not a universal replacement for coordinate calculus.

There is a compatible mixed construction when it is useful (source 01). For a given Hahn field with smooth real coefficient functions, define

$$\partial_{\text{BM}} F = \sum_{\gamma} f_{\gamma} \partial_{\text{BM}}(t^{\gamma}).$$

Strong additivity of the scalar derivation provides a well-ordered support union and finite contributions. Coefficients are finite real linear combinations of the f_{γ} , so they are smooth. The image may require a larger workspace containing the derivative supports. On such admissible expressions,

$$[D_i, \partial_{\text{BM}}]F = 0,$$

because the D_i act only on the real coefficient functions, while ∂_{BM} acts only on the monomials.

An additional connection direction can be written $\nabla_{\text{BM}} = \partial_{\text{BM}} + B$, with the semilinear Leibniz rule $\nabla_{\text{BM}}(aY) = \partial_{\text{BM}}(a)Y + a\nabla_{\text{BM}}Y$. Together with $\nabla_i = D_i + \Gamma_i$ its mixed curvature is

$$[\nabla_{\text{BM}}, \nabla_i] = \partial_{\text{BM}} \Gamma_i - D_i B + [B, \Gamma_i].$$

This is a differential-module extension of the theory. Interpreting it as an extra physical time or scale direction would be a further modelling choice; it is not forced by the definition of a surreal-valued tensor field.

18.2 Explicit scalar derivations and their support

For $\Gamma \subseteq \mathbb{R}$ containing 1 and the normalization $\partial_{\text{BM}} \omega = 1$, so that $\partial_{\text{BM}} t = -t^2$, one has

$$\partial_{\text{BM}}(t^{\gamma}) = -\gamma t^{\gamma+1} \quad (\gamma \in \Gamma \subseteq \mathbb{R})$$

(sources 02 and 03). The formula must not be used unchanged for arbitrary surreal exponents, which can themselves have nonzero scalar derivative, and closure of the workspace under the shift by 1, together with strong summation of the result, must be ensured. On general surreal exponent workspaces a strongly acting scalar derivation needs more structure than this formula [16, 15].

A simpler support-preserving derivation exists at every rank (source 03). For an additive character $\chi : \Gamma \rightarrow \mathbb{R}$ put

$$\mathcal{D}_{\chi} \left(\sum_{\gamma} t^{\gamma} F_{\gamma} \right) = \sum_{\gamma} t^{\gamma} \chi(\gamma) F_{\gamma}.$$

It is a derivation because χ is additive. It commutes with every D_i and is $\mathbb{R}$ -linear but not K -linear. For $\Gamma \subseteq \mathbb{R}$ and $\chi = \text{id}$ it is the Euler scale derivation $t \partial_t$, with $\mathcal{D}_{\text{id}} t = t$; this is source 02's derivation δ on $\mathbb{R}((t^{\mathbb{Q}}))$.

18.3 Scale connections

Let $a = a_i dx^i$ be a real smooth one-form and put

$$\mathcal{D}_i = D_i + a_i \mathcal{D}_\chi$$

(source 03; source 02's coupled scale connection $\partial_\mu + a_\mu \mathcal{D}_{\text{id}}$ is the case $\chi = \text{id}$, $n = 4$). These are real-linear derivations of $\mathcal{A}_\Gamma(U)$ modelling a chosen comparison rule for scale-dependent quantities. They are not the coordinate derivatives of the scalar-constant geometry unless the connection is trivialized.

Proposition 18.1 (Scale curvature; sources 02 and 03). *Put $b_{ij} = D_i a_j - D_j a_i$. Then*

$$[\mathcal{D}_i, \mathcal{D}_j] = b_{ij} \mathcal{D}_\chi.$$

If $\chi \neq 0$, the $\mathcal{D}_i$ commute exactly when $da = 0$. If the a_i are allowed to be Hahn-valued, the curvature coefficient becomes $D_i a_j - D_j a_i + a_i \mathcal{D}_\chi a_j - a_j \mathcal{D}_\chi a_i$.

Proof. Expand the commutator. The terms $[D_i, a_j \mathcal{D}_\chi] = (D_i a_j) \mathcal{D}_\chi$ use that $\mathcal{D}_\chi$ commutes with D_i . The term $[a_i \mathcal{D}_\chi, a_j \mathcal{D}_\chi]$ equals $(a_i \mathcal{D}_\chi a_j - a_j \mathcal{D}_\chi a_i) \mathcal{D}_\chi$, because the second-order terms $a_i a_j \mathcal{D}_\chi^2$ cancel; it vanishes for real a_i , which $\mathcal{D}_\chi$ annihilates. If $\chi(\gamma) \neq 0$, evaluating a commutator on t^γ proves the equivalence with $da = 0$. $\square$

Source 03 states the first two assertions for $n = 3$, with the vector identities of [theorem 27.1](#); the commutator computation is the same in every dimension. Source 02 states the Hahn-valued formula for $n = 4$ and $\chi = \text{id}$. When $a = d\psi$ with real smooth ψ , the support-preserving algebra automorphism

$$\mathcal{E}_\psi \left(\sum_\gamma t^\gamma F_\gamma \right) = \sum_\gamma t^\gamma e^{\chi(\gamma)\psi} F_\gamma$$

satisfies $\mathcal{D}_i = \mathcal{E}_\psi^{-1} D_i \mathcal{E}_\psi$. A locally exact scale connection is thus a change of scale convention, and nonzero da is an integrability obstruction (source 03). Each of these is a possible additional model, not the flat derivative of the theory already constructed: changing the derivative rules changes the equations (sources 01 and 02).

19 Boundaries of the coherent calculus

Two explicit obstructions show why the support hypotheses cannot be dropped, and a third subsection records how far distributions go. All three concern the chosen coefficient algebra $\mathcal{A}_\Gamma$, not every conceivable function class.

19.1 No coherent oscillation at a negative valuation

Let $\mathcal{A}_\Gamma(I; \mathbb{C}) = C^\infty(I, \mathbb{C})((t^\Gamma))$ for a nonempty real open interval I . By [\(4.8\)](#), differentiation cannot lower the least exponent of a coherent series.

Theorem 19.1 (No coherent pure oscillation at a negative valuation; sources 02 and 03). *Let $\theta > 0$ in Γ . The equation*

$$u' = it^{-\theta} u \tag{19.1}$$

has no nonzero solution in $\mathcal{A}_\Gamma(I; \mathbb{C})$. Likewise,

$$u'' + t^{-2\theta}u = 0 \quad (19.2)$$

has no nonzero coherent solution. More generally, if a real finite-order differential operator P_0 preserves exponent support and $c \in K[i]$ has $v(c) < 0$, then $P_0u + cu = 0$ has no nonzero coherent scalar solution.

Proof. If $\alpha = v_H(u)$, then $v_H(u') \geq \alpha$, while $v_H(it^{-\theta}u) = \alpha - \theta$. The nonzero leading coefficient at that smaller exponent cannot be matched. The second-order statement is identical. For the general statement, multiplication by a nonzero constant has exact order $v_H(cu) = v(c) + \alpha$, even though multiplication by an arbitrary smooth coefficient need not have exact order. This order is strictly smaller than any nonzero exponent in P_0u . $\square$

The theorem is a limitation of the *chosen coherent amplitude algebra*, not a proof that an oscillatory extension is impossible. One can adjoin phase symbols, work with oscillatory distributions, or construct a larger transseries-type field. Each approach must state its multiplication, derivative, summation, and real-observable rules. The usual exponential symbol by itself does not provide those rules.

Source 03 proves the special cases $Y'' + t^{-2}Y = 0$ and $Y' + t^{-1}Y = 0$ (for $1 \in \Gamma$) by the same least-exponent argument. It adds that the formal series $\sum_{m \geq 0} (-1)^m \tau^{2m} t^{-2m} / (2m)!$ solves the second equation in $K[[\tau]]$, but its exponents are unbounded below and it does not define a Hahn-smooth field of ordinary time; in the fast-time chart $\tau' = \tau/t$ the ordinary function $Y = \cos \tau'$ solves the rescaled equation. Fast dynamics therefore need a fast-time domain, an oscillatory enrichment, or another function class; the same issue arises in thin boundary layers and rapidly varying material laws. The proof of [theorem 19.1](#) uses only (4.8) and the exact order of multiplication by a nonzero constant, so it applies verbatim on any real open domain in place of I ; source 02 states it on an interval.

19.2 Every real time is not every surreal time

The initial value problem

$$y' = y^2, \quad y(0) = t \quad (19.3)$$

has the coherent solution

$$y(\tau) = \sum_{n \geq 0} t^{n+1} \tau^n = \frac{t}{1 - t\tau} \quad (\tau \in \mathbb{R}). \quad (19.4)$$

The formal series is well defined at every real time and differentiates coefficientwise to y^2 . But the rational expression has no value at the surreal time $\tau = t^{-1}$, where its denominator is zero. At that argument the displayed series would contribute infinitely many copies of the same exponent and is not strongly summable either.

Thus a common-domain coefficient solution on $\mathbb{R}$ is not a theorem of existence at every K -valued time. In this example the obstruction is completely explicit. Naming the blow-up time as a surreal number does not supply a continuation through it.

19.3 Distributional coefficients

Let $\mathcal{D}'(M, E)$ be the real space of distributional sections of a real bundle. Its Hahn extension $\mathcal{D}'(M, E)((t^\Gamma))$ has coefficientwise derivatives and pairings with fixed real test sections, and fixed real-linear distributional operators extend to it without enlarging support. Multiplication by a Hahn-smooth function is defined by convolution. Linear equations, weak divergences and smooth-coefficient differential operators therefore have distributional versions in the same workspace (sources 02 and 03). A general product of two distributions is not thereby defined: the ambiguity of δ_0^2 is not removed by changing the scalar field, and nonlinear stress, energy or convective terms formed from unrelated distributional fields need a further product theory or regularity hypothesis.

For an infinitesimal displacement $h \in \mathfrak{m}_K^n$ the family

$$\delta_h^{\text{jet}} = \sum_{\alpha \in \mathbb{N}^n} \frac{(-h)^\alpha}{\alpha!} \partial^\alpha \delta_0$$

is strongly summable, and its value on a real test function φ is $\sum_\alpha h^\alpha \partial^\alpha \varphi(0)/\alpha! = (\text{Ev } \varphi)(h)$. It is a precisely defined infinitesimally shifted jet distribution, not an ordinary distribution supported at a nonreal point, and it inherits the flat-function limitation of [theorem 5.2](#). Source 03 states it for $n = 3$; its support argument is that of [theorem 5.2](#) in any dimension. The three-dimensional identity $\text{div}(q x/r^3) = 4\pi q \delta_0$ is in [section 25.4](#).

20 Computation, proof boundaries, and implementation

20.1 Representations and exactness

A practical implementation should keep four objects separate: finite tensors over an exact scalar field; sparse finite Hahn truncations; support-certified infinite Hahn expressions; and coefficient operators on smooth or formal real functions. A truncation is not an exact infinite series unless its tail is separately represented or certified.

For a grid generated by positive exponents $\eta_1, \dots, \eta_m$, a useful normal form records a finite exponent vector, a real symbolic coefficient, and a support certificate for allowed sums. Negative initial shifts are stored separately. At arbitrary ordered-group rank, a cutoff $\gamma < B$ need not leave finitely many terms: for $0 < \eta \ll B$, all $m\eta$ lie below B . Algorithms must therefore use an explicit finite set of target coefficients, a degree bound, or a proved left-finiteness condition. This restriction is consistent with the repository's distinction between exact denotation and effective support access [\[1\]](#).

Source 03 lists what a field object must carry: base domain, tensor type, coefficient representation, exponent group, support certificate and derivative convention. A metric additionally needs an inverse and a volume-density certificate, and a chain needs an orientation, a parameter domain, a pullback rule and an integral convention. For infinite fields a coefficient oracle alone is insufficient; useful restricted formats include bounded-denominator Puiseux series, nonnegative integer power series, and positive-monoid expressions with a proved summability construction. A safe implementation never treats a failed support check as zero. It reports an unsupported operation, as for inversion across a degenerating leading coefficient, evaluation of a formal series at an argument of nonpositive valuation, or a threshold truncation that is not finite. Already in $\Gamma = \mathbb{Q}$ the support $\{1 - 1/m : m \geq 2\}$ is well ordered with infinitely many exponents below 1, so “retain all terms below a threshold” need not give a finite representation.

20.2 A minimal implementation architecture

A suitable dependency order is: finite tensor algebra over an abstract real closed field; Hahn support and strong evaluation; smooth coefficient operators; the prolongation bridge; admissible inverses and metrics; tensor calculus and curvature; coefficientwise chain integration; and finally support-controlled equation solvers. A theorem about a metric inverse should consume a unit certificate. A theorem about a substitution should consume a joint support certificate. These are mathematical data, not comments that a simplifier may silently discard.

Source 03 expresses the separation of responsibilities as follows; the words “require” and “certified” indicate obligations to be discharged, not checks available for arbitrary infinite objects.

```
Field(domain, tensor_type, exponent_group,
      coefficient_family, support_certificate)

spatial_derivative(F, i):
  return map_coefficients(F, real_partial_derivative(i))

tensor_product(F, G):
  require compatible_domains_and_exponent_groups(F, G)
  certificate = well_ordered_sum(F.support, G.support)
  return finite_at_each_exponent_convolution(F, G, certificate)

inverse_regular_operator(L0_inverse, positive_P, forcing):
  require positive_well_ordered_support(positive_P)
  return certified_neumann_expression(L0_inverse, positive_P, forcing)

integrate_form(alpha, chain):
  pulled_back = certified_pullback(alpha, chain)
  return map_coefficients(pulled_back, ordinary_chain_integral)
```

For the Lorentzian part, source 02 proposes the following order of layers.

Layer	Objects and proofs	Main dependency
Finite Lorentz algebra	Metric, cones, explicit boosts, inverse formulas, tensor actions, Hodge square	Ordered real-closed field and finite matrices
Electromagnetic algebra	Invariants, stress tensor, Rainich identity, dominant energy, null decomposition	Finite Lorentz algebra and sums of squares
Hahn tensor modules	Finite-component supports, minimum order, bounded-frame invariance	Hahn summable families and coefficient maps
Reduction and halo jets	Split bounded group, matrix exp/log, Taylor prolongation, chain rule	Positive-support substitution and real smooth jets
Differential coefficient algebra	Commuting derivations, forms, Stokes lifting, Euler–Lagrange identities	Coefficientwise maps and ordinary calculus
Hyperbolic lifting	Cauchy operator, retarded inverse, nonlinear recursion	Precisely packaged real existence and uniqueness theorems

For Lean, the finite part can be formulated over an abstract ordered field, with additional square-root or real-closed hypotheses where needed. The Hahn part should be proved separately and then instantiated through a verified normal-form bridge. A global class of all surreals should

not be forced into a smaller universe by asserting that all its cuts or all its coefficients form a set. Our proofs describe this layering; no new Lean implementation or kernel verification is claimed here.

Finite Lorentz statements should be proved over an abstract real-closed field and only then instantiated with a surreal workspace. Source 03 proposes the same staging for three dimensions: finite vector identities over an ordered field with the needed square roots, then tensor type and convolution extension, then an abstract differential algebra with commuting derivations, then coefficientwise smooth fields and the support-preserving operator lemmas. The nontrivial real analytic inputs, namely the real Hodge theorem, the torus inverses and the Leray projection, and the real Cauchy and Green theorems of Part III, should be explicit interfaces with documented hypotheses; proving a lift is not the same as formalizing the underlying real theory. No Lean source is delivered with any of the three manuscripts, the repository was not modified by them, and no Lean build was performed for them. Since their pin, the repository's Lean Hahn layer has grown ([section 1.4](#)), but it contains no counterpart of the field-level theorems here.

20.3 Included finite verification programs

Three programs are shipped under `code/`, byte-identical to the delivered ones; their recorded outputs are under `data/`.

code/01-vector-tensor-fields-n-verify.py checks the warped-metric curvature, the first-order metric inverse, the Lie-bracket Jacobi identity for sample polynomial fields, $d^2 = 0$ for polynomial forms, Hodge-star-square signs in dimensions 1 through 6 for every diagonal signature, a radial homotopy identity, and six coefficients of the nonlinear periodic example: seven check groups and 5,533 exact assertions, recorded in `data/01-vector-tensor-fields-n-verification.txt`.

code/02-minkowski-check_identities.py performs 42 named exact checks: boosts (metric preservation, determinant, null eigenvectors, the infinite-boost example); electromagnetic tensors (both invariants, Hodge signs, the symmetric traceless stress and its blocks, the dominant-energy identity, the Rainich square, null radiation stress); the Hodge square on all 16 basis wedges in dimension four; current conservation and the four stress-divergence components for a general potential; the null nilpotent; and the cubic-oscillator truncation. Records: `data/02-minkowski-ki-verification_log.txt` and `data/02-minkowski-verification_results.json`.

code/03-three-dimensional-fields-verify_examples.py runs 11 check groups: finite vector identities and the anisotropic cross-product rule, grad/div/curl identities, the polynomial box, radial potentials, conformal curvature from Christoffel symbols, scale-connection defects, elastic energy, the Beltrami field, the quadratic recursion through degree 12, the softened monopole density, and the worked three-scale field. Record: `data/03-three-dimensional-fields-verification_results.txt`.

For this merge all three were rerun on a copy of the directory with Python 3.14.4 and SymPy 1.14.0. Every check passed, and the outputs agree with the records apart from line endings and the Python version printed by programs 01 and 03 (the records used Python 3.13.5). Program 02 writes `verification_results.json` next to itself, so it should be run on a copy.

These are exact finite symbolic checks. They do not prove real closedness, the support lemma, arbitrary-rank summability, transfinite recursion, PDE existence, the prolongation theorem, the operator theorems, the imported real Hodge and Cauchy theorems, or global cohomology; those are

proved or explicitly imported in the text. Successful typesetting has no bearing on mathematical correctness.

Part II

Three-dimensional fields

21 Scope of Part II

This part is source 03's three-dimensional development; its dimension-free results have been moved to Part I and are cited from there. The analytic model is

$$V(x) = \sum_{\gamma \in S} t^\gamma V_\gamma(x), \quad x \in M, \quad V_\gamma \in C^\infty(M, TM), \quad (21.1)$$

where M is a real smooth three-manifold or a domain of $\mathbb{R}^3$ and S is well ordered. The value $V(x)$ is an actual surreal vector, not a numerical approximation, and spatial differentiation acts on the V_γ , not on t^γ . A nonlinear equation for V becomes a support-controlled family of real equations, and the resulting series solves it as an exact algebraic identity. Not every pointwise surreal-valued assignment admits a common well-ordered support of smooth coefficient fields ([section 4](#)). Everything in Part I applies with $n = 3$, with the Euclidean metric unless another metric is named, and with $\partial_i = D_i$. Source 03 organizes the layers as in [section 1](#), with polynomial and rational fields on K^3 as a separate layer carrying coordinate derivatives, algebraic forms and polynomial-cell integration ([sections 6](#) and [11](#)).

22 Vectors and tensors in three dimensions

22.1 Euclidean vector algebra over surreal scalars

Let K be any real-closed ordered field. On $V = K^3$, define

$$u \cdot v = \sum_{i=1}^3 u_i v_i, \quad \|u\| = \sqrt{u \cdot u}, \quad (u \times v)_i = \sum_{j,k=1}^3 \epsilon_{ijk} u_j v_k,$$

where ϵ_{ijk} is the alternating symbol with $\epsilon_{123} = 1$. The positive square root exists in K . No Archimedean assumption appears in these definitions.

Theorem 22.1 (Finite vector identities). *For $u, v, w \in K^3$,*

$$u \cdot (v \times w) = \det[u \ v \ w], \quad (22.1)$$

$$u \times (v \times w) = v(u \cdot w) - w(u \cdot v), \quad (22.2)$$

$$\|u \times v\|^2 = \|u\|^2 \|v\|^2 - (u \cdot v)^2, \quad (22.3)$$

$$|u \cdot v| \leq \|u\| \|v\|, \quad \|u + v\| \leq \|u\| + \|v\|. \quad (22.4)$$

If $u \neq 0$, then $\|u\| > 0$, even when its norm is infinitesimal. Orthogonal projection onto Ku is $P_u(v) = u(u \cdot v)/(u \cdot u)$.

Proof. The first three identities follow by finite expansion of the alternating symbol. The right side of (22.3) is a sum of three squares, hence nonnegative. Taking nonnegative square roots gives Cauchy–Schwarz. Expanding $\|u + v\|^2$ and using that inequality gives the triangle inequality. A sum of squares in an ordered field is zero only when every term is zero; this proves positive definiteness and justifies the projection denominator. $\square$

The groups $O(3, K)$ and $SO(3, K)$ consist of matrices satisfying $Q^T Q = I$ and, in the second case, $\det Q = 1$. Cross product is equivariant under $SO(3, K)$; an orientation-reversing orthogonal map changes its sign. These are algebraic rotation groups. Parametrizing all their elements by an arbitrarily large surreal angle would require a separate trigonometric convention. The collection’s report `docs/surreal/euclidean-three-space`, placed with this one, treats rotations of $\mathbf{No}^3$ with finite angles, Rodrigues rotations and quaternions, and the spin description of $SO(3, F)$ over a real-closed field is in `docs/surquaternions/surquaternions`. For pure quaternions u, v one has $uv = -u \cdot v + u \times v$, so the cross product is the vector part of their product (source 03).

Tensors of type (r, s) on K^3 have 3^{r+s} components, and all the finite operations of section 3 apply.

Repeated coordinate indices run from 1 to 3 and are summed. A $(1, 1)$ tensor is an endomorphism; a $(0, 2)$ tensor is a bilinear form. Their matrices can coincide in an orthonormal frame but have different transformation laws.

Proposition 22.2 (Symmetric spectral theorem). *Every real symmetric 3×3 matrix over a real-closed field has an orthonormal eigenbasis and real eigenvalues in that field.*

Proof. The algebraic closure is $K(i)$. For an eigenpair $Az = \lambda z$ there, $z^* z > 0$ and $z^* Az$ is fixed by conjugation; hence $\lambda = (z^* Az)/(z^* z) \in K$. The nullspace of $A - \lambda I$ has a nonzero vector over K . Normalize it. Its orthogonal complement is A -invariant, so induction on dimension completes the proof. $\square$

This is a pointwise theorem, not a guarantee of globally smooth eigenvector fields across repeated eigenvalues. The repository’s spectral report supplies a broader treatment [5].

The proof is by induction on the dimension and works for symmetric $n \times n$ matrices; source 03 states it for $n = 3$. In Lorentzian signature the analogous statement fails ((31.3) in Part III).

22.2 Leading directions and multiscale rank

Proposition 22.3 (Leading geometry). *For nonzero $u \in K^3$, $v(\|u\|) = v_V(u)$. If $a = v_V(u)$ and $b = v_V(v)$, then*

$$v_V(u \times v) \geq a + b,$$

with equality exactly when the leading real vectors $\text{st}(t^{-a}u)$ and $\text{st}(t^{-b}v)$ are linearly independent. The analogous statement for a determinant says that its leading valuation is the sum of the three column valuations exactly when their normalized leading columns are independent.

Proof. The coefficient of t^{2a} in $u \cdot u$ is the positive real sum of squares of the normalized leading coordinates. Thus it cannot cancel. For the cross product, the coefficient of t^{a+b} is precisely the cross product of the two leading real vectors. It vanishes exactly when they are dependent. The determinant argument is identical with three columns. $\square$

For example, $u = (1, 0, 0)$ and $v = (1, t^\eta, 0)$ are independent for $\eta > 0$, but their standard parts coincide. Their area is t^η , not zero. Infinitesimal linear independence is genuine independence that ordinary reduction loses.

For a matrix A , valuations of its minors encode a hierarchy of rank. The ordinary rank of $\text{st } A$ records only scale zero. A higher-order nonzero minor may restore the full exact rank. This is often the relevant distinction in nearly incompressible materials, weakly coupled modes, and anisotropic metrics.

23 Vector calculus in three dimensions

On a real coordinate domain $U \subseteq \mathbb{R}^3$ the coefficientwise derivatives $\partial_i = D_i$ of [proposition 4.1](#) are K -linear commuting derivations annihilating every constant of K ; under a change of ordinary real coordinates the real Jacobian factors do not enlarge scale support, so the construction is intrinsic on real manifolds although local components are surreal.

23.1 Gradient, divergence, and curl in a Cartesian frame

With the Euclidean metric,

$$(\text{grad } F)^i = \partial_i F, \quad \text{div } V = \partial_i V^i, \quad (\text{curl } V)^i = \epsilon_{ijk} \partial_j V^k, \quad (23.1)$$

$$\Delta F = \sum_i \partial_i^2 F, \quad ((V \cdot \nabla)W)^i = V^j \partial_j W^i. \quad (23.2)$$

Here Δ is the usual spatial Laplacian, with nonpositive Fourier eigenvalues on a flat torus. Later, the Hodge Laplacian Δ_H has the opposite sign on functions.

Theorem 23.1 (Vector differential identities). *For Hahn-smooth scalar fields F, G and vector fields V, W ,*

$$\text{curl grad } F = 0, \quad \text{div curl } V = 0, \quad (23.3)$$

$$\text{curl curl } V = \text{grad div } V - \Delta V, \quad (23.4)$$

$$\text{div}(FV) = \text{grad } F \cdot V + F \text{div } V, \quad (23.5)$$

$$\text{curl}(FV) = \text{grad } F \times V + F \text{curl } V, \quad (23.6)$$

$$\text{div}(V \times W) = W \cdot \text{curl } V - V \cdot \text{curl } W, \quad (23.7)$$

$$\text{curl}(V \times W) = V \text{div } W - W \text{div } V + (W \cdot \nabla)V - (V \cdot \nabla)W, \quad (23.8)$$

$$(V \cdot \nabla)V = \text{grad}(\|V\|^2/2) - V \times \text{curl } V. \quad (23.9)$$

Proof. The first two identities contract a symmetric second derivative with an alternating symbol. For (23.4), use $\epsilon_{ijk}\epsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$ and commute derivatives. The product identities follow from Leibniz's rule and the same finite contractions. Every coefficient in a nonlinear expression is a finite sum, so [theorem 4.2](#) justifies the calculation at arbitrary rank. $\square$

23.2 Forms explain the three vector operators

Lie derivatives, forms and covariant derivatives of arbitrary tensor type are those of [sections 7](#) and [8](#) with $n = 3$. The Lie derivative is a formal infinitesimal transport operator; its existence does not prove that an arbitrary surreal vector field has a globally defined flow.

For the Euclidean Hodge star $*$ and musical identifications,

$$(\text{grad } F)^b = dF, \quad (\text{curl } V)^b = *d(V^b), \quad d(\iota_V \text{vol}) = (\text{div } V) \text{vol}.$$

Thus $\text{curl grad} = 0$ and $\text{div curl} = 0$ are the degree-one and degree-two instances of $d^2 = 0$.

23.3 Metric vector calculus

Let $\mu_g = \sqrt{\det(g_{ij})}$ in an oriented chart. Then

$$(\text{grad}_g F)^i = g^{ij} \partial_j F, \quad (23.10)$$

$$\text{div}_g V = \mu_g^{-1} \partial_i (\mu_g V^i), \quad (23.11)$$

$$(\text{curl}_g V)^i = \mu_g^{-1} \epsilon^{ijk} \partial_j (g_{k\ell} V^\ell), \quad (23.12)$$

$$\Delta_g F = \mu_g^{-1} \partial_i (\mu_g g^{ij} \partial_j F). \quad (23.13)$$

The raised alternating symbol in (23.12) denotes the numerical symbol, not the result of raising three indices on a tensor without its density factor. The invariant definition is $(\text{curl}_g V)^b = *_g d(V^b)$.

For k -forms in dimension three,

$$\delta_g = (-1)^k *_g d *_g, \quad \Delta_{H,g} = d\delta_g + \delta_g d.$$

In particular $\delta_g(V^b) = -\text{div}_g V$ and $\Delta_{H,g} F = -\Delta_g F$. The curved vector identity is

$$\text{curl}_g \text{curl}_g V = \text{grad}_g \text{div}_g V - \nabla^j \nabla_j V + \text{Ric}^\sharp(V). \quad (23.14)$$

It follows by commuting the two covariant derivatives in the alternating-symbol calculation. Omitting the Ricci term would be incorrect on a curved metric, surreal or real.

23.4 Scale charts in three dimensions

The linear scale charts $x = c + Ly$ of [section 6](#) transform gradients and divergences as stated there. Curl must be transformed with the metric and orientation, not as three unrelated scalar derivatives; the finite identity

$$(Lu) \times (Lv) = (\det L) L^{-T}(u \times v) \quad (L \in \text{GL}_3(K)) \quad (23.15)$$

encodes the required area transformation.

Let $L = \text{diag}(t^{-1}, t, t^2)$ and $f(y) = y_1^2 + y_2^2 + y_3^2$. Then

$$F(x) = t^2 x_1^2 + t^{-2} x_2^2 + t^{-4} x_3^2, \quad \text{grad}_x F = (2t^2 x_1, 2t^{-2} x_2, 2t^{-4} x_3).$$

A function of order one in scaled coordinates has very large derivatives in thin directions and a small derivative in the long direction. This is an exact identity, not a limit as a real parameter tends to zero.

24 Two metric examples

24.1 Conformal infinitesimal curvature

Take $g = e^{2t\psi}\delta$ on a real domain, with ψ smooth. The exponential is the strong series in the positive-order field $t\psi$. In three dimensions,

$$\Gamma_{ij}^k = t(\delta_i^k \partial_j \psi + \delta_j^k \partial_i \psi - \delta_{ij} \partial_k \psi), \quad (24.1)$$

$$\text{Ric}_{ij} = -t \partial_i \partial_j \psi + t^2 \partial_i \psi \partial_j \psi - (t \Delta \psi + t^2 |\text{grad } \psi|^2) \delta_{ij}, \quad (24.2)$$

$$\text{Scal}_g = e^{-2t\psi} (-4t \Delta \psi - 2t^2 |\text{grad } \psi|^2). \quad (24.3)$$

For $\psi = |x|^2/2$ this becomes

$$\text{Scal}_g = e^{-t|x|^2} (-12t - 2t^2 |x|^2).$$

The standard metric is flat, but exact curvature is nonzero at infinitesimal order. The included symbolic checks verify this example directly from (8.3).

24.2 Anisotropic nonregular metrics

The constant metric

$$g = \text{diag}(1, t^2, t^4)$$

is positive definite, flat, and has admissible inverse $\text{diag}(1, t^{-2}, t^{-4})$ and density $\mu_g = t^3$. It is not regular in the sense of (8.1) because its standard part is degenerate. Its scalar Laplacian is

$$\Delta_g = \partial_1^2 + t^{-2} \partial_2^2 + t^{-4} \partial_3^2.$$

For $F = t^4 x_3$, the scalar field has zero standard part while $\text{grad}_g F = (0, 0, 1)$. Thus raising indices and taking standard part do not commute for this metric. The exact geometry is valid; its naive real reduction is not.

25 Integral theorems in three dimensions

The integrals are those of section 11: coefficientwise over compact real chains, algebraic over polynomial chains, and by pullback over admissibly deformed chains.

By theorem 11.1, for a compact oriented real surface S and region D ,

$$\int_S \text{curl } V \cdot n \, dS = \int_{\partial S} V \cdot d\ell, \quad (25.1)$$

$$\int_D \text{div } V \, dx = \int_{\partial D} V \cdot n \, dS. \quad (25.2)$$

For regular Riemannian metrics use the flux form $\iota_V \text{vol}_g$. The normal and area density are defined from that metric with their admissible square roots, so the same statement becomes $\int_D \text{div}_g V \, \text{vol}_g = \int_{\partial D} \iota_V \text{vol}_g$.

25.1 Polynomial boxes at arbitrary scale

With the algebraic moments (11.2), polynomial cells can be integrated at any scale.

For a box with side lengths $a, b, c \in K_{>0}$,

$$\int_0^a \int_0^b \int_0^c x^p y^q z^r \, dz \, dy \, dx = \frac{a^{p+1} b^{q+1} c^{r+1}}{(p+1)(q+1)(r+1)}.$$

For $V = (x^2, y^2, z^2)$, its divergence integral and outward flux are both

$$abc(a + b + c).$$

The equality remains exact when $a = t^{-1}$, $b = t$, and $c = t^2$. The coordinate volume then equals t^2 although one side is infinite.

25.2 A monopole whose flux survives every scale

On $K^3 \setminus \{0\}$ consider the algebraic-radical field

$$B(x) = q \frac{x}{r^3}, \quad r = \sqrt{x \cdot x}, \quad q \in K.$$

Direct differentiation gives $\operatorname{div} B = 0$ away from zero. For an oriented sphere of any positive radius $R \in K$, use the scaled ordinary parametrization $x = Rn$, $n \in S^2(\mathbb{R})$. Then

$$B \cdot n = \frac{q}{R^2}, \quad dS = R^2 d\Omega, \quad \int_{S_R} B \cdot n \, dS = 4\pi q.$$

Thus an infinitesimal sphere carries the same exact flux as a unit sphere; this is the case $n = 3$ of [section 14.3](#). When q is a nonzero infinitesimal, the flux has zero standard part but is not zero.

On the real punctured space with Hahn coefficients, no global admissible vector potential A can satisfy $\operatorname{curl} A = B$ for $q \neq 0$: Stokes on the closed sphere would force its flux to vanish. This is a topological obstruction, not an analytic failure at a small scale. The scalar potential q/r still satisfies $B = -\operatorname{grad}(q/r)$; scalar and vector potentials solve different problems.

25.3 Radial scalar and vector potentials

Specializing the radial homotopy of [theorem 11.3](#) to $n = 3$ gives explicit formulas (source 03).

Proposition 25.1 (Radial scalar and vector potentials). *If $\operatorname{curl} V = 0$, then*

$$\phi(x) = \int_0^1 x \cdot V(sx) \, ds \quad \text{satisfies} \quad \operatorname{grad} \phi = V.$$

If $\operatorname{div} B = 0$, then

$$A(x) = -x \times \int_0^1 sB(sx) \, ds \quad \text{satisfies} \quad \operatorname{curl} A = B.$$

Both potentials have support contained in the original field's support.

Proof. For the first formula, $\partial_i \phi = \int_0^1 [V_i(sx) + sx_j \partial_i V_j(sx)] ds$. The curl-free condition replaces $\partial_i V_j$ by $\partial_j V_i$, and the integrand is $\frac{d}{ds}(sV_i(sx))$. For the second formula apply H to the closed two-form $\iota_B \operatorname{vol}$; its one-form components are exactly the displayed A . Equivalently a direct cross-product calculation differentiates $s^2 B(sx)$. Coefficient integration does not enlarge support. $\square$

These are specializations of the support-preserving Poincaré construction already present in [4], included to make the vector theory usable without consulting a different dimensional notation.

25.4 The weak form of the monopole

On ordinary $\mathbb{R}^3$, the locally integrable real vector field x/r^3 has distributional divergence $4\pi\delta_0$. Testing against a compactly supported smooth function and applying the divergence theorem outside a small real sphere proves this: the boundary term tends to 4π times the test function's value at zero, while the ordinary divergence vanishes away from zero. Multiplication by $q \in K$ gives the exact Hahn-distribution identity

$$\operatorname{div} \left(q \frac{x}{r^3} \right) = 4\pi q \delta_0.$$

This is the weak counterpart of the flux computation, and distinguishes an infinitesimal source from no source.

Distributional coefficients in general, and the limits of products of distributions, are discussed in [section 19.3](#).

26 Exact inversion on the three-torus

26.1 An explicit Helmholtz decomposition on the three-torus

Let $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ with flat metric. On ordinary mean-zero smooth functions, Δ^{-1} is defined by Fourier multipliers $-1/|k|^2$ for $k \in \mathbb{Z}^3 \setminus \{0\}$. Smoothness follows because smooth Fourier coefficients decay faster than any power, and division by $|k|^2$ preserves that property. Extend this real operator coefficientwise.

For $V \in \mathfrak{X}_\Gamma(\mathbb{T}^3)$, let h be its coefficientwise spatial mean and set

$$\phi = \Delta^{-1} \operatorname{div} V, \quad A = -\Delta^{-1} \operatorname{curl} V.$$

Then

$$V = h + \operatorname{grad} \phi + \operatorname{curl} A, \quad \operatorname{div} A = 0, \quad \int_{\mathbb{T}^3} \phi = 0, \quad \int_{\mathbb{T}^3} A = 0. \quad (26.1)$$

Indeed, (23.4) gives $\operatorname{curl} A = -\Delta^{-1}(\operatorname{grad} \operatorname{div} V - \Delta V) = V - h - \operatorname{grad} \phi$. These conditions uniquely fix both potentials: a mean-zero harmonic scalar is zero, and a mean-zero vector with zero divergence and curl is zero. All supports stay inside $\operatorname{supp} V$.

If $V = t^{-2} \operatorname{grad} f + t \operatorname{curl} A_0 + t^3 h_0$ with real mean-zero f , real divergence-free mean-zero A_0 , and real constant h_0 , the decomposition separates an infinite irrotational part, an infinitesimal solenoidal part, and a still smaller topological harmonic part. No relative numerical smallness threshold has to be selected.

26.2 An elliptic example with an explicit reference inverse

On $\mathbb{T}^3$ take $L_0 = 1 - \Delta : C^\infty \rightarrow C^\infty$. Its inverse has Fourier multiplier $(1 + |k|^2)^{-1}$. Let $A = I + H(x, t)$ be a symmetric matrix with positive-support H , and set

$$Lu = u - \operatorname{div}(A \operatorname{grad} u).$$

Then $P = -\operatorname{div}(H \operatorname{grad} \cdot)$ has positive support, so by [theorem 4.3](#) every $f \in \mathscr{A}_\Gamma(\mathbb{T}^3)$ has one exact Hahn-smooth solution. The metric or material perturbation can contain infinitely many positive exponents in an arbitrary-rank group. Each coefficient of the solution is obtained by a finite number of ordinary smooth operations and applications of $(1 - \Delta)^{-1}$.

For instance, if $H = tH_1$ and $f = f_0 + tf_1$, then

$$\begin{aligned} u_0 &= L_0^{-1} f_0, \\ u_1 &= L_0^{-1} (f_1 + \operatorname{div}(H_1 \operatorname{grad} u_0)), \\ u_n &= L_0^{-1} \operatorname{div}(H_1 \operatorname{grad} u_{n-1}) \quad (n \geq 2). \end{aligned}$$

The output is exact in $C^\infty((t^\Gamma))$, even if its coefficients grow too rapidly for convergence at any nonzero real parameter.

26.3 Stationary incompressible flow on $\mathbb{T}^3$

Let

$$E = \{u \in C^\infty(\mathbb{T}^3, \mathbb{R}^3) : \operatorname{div} u = 0, \langle u \rangle = 0\}.$$

Let Π be the real Leray projection onto this space. In Fourier variables,

$$\widehat{\Pi f}(k) = \begin{cases} \left(I - kk^T/|k|^2\right) \widehat{f}(k), & k \neq 0, \\ 0, & k = 0. \end{cases}$$

It preserves smoothness. Fix a *real* viscosity $\nu > 0$. The map $L_0 = -\nu\Delta : E \rightarrow E$ is an isomorphism with multiplier $1/(\nu|k|^2)$ at nonzero modes. Let

$$B(u, v) = \Pi((u \cdot \nabla)v).$$

It is a bilinear map $E \times E \rightarrow E$. The convective field has zero spatial mean because periodic integration by parts and $\operatorname{div} u = 0$ give $\int u_j \partial_j v_i = 0$.

Corollary 26.1 (Infinitesimally forced steady flow). *For every positive-support, coefficientwise mean-zero forcing $f \in C^\infty(\mathbb{T}^3, \mathbb{R}^3)((t^\Gamma))$, there is a unique positive-support velocity $u \in E((t^\Gamma))$ and a unique positive-support mean-zero pressure p solving*

$$-\nu\Delta u + (u \cdot \nabla)u + \operatorname{grad} p = f, \quad \operatorname{div} u = 0. \quad (26.2)$$

Both supports are contained in the positive monoid generated by $\operatorname{supp} f$.

Proof. Apply [theorem 16.2](#) to forcing Πf , omitting its vanishing coefficients. Then define

$$p = \Delta^{-1} \operatorname{div}(f - (u \cdot \nabla)u)$$

with zero mean. Helmholtz decomposition reconstructs [\(26.2\)](#). Any other positive-support solution projects to the same velocity equation and therefore has the same velocity; its mean-zero pressure then agrees. The support estimate follows from the theorem, bilinear convolution, and support-preserving real operators. If $\Pi f = 0$, the velocity is zero and the pressure carries the gradient forcing. $\square$

For $f = tf_1$ and $u = \sum_{n \geq 1} t^n u_n$, the recursion is

$$u_1 = (-\nu\Delta)^{-1} \Pi f_1, \quad (26.3)$$

$$u_n = -(-\nu\Delta)^{-1} \Pi \sum_{a=1}^{n-1} (u_a \cdot \nabla) u_{n-a} \quad (n \geq 2). \quad (26.4)$$

There are finitely many summands at each order. The theorem is stationary, periodic, and restricted to infinitesimal data around the zero solution. It makes no assertion about unrestricted time-dependent three-dimensional Navier–Stokes regularity.

26.4 What changes when the leading operator degenerates

If the viscosity itself is $t^\eta \nu_0$ with $\eta > 0$, the leading linear operator in (26.2) is no longer the real isomorphism used above. Rescaling may restore a useful balance, but the theorem cannot be applied unchanged. Similarly, linearization about a background with a kernel requires compatibility conditions or a finite-dimensional reduction. Noninvertible leading tensors are not a minor technical inconvenience: they change which scale controls the equation.

27 The scale connection in three dimensions

Let $\mathcal{D}_i = \partial_i + a_i \mathcal{D}_\chi$ be a scale connection with a real smooth one-form a (theorem 18.1), and define $\text{grad}_\mathcal{D}$, $\text{div}_\mathcal{D}$ and $\text{curl}_\mathcal{D}$ by replacing each ∂_i in (23.1) with $\mathcal{D}_i$.

Corollary 27.1 (Modified vector identities; source 03). *With $b = \text{curl } a$,*

$$\text{curl}_\mathcal{D} \text{grad}_\mathcal{D} F = b \mathcal{D}_\chi F, \quad (27.1)$$

$$\text{div}_\mathcal{D} \text{curl}_\mathcal{D} V = b \cdot \mathcal{D}_\chi V. \quad (27.2)$$

Proof. The first component of (27.1) is $[\mathcal{D}_2, \mathcal{D}_3]F = b_{23} \mathcal{D}_\chi F$ by theorem 18.1, and b_{23} is the first component of $\text{curl } a$; cyclic permutations give the rest. Expanding the divergence of the curl gives $[\mathcal{D}_1, \mathcal{D}_2]V^3 + [\mathcal{D}_2, \mathcal{D}_3]V^1 + [\mathcal{D}_3, \mathcal{D}_1]V^2$, which proves (27.2). $\square$

Thus $\text{curl grad} = 0$ and $\text{div curl} = 0$ hold for the scale-covariant operators exactly when the scale connection is flat. The failure of these identities has a precise source, the integrability obstruction $da \neq 0$; it is not a property of surreal numbers as such.

28 Continuum applications

28.1 Maxwell fields and exact conservation

Let τ be an ordinary time coordinate, distinct from the infinitesimal scale t . Work on a fixed real spacetime domain $I \times M$, with common support for all coefficients. In a flat spatial chart, take spatially and temporally constant material scalars $\kappa_E, \kappa_B \in K_{>0}$ and write

$$\partial_\tau B = -\text{curl } E, \quad \text{div } B = 0, \quad (28.1)$$

$$\kappa_E \partial_\tau E = \kappa_B^{-1} \text{curl } B - J, \quad \kappa_E \text{div } E = \rho. \quad (28.2)$$

All operations are meaningful for admissible fields. Divergence of the evolution equations shows that the magnetic constraint is propagated and that electric constraint propagation is equivalent to

$$\partial_\tau \rho + \text{div } J = 0.$$

These are exact identities over K .

Proposition 28.1 (Poynting balance; source 03). *Every admissible solution of (28.2) satisfies*

$$\partial_\tau \left(\frac{\kappa_E}{2} \|E\|^2 + \frac{1}{2\kappa_B} \|B\|^2 \right) + \text{div} \left(\frac{1}{\kappa_B} E \times B \right) = -J \cdot E.$$

Proof. Dot the electric evolution equation with E and the magnetic one with B/κ_B , add, and use (23.7). Commutativity of the scalar coefficients and constancy of the material parameters are the only additional algebraic requirements. $\square$

Integration over a compact region gives the exact energy balance with boundary flux. If the material parameters are real, any specified real linear Maxwell solution operator on a common time interval extends coefficientwise. If the parameters themselves have singular scale orders, the identity still holds for admissible solutions, but existence in this same function space must be re-established.

With $\kappa_E = \kappa_B = 1$ the balance law is the time component of the stress balance of [theorem 35.1](#), which source 02 proves in covariant form; the two statements are kept because their hypotheses differ. The fast-time obstruction that source 03 proves at this point is printed with source 02's general version in [section 19](#).

28.2 Cauchy stress and momentum balance

A Cauchy stress is conveniently a symmetric contravariant two-tensor σ^{ij} , with traction on a unit conormal n_j given by

$$T^i = \sigma^{ij} n_j.$$

Its divergence is $(\operatorname{div}_g \sigma)^i = \nabla_j \sigma^{ij}$. In Cartesian coordinates, $\nabla = \partial$. Momentum balance for velocity u , density ρ , and force density f is

$$\rho(\partial_\tau u + \nabla_u u) = \operatorname{div}_g \sigma + f, \quad \partial_\tau \rho + \operatorname{div}_g(\rho u) = 0$$

when the metric is time independent. Each equation is a tensor identity in the admissible field algebra. The divergence theorem converts local linear momentum balance into the integral balance over a fixed region, with the usual transport terms when appropriate.

In Euclidean coordinates, the torque density from stress has an extra term proportional to the antisymmetric part of σ . Indeed,

$$\partial_j(\epsilon_{kil} x_i \sigma^{\ell j}) = \epsilon_{kjl} \sigma^{\ell j} + \epsilon_{kil} x_i \partial_j \sigma^{\ell j}.$$

For symmetric stress the first term vanishes. Thus the usual link between stress symmetry and local angular-momentum balance is unchanged at infinitesimal or infinite coefficient scales.

28.3 Isotropic linear elasticity as a fourth-order tensor

For a displacement u , the infinitesimal strain is

$$e_{ij} = \frac{1}{2}(\nabla_i u_j + \nabla_j u_i).$$

A general elasticity tensor C^{ijkl} has the minor symmetries $C^{ijkl} = C^{jikl} = C^{ijlk}$ and, for an energy law, the major symmetry $C^{ijkl} = C^{klij}$. It acts by $\sigma^{ij} = C^{ijkl} e_{kl}$. In an orthonormal frame the isotropic law is

$$C^{ijkl} = \lambda_L \delta^{ij} \delta^{kl} + \mu_L (\delta^{ik} \delta^{jl} + \delta^{il} \delta^{jk}), \quad \sigma = \lambda_L (\operatorname{tr} e) I + 2\mu_L e.$$

Here λ_L, μ_L may be surreal constants or admissible coefficient fields. Write $e = e_0 + \frac{1}{3}(\operatorname{tr} e)I$, where $\operatorname{tr} e_0 = 0$. The energy density is

$$W(e) = \frac{\lambda_L}{2} (\operatorname{tr} e)^2 + \mu_L \operatorname{tr}(e^2) = \mu_L \|e_0\|^2 + \frac{3\lambda_L + 2\mu_L}{6} (\operatorname{tr} e)^2. \quad (28.3)$$

Therefore W is positive definite on symmetric strains exactly when $\mu_L > 0$ and $3\lambda_L + 2\mu_L > 0$. This proof uses only ordered-field positivity. A modulus can be positive infinitesimal without being zero, so very soft modes remain genuine elastic modes. Analytic coercivity estimates on a chosen real function space, however, cannot be inferred merely from this algebraic positivity.

28.4 A Beltrami field at arbitrary amplitude scale

On $\mathbb{T}^3$, let

$$u = a(t)(\sin z, \cos z, 0), \quad a(t) \in K.$$

Then

$$\operatorname{div} u = 0, \quad \operatorname{curl} u = u, \quad \Delta u = -u, \quad (u \cdot \nabla)u = 0.$$

Its kinetic density is $a^2/2$ and helicity density is a^2 . For constant real viscosity it is an exact stationary solution with force $f = \nu u$ and pressure zero. Choosing $a = t^{-1} + t^2$ creates interacting energy scales $a^2 = t^{-2} + 2t + t^4$, even though the velocity itself has only two scales. This simple example illustrates why nonlinear observables use convolution rather than independent coefficientwise multiplication.

28.5 Singular fields and smoothing are different constructions

Replacing a small real radius by a nonzero infinitesimal permits evaluating $1/r$ and $1/r^2$ there, but does not define either at $r = 0$. It also does not prove finite energy or remove a singular source. This agrees with the repository's distinction between scalar enrichment and physical regularization [6].

For comparison, a *changed* field law

$$B_a(x) = q \frac{x}{(r^2 + a^2)^{3/2}}, \quad a > 0,$$

is defined even at zero and satisfies

$$\operatorname{div} B_a = \frac{3qa^2}{(r^2 + a^2)^{5/2}}, \quad \int_{S_R} B_a \cdot n \, dS = 4\pi q \frac{R^3}{(R^2 + a^2)^{3/2}}.$$

It has replaced a point source by an extended density; that is extra modeling content. If $a = t$, its value and derivatives near zero live at different orders from their expansions on $r > 0$. A single common-domain expansion cannot be presumed to cross this boundary layer. Scaled core charts $x = ay$ and exterior charts serve different purposes.

Surreal coefficients thus provide a rigorous language for exact scale hierarchies. They do not, by themselves, establish a new experimentally validated field theory or a resolution of gravitational or hydrodynamic singularities.

28.6 A worked three-scale field

Let $t = \omega^{-1}$, $\phi = x^2y$, and

$$A_0 = (0, 0, xyz), \quad h = (1, 2, 3).$$

Define

$$V = t^{-1} \operatorname{grad} \phi + t \operatorname{curl} A_0 + t^2 h.$$

Since

$$\text{grad } \phi = (2xy, x^2, 0), \quad \text{curl } A_0 = (xz, -yz, 0),$$

we obtain

$$V = (2t^{-1}xy + txz + t^2, \quad t^{-1}x^2 - tyz + 2t^2, \quad 3t^2).$$

Direct calculation gives

$$\text{div } V = 2t^{-1}y, \quad \text{curl } V = t(y, x, 0), \quad \text{div curl } V = 0.$$

Thus the infinite irrotational component controls divergence, the infinitesimal solenoidal component controls curl, and the spatial constant component is invisible to both. This is an explicit example of why divergence and curl alone do not reconstruct a field without global or boundary information.

The dot product of the first two ordinary component fields is

$$(2xy, x^2, 0) \cdot (xz, -yz, 0) = x^2yz.$$

Consequently the squared norm contains a scale-zero mixed term $2x^2yz$ arising from $t^{-1} \cdot t$. Taking standard part of the individual vector components is not defined here because of their negative support. It would be wrong to discard their infinite parts first and then compute the energy. One must normalize or otherwise select an appropriate finite observable before using standard part.

Part III

Minkowski space

29 Scope and conventions of Part III

This part is source 02's Lorentzian development. Its dimension-free material has been moved to Part I and is cited from there. As stated in [section 1.2](#), it is mathematics of fields on Minkowski space: physical vocabulary names objects in the conventions fixed below, and nothing in it is a claim about the physical world.

There are two different readings of “Minkowski space with surreal components.” In the first, the points themselves have surreal coordinates: an event is an element of K^4 , and the quadratic form is defined over K . In the second, events remain ordinary real points, while vectors, tensors, currents, amplitudes, and coupling parameters take values in K . These readings share the same finite tensor algebra but not the same analytic foundations.

We develop both. The first supplies unrestricted algebraic Lorentz geometry. The second supplies a rigorous, useful field theory with differential equations, ordinary causal domains, and coefficient-wise integration. A third, intermediate construction evaluates coherent fields on infinitesimal neighborhoods of real events. It permits bounded surreal coordinate changes without pretending that every real smooth function has a globally defined value at every surreal argument.

Three carriers, not one implicit substitution.

$$\underbrace{\mathbb{M}_{\mathbb{R}} = \mathbb{R}^4}_{\text{real events}} \quad \underbrace{\mathcal{O}_K^4 \supset \mathbb{R}^4}_{\text{finite surreal events}} \quad \underbrace{\mathbb{M}_K = K^4}_{\text{all events in the workspace}}.$$

The full Lorentz group acts algebraically on K^4 . Real Lorentz transformations act on ordinary spacetime and coherent fields. Bounded surreal transformations act on finite halos by Taylor prolongation. The assertions are different and will not be conflated.

29.1 Conventions

Throughout, K is a specified set-sized real-closed ordered subfield of $\mathbf{No}$; our main choice is a Hahn field described in [section 2](#). We use units $c = 1$, rationalized electromagnetic units, and the signature

$$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1), \quad x = (x^0, x^1, x^2, x^3) = (\tau, \mathbf{x}), \quad q_\eta(x) = \eta_{\mu\nu} x^\mu x^\nu. \quad (29.1)$$

Greek indices range from 0 to 3; Latin spatial indices range from 1 to 3. Repeated tensor indices are summed once up and once down. The orientation is

$$\text{vol} = dx^0 \wedge dx^1 \wedge dx^2 \wedge dx^3, \quad \epsilon_{0123} = 1, \quad \epsilon^{0123} = -1. \quad (29.2)$$

The symbol $\epsilon_{\mu\nu\rho\sigma}$ denotes the alternating tensor; the monomial $t = t^1 = \omega^{-1}$ (source 02's ε) is the basic positive infinitesimal, and $1 \in \Gamma$ is assumed throughout Part III. Ordinary time is τ , never the Hahn monomial.

The wave operator and electromagnetic conventions are

$$\square = \partial_\mu \partial^\mu = \partial_\tau^2 - \Delta, \quad (29.3)$$

$$F = dA, \quad F_{0i} = E_i, \quad F_{ij} = -\epsilon_{ijk} B_k, \quad \partial_\mu F^{\mu\nu} = J^\nu. \quad (29.4)$$

All spacetime derivatives annihilate constant elements of K . They are *not* the Berarducci–Mantova scalar derivation.

29.2 Provenance and the main results

Conway normal forms, real closedness, and the strong-summation calculus are classical inputs; the relation to derivations is discussed in [\[9, 10, 15, 16\]](#). Ordered-field relativity also has an established literature; [\[21\]](#) is a relevant precedent. Our ordinary wave-equation input is the real smooth Cauchy and Green theory summarized in [\[25, 26\]](#). The conventional electromagnetic formulas can be compared with [\[27, 28\]](#); we derive the signs used here.

At the pin, the repository's documentation already emphasized set-sized workspaces, distinct derivative notions, strong versus coefficientwise operations, and the fact that scalar extension does not regularize a singular spacetime [\[1, 6\]](#). Source 02 specializes these foundations to Minkowski fields. Its substantive outputs include the following.

Construction or result	Precise content	Location
Lorentz geometry	Cones, boosts, frames, tensors over K ; no analytic transfer assumed	sections 30 and 31
Bounded reduction	A split reduction map to the real Lorentz group; its kernel is the exponential of infinitesimal Lorentz generators	section 32
Coherent calculus	The coefficient algebra of Part I on real spacetime, bounded Poincaré action on halos	sections 5 and 33
Causal dynamics	Coefficientwise Cauchy solutions and retarded inverses, including positive-valuation operator perturbations	section 36
Maxwell scale detection	Positive energy, dominant energy, null stress, and exact doubling of the leading field valuation	sections 34 and 35
Nonlinear lifting	A unique positive-order deformation of a given real cubic-wave solution on its fixed time slab	section 38
Obstructions	Infinite boosts, infinite frequencies, scalar-derivative confusion, and singular-domain limits	sections 19 and 40

These are theorems of the explicitly defined framework, not assertions that all are new in the literature. The classical algebraic identities are not claimed as new merely because their coefficient field has changed.

30 Algebraic Minkowski geometry over a surreal field

30.1 Vectors, cones, and the reverse triangle inequality

Put $V = K^4$. The metric η identifies V with its algebraic dual $V^* = \text{Hom}_K(V, K)$ by $u^\flat(v) = \eta(u, v)$. A nonzero vector is timelike, null, or spacelike according as $q_\eta(u) > 0$, $q_\eta(u) = 0$, or $q_\eta(u) < 0$. Future causal means

$$u^0 > 0, \quad q_\eta(u) \geq 0. \quad (30.1)$$

The zero vector is included when discussing the closed future cone C^+ .

Spatial norms are algebraic: $|\mathbf{u}| = (\sum_i (u^i)^2)^{1/2}$, using real closedness. The identity

$$|\mathbf{a}|^2 |\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2 = \sum_{i < j} (a_i b_j - a_j b_i)^2 \quad (30.2)$$

proves Cauchy–Schwarz over K without compactness or limits. Hence future causality is equivalent to $u^0 \geq |\mathbf{u}|$ with $u^0 \geq 0$, and C^+ is closed under addition and nonnegative scalar multiplication.

Proposition 30.1 (Reverse Cauchy–Schwarz). *For future timelike vectors u, w ,*

$$\eta(u, w) \geq \sqrt{q_\eta(u) q_\eta(w)}. \quad (30.3)$$

Equality holds precisely when u, w are positive scalar multiples. Moreover,

$$\sqrt{q_\eta(u + w)} \geq \sqrt{q_\eta(u)} + \sqrt{q_\eta(w)}. \quad (30.4)$$

Proof. Divide u by $\sqrt{q_\eta(u)}$ and write the resulting unit vector as $(a, \mathbf{b})$, with $a > 0$ and $a^2 - |\mathbf{b}|^2 = 1$. The explicit Lorentz transformation in [equation \(30.7\)](#) below takes this vector to e_0 after inversion and preserves future causality. In that frame the desired inequality is $w^0 \geq \sqrt{(w^0)^2 - |\mathbf{w}|^2}$, with equality exactly when $\mathbf{w} = 0$. Squaring the second assertion and expanding $q_\eta(u + w)$ gives the first assertion. All square roots are of nonnegative elements. $\square$

This argument is algebraic; “frame” here means a basis change, not an appeal to connectedness of a non-Archimedean topological group.

30.2 Lorentz transformations and explicit rest frames

Define

$$O(1, 3; K) = \{\Lambda \in \text{GL}_4(K) : \Lambda^T \eta \Lambda = \eta\}, \quad (30.5)$$

and

$$\mathcal{G} = \text{SO}^+(1, 3; K) = \{\Lambda \in O(1, 3; K) : \det \Lambda = 1, \Lambda^0_0 > 0\}. \quad (30.6)$$

The superscript $+$ is an algebraic time-orientation condition, not the assertion that $\mathcal{G}$ is a topological identity component.

For a unit future timelike $u = (a, \mathbf{b})$ define

$$B_u = \begin{pmatrix} a & \mathbf{b}^T \\ \mathbf{b} & I_3 + \frac{\mathbf{b}\mathbf{b}^T}{a+1} \end{pmatrix}. \quad (30.7)$$

Since $a \geq 1$, the denominator is nonzero. Multiplication, using $|\mathbf{b}|^2 = a^2 - 1$, proves $B_u^T \eta B_u = \eta$, $B_u e_0 = u$, and $B_u^{-1} = B_{(a, -\mathbf{b})}$. Its determinant is 1: on the spatial orthogonal complement of $\mathbf{b}$ it is the identity, and the remaining two-dimensional block has determinant $a^2 - |\mathbf{b}|^2 = 1$; $\mathbf{b} = 0$ is immediate.

To see that it preserves the future cone, its time component on $w \in C^+$ satisfies $aw^0 + \mathbf{b} \cdot \mathbf{w} \geq (a - |\mathbf{b}|)w^0 \geq 0$, strictly for nonzero w . Spatial rotations preserve that cone too. More generally, if $\Lambda \in \mathcal{G}$, its first column is future unit timelike; $B_{\Lambda e_0}^{-1} \Lambda$ fixes e_0 and has an $\text{SO}_3(K)$ spatial block. Thus every element of $\mathcal{G}$ preserves C^+ and decomposes into a boost and a rotation.

30.3 Boosts without a globally available rapidity

A useful parametrization uses any $\lambda \in K_{>0}$:

$$B(\lambda) = \begin{pmatrix} (\lambda + \lambda^{-1})/2 & (\lambda - \lambda^{-1})/2 & 0 & 0 \\ (\lambda - \lambda^{-1})/2 & (\lambda + \lambda^{-1})/2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (30.8)$$

This is an active boost. The passive coordinate boost uses its inverse. It satisfies

$$B(\lambda)B(\mu) = B(\lambda\mu), \quad B(\lambda)k_+ = \lambda k_+, \quad B(\lambda)k_- = \lambda^{-1}k_-, \quad k_{\pm} = (1, \pm 1, 0, 0). \quad (30.9)$$

No logarithm of λ is required inside the fixed Hahn workspace. If a suitable exponential extension has been selected, one may set $\lambda = e^r$ and recover rapidity.

For $\lambda = t^{-1}$,

$$\gamma = \frac{t^{-1} + t}{2}, \quad \beta = \frac{1 - t^2}{1 + t^2}, \quad 1 - \beta^2 = \frac{4t^2}{(1 + t^2)^2} > 0. \quad (30.10)$$

The velocity remains strictly subluminal, yet the Lorentz factor is unlimited. This is an exact algebraic fact, not a physical claim about attainable observers.

30.4 Four-momentum and the mass shell

For a positive rest mass $m \in K$ and future unit velocity U , define the four-momentum $p = mU$. Then

$$q_{\eta}(p) = m^2, \quad p^0 = \sqrt{m^2 + |\mathbf{p}|^2}, \quad \mathbf{v} = \frac{\mathbf{p}}{p^0}, \quad |\mathbf{v}| < 1. \quad (30.11)$$

A future null momentum has $|\mathbf{v}| = 1$. The energy measured by a unit future observer u is $E_u = \eta(p, u)$; reverse Cauchy-Schwarz gives $E_u \geq m$, with equality exactly in the rest frame. These are algebraic kinematic relations, valid for finite, infinitesimal, and unlimited positive m .

Collinear velocity addition also remains exact. For $|v|, |w| < 1$, put

$$v \oplus w = \frac{v + w}{1 + vw}, \quad 1 - (v \oplus w)^2 = \frac{(1 - v^2)(1 - w^2)}{(1 + vw)^2} > 0. \quad (30.12)$$

The denominator is positive because $|vw| < 1$. No limiting process is needed to prove that a composition remains subluminal.

In reference units, take $m = t$ and boost its rest momentum by $B(t^{-1})$. The resulting momentum is

$$p = \left(\frac{1+t^2}{2}, \frac{1-t^2}{2}, 0, 0 \right), \quad q_\eta(p) = t^2. \quad (30.13)$$

It is timelike and finite but has the nonzero null shadow $(1/2, 1/2, 0, 0)$. This illustrates how a real massless approximation can coexist with an exact positive infinitesimal mass, without identifying the two exact causal types.

30.5 Affine events and causal order

The Poincaré group $K^4 \rtimes \mathcal{G}$ acts on $\mathbb{M}_K$ by $x \mapsto \Lambda x + a$. Define $x \preceq y$ if $y - x \in C^+$. Cone closure gives transitivity, and $C^+ \cap (-C^+) = \{0\}$ gives antisymmetry. Thus this is a partial order invariant under the time-oriented Poincaré group.

If $y - x$ is timelike, the proper length of the affine segment is $\sqrt{q_\eta(y - x)}$. That is an algebraic invariant. It does not by itself define integrals along arbitrary K -parameterized curves or prove any causal compactness theorem for the order topology of K^4 .

31 Tensor algebra, exterior algebra, and Hodge duality

31.1 Tensor types and their transformations

A tensor of type (r, s) is an element of

$$T^r_s(V) = V^{\otimes Kr} \otimes_K (V^*)^{\otimes Ks}, \quad \dim_K T^r_s(V) = 4^{r+s}. \quad (31.1)$$

Tensor products, permutation symmetries, contractions, traces and metric raising and lowering are finite operations and require no summability. Under $x' = \Lambda x$, upper indices transform with Λ and lower indices with Λ^{-1} ; this is (3.1) with $P = \Lambda^{-1}$.

The dual map is inverse transpose in column-vector notation. Confusing a covector with a vector merely because both have four components is especially dangerous when some entries are infinite.

A rank-two covariant tensor has the direct decomposition

$$T_{\mu\nu} = T_{(\mu\nu)}^{\text{tf}} + \frac{1}{4} \eta_{\mu\nu} \eta^{\alpha\beta} T_{\alpha\beta} + T_{[\mu\nu]}, \quad (31.2)$$

with dimensions $9 + 1 + 6 = 16$. The formula remains valid over K .

31.2 Indefinite self-adjointness is not Euclidean spectral theory

The adjoint of an endomorphism is $A^\dagger = \eta^{-1} A^T \eta$. A Lorentz-self-adjoint operator need not be diagonalizable. For a nonzero null vector k , let

$$N(v) = k\eta(k, v). \quad (31.3)$$

Then $N^\dagger = N$, $N \neq 0$, and $N^2 = 0$. Therefore no theorem saying that every symmetric tensor has an orthonormal eigenbasis can be imported from positive-definite geometry. This counterexample already exists over $\mathbb{R}$ and persists over the surreals.

For a unit timelike observer u , however, the orthogonal space $u^\perp$ is three-dimensional and $-\eta$ is positive definite on it. All ordinary finite-dimensional Euclidean algebra works there over the real-closed field K . The projector

$$h^\mu{}_\nu = \delta^\mu{}_\nu - u^\mu u_\nu \quad (31.4)$$

gives the observer's time-space decomposition $v = (u \cdot v)u + h(v)$.

31.3 Forms and the Lorentzian Hodge operator

The exterior powers have dimensions 1, 4, 6, 4, 1. The metric gives the pairing

$$\langle \alpha, \beta \rangle = \frac{1}{p!} \alpha_{\mu_1 \dots \mu_p} \beta^{\mu_1 \dots \mu_p} \quad \text{on } \Lambda^p V^*. \quad (31.5)$$

Define $*$ uniquely by

$$\alpha \wedge * \beta = \langle \alpha, \beta \rangle \text{vol}. \quad (31.6)$$

For a two-form,

$$(*F)_{\mu\nu} = \frac{1}{2} \epsilon_{\mu\nu\rho\sigma} F^{\rho\sigma}. \quad (31.7)$$

On a basis wedge indexed by I , $*$ sends it to the complementary wedge, multiplied by the product of metric signs on I and the permutation sign of (I, I^c) . Applying this rule twice proves

$$*^2 = (-1)^{p(4-p)+3} \text{id}. \quad (31.8)$$

Thus $*^2 = -1$ on two-forms, and $*^2 = 1$ on one- and three-forms; this is (10.2) with $n = 4$ and $q = 3$ (theorem 10.1). The construction commutes with orientation-preserving Lorentz transformations.

There is no direct four-dimensional replacement for the three-dimensional cross product of *two* vectors with vector output: $u \wedge v$ has six components. The natural output is a bivector. The Hodge dual of a wedge of *three* covectors does have one-form degree; after raising its index it is orthogonal to all three original vectors. An observer u and an orientation of $u^\perp$ recover an ordinary spatial cross product within that three-space.

31.4 Surcomplex self-dual fields

After extending coefficients to $K[i]$, set

$$F_+ = \frac{1}{2}(F - i * F), \quad F_- = \frac{1}{2}(F + i * F). \quad (31.9)$$

Then $*F_+ = iF_+$, $*F_- = -iF_-$, and each eigenspace has dimension three over $K[i]$. For a real K -valued field, conjugation exchanges F_+ and F_- . This is a representation-theoretic decomposition, not an invocation of surcomplex contour integration.

32 Bounded Lorentz transformations and real shadows

32.1 The reduction group

Write $\mathcal{G} = \text{SO}^+(1, 3; K)$ for the group of section 30, and define

$$\mathcal{G}_{\mathcal{O}_K} = \mathcal{G} \cap \text{Mat}_4(\mathcal{O}_K), \quad \mathcal{G}_1 = \mathcal{G} \cap (I + \text{Mat}_4(\mathfrak{m}_K)). \quad (32.1)$$

The word *bounded* means entrywise finite; it does not prescribe one common real bound for every matrix in the group. The inverse formula $\Lambda^{-1} = \eta^{-1} \Lambda^\top \eta$ shows that $\mathcal{G}_{\mathcal{O}_K}$ is a group.

Theorem 32.1 (Bounded Lorentz reduction). *Entrywise standard part gives a split exact sequence*

$$1 \longrightarrow \mathcal{G}_1 \longrightarrow \mathcal{G}_{\mathcal{O}_K} \xrightarrow{\text{st}} \text{SO}^+(1, 3; \mathbb{R}) \longrightarrow 1. \quad (32.2)$$

Let

$$\mathfrak{g}_1 = \{X \in \text{Mat}_4(\mathfrak{m}_K) : X^\top \eta + \eta X = 0\}. \quad (32.3)$$

The strongly summable matrix series $\exp$ and $\log$ define mutually inverse bijections

$$\exp : \mathfrak{g}_1 \longrightarrow \mathcal{G}_1, \quad \log : \mathcal{G}_1 \longrightarrow \mathfrak{g}_1. \quad (32.4)$$

Consequently every bounded Lorentz matrix has a unique expression

$$\Lambda = \exp(X) L_0, \quad L_0 = \text{st } \Lambda \in \text{SO}^+(1, 3; \mathbb{R}), \quad X \in \mathfrak{g}_1. \quad (32.5)$$

Proof. Reduction preserves the defining polynomial equations, including determinant one. The first column of a Lorentz matrix satisfies $(\Lambda^0_0)^2 = 1 + \sum_i (\Lambda^i_0)^2$; time orientation implies $\Lambda^0_0 \geq 1$. Hence its standard part is at least one and the reduced matrix is orthochronous. The real constant matrices give a section, and the kernel is exactly $\mathcal{G}_1$.

For an infinitesimal matrix, the finite union of the supports of its entries lies in $\Gamma_{>0}$. The Neumann lemma makes every coefficient of each matrix series a finite sum, even when the value group has arbitrary rank. Formal one-variable exponential and logarithm identities therefore apply.

Put $A^\dagger = \eta^{-1} A^\top \eta$. If $X^\dagger = -X$, then $(\exp X)^\dagger \exp X = \exp(-X) \exp X = I$. Thus $\det(\exp X)^2 = 1$. Its determinant has standard part one, so it equals one, rather than minus one. Its 00 entry also has standard part one, so it is orthochronous. Conversely, if $\Lambda = I + H \in \mathcal{G}_1$, then $\Lambda^\dagger = \Lambda^{-1}$, and the formal identities give

$$(\log \Lambda)^\dagger = \log(\Lambda^\dagger) = \log(\Lambda^{-1}) = -\log \Lambda.$$

The identities $\exp \log(I + H) = I + H$ and $\log \exp X = X$ prove the bijection. Finally $\Lambda L_0^{-1} \in \mathcal{G}_1$, which gives [equation \(32.5\)](#) and its uniqueness. $\square$

As groups, [equation \(32.2\)](#) gives $\mathcal{G}_{\mathcal{O}_K} \simeq \mathcal{G}_1 \rtimes \text{SO}^+(1, 3; \mathbb{R})$, with the real group acting by conjugation. The logarithmic coordinate on $\mathcal{G}_1$ is *not* additive: multiplying exponentials uses the Baker–Campbell–Hausdorff series, again with positive support. The theorem uses only an infinitesimal exponential; it does not require a globally defined exponential on arbitrary Hahn matrices.

32.2 Tensor valuations and bounded-frame invariance

For a nonzero finite-component tensor, define its component valuation by

$$v_V(T) = \min\{v(T^{\mu_1 \dots \mu_r}_{\nu_1 \dots \nu_s})\}, \quad v_V(0) = +\infty. \quad (32.6)$$

This minimum refers to a specified frame, not to the scalar contractions of T .

Proposition 32.2 (Frame invariance and reduction). *If $\Lambda \in \mathcal{G}_{\mathcal{O}_K}$, then $v_V(\Lambda \cdot T) = v_V(T)$ for every tensor type. For tensors with finite components,*

$$\text{st}(\Lambda \cdot T) = (\text{st } \Lambda) \cdot \text{st}(T). \quad (32.7)$$

Proof. Every coefficient in the tensor transformation matrix is a finite product of entries of Λ and Λ^{-1} , and therefore has nonnegative valuation. The finite transformation sums cannot lower the minimum component valuation. Applying the same argument to the inverse gives the opposite inequality. The reduction identity follows from multiplicativity and additivity of standard part on $\mathcal{O}_K$. $\square$

This is the invariance recorded in [section 3](#), for the bounded Lorentz group.

Example 32.3 (An infinite boost reveals an infinitesimal vector). Let $k_+ = (1, 1, 0, 0)$ and let $B(t^{-1})$ be the null-eigenvalue boost of [equation \(30.8\)](#). Then

$$V = tk_+, \quad \text{st } V = 0, \quad B(t^{-1})V = k_+, \quad \text{st}(B(t^{-1})V) = k_+. \quad (32.8)$$

Both the input and output vectors have finite components. No real linear transformation can send the zero shadow to this nonzero shadow. Thus there is no extension of the standard-part equivariance statement to the full Lorentz group. The same boost sends the unit rest vector to a vector with unlimited components.

32.3 Near-null geometry and a quantitative stability criterion

The causal type of a finite vector need not be visible in its real shadow. The three vectors

$$(1, 1 - t, 0, 0), \quad (1, 1, 0, 0), \quad (1, 1 + t, 0, 0) \quad (32.9)$$

are respectively timelike, null, and spacelike, but all have the same standard part. Their squared lengths are $2t - t^2$, 0 , and $-2t - t^2$.

For $V \neq 0$, put $\alpha = v_V(V)$. If $q_\eta(V) \neq 0$, define its *null depth*

$$d(V) = v(q_\eta(V)) - 2\alpha \in \Gamma_{\geq 0}; \quad (32.10)$$

put $d(V) = +\infty$ for a null vector. A positive depth records cancellation of the leading quadratic form, not smallness of every component.

Theorem 32.4 (Causal stability beyond the null depth). *For $q_\eta(V) \neq 0$, let $d = d(V)$. If*

$$v_V(W) > \alpha + d, \quad (32.11)$$

then $q_\eta(V + W)$ has the same leading exponent, leading coefficient, and sign as $q_\eta(V)$. Both $d(V)$ and condition [equation \(32.11\)](#) are invariant under bounded Lorentz changes of frame.

Proof. Let $\beta = v_V(W)$. The bilinear expression obeys $v(\eta(V, W)) \geq \alpha + \beta$ and $v(q_\eta(W)) \geq 2\beta$. Since $d \geq 0$ and $\beta > \alpha + d$, both bounds are strictly greater than $2\alpha + d = v(q_\eta(V))$. In $q_\eta(V + W) = q_\eta(V) + 2\eta(V, W) + q_\eta(W)$, the leading term therefore survives. Frame invariance follows from [theorem 32.2](#) and the exact invariance of q . $\square$

The criterion is sufficient, not necessary. A much larger perturbation tangent to an appropriate level set can preserve the sign. Its significance is that it works without an Archimedean comparison of scales. For an exactly null vector there is no finite depth protecting the zero value against arbitrary transverse perturbations.

33 Fields on real Minkowski spacetime

On an open $U \subset \mathbb{M}_{\mathbb{R}}$ the coefficient algebra is $\mathcal{A}_{\Gamma}(U)$ of [section 4](#); a tensor field of type (r, s) is an element of $\mathcal{A}_{\Gamma}(U)^{4^{r+s}}$ with the transition laws of the tensor representation, forms are $\Omega_{\Gamma}^p(U) = \Omega^p(U; \mathbb{R})((t^{\Gamma}))$, and ∂_{μ} is the coefficientwise derivative. These are modules over a coefficient algebra, not finite-dimensional vector spaces over K , and they contain much more than $K \otimes_{\mathbb{R}} C^{\infty}(U)$ ([theorem 4.4](#)). Everything in [sections 4, 5, 7, 11](#) and [12](#) applies with $n = 4$. A vector with nonreal entries is not literally an ordinary real tangent vector to U . In Minkowski Cartesian coordinates the flat connection is $\nabla_{\mu} = \partial_{\mu}$, and curvature and torsion vanish identically, over any coefficient workspace (source 02).

33.1 Moving frames and invertibility

An admissible moving frame is a matrix $E \in \mathrm{GL}_4(\mathcal{A}_{\Gamma}(U))$. Pointwise invertibility alone does not certify that its inverse has one coherent common-domain expansion. A useful sufficient condition is

$$E = E_0 + H, \quad E_0 \in C^{\infty}(U, \mathrm{GL}_4(\mathbb{R})), \quad H \in \mathrm{Mat}_4(\mathcal{A}_{>0}(U)). \quad (33.1)$$

Then

$$E^{-1} = \sum_{n \geq 0} (-E_0^{-1} H)^n E_0^{-1} \quad (33.2)$$

is coherent by the positive-support lemma. In the moving frame the connection one-form is $\omega = E^{-1} dE$. Direct differentiation of $E^{-1} E = I$ proves

$$d\omega + \omega \wedge \omega = 0. \quad (33.3)$$

Thus a varying frame need not have zero connection coefficients, although the underlying connection is flat. If $E^{\top} \eta E = \eta$, differentiating this identity proves $\omega^{\top} \eta + \eta \omega = 0$. These formulas allow accelerated tetrads and nonconstant observer fields while retaining the metric and support conditions explicitly.

33.2 Bounded Poincaré transformations of coherent fields

Let

$$g(z) = \Lambda z + b, \quad \Lambda \in \mathcal{G}_{\mathcal{O}_K}, \quad b \in \mathcal{O}_K^4, \quad (33.4)$$

and let $g_0(x) = (\mathrm{st} \Lambda)x + \mathrm{st} b$. For real $x \in g_0^{-1}U$, define

$$(g^* f)(x) = \mathrm{Ev} f(g(x)). \quad (33.5)$$

The positive-support displacement $(\Lambda - \mathrm{st} \Lambda)x + (b - \mathrm{st} b)$ has coefficient functions polynomial in x and an exponent support independent of x . The proof of [theorem 5.2](#) therefore shows that [equation \(33.5\)](#) is a coherent smooth series on $g_0^{-1}U$.

The formal chain rule proves

$$\partial_{\mu}(g^* f) = \Lambda^{\nu}_{\mu} g^*(\partial_{\nu} f), \quad \square(g^* f) = g^*(\square f). \quad (33.6)$$

Composition obeys $(g \circ h)^* = h^* g^*$. The derivative statement extends to covariant and contravariant tensors with their respective transformation matrices; thus the flat tensor calculus and the field

equations below are covariant under this bounded halo action. For finite fields, reduction commutes with the action:

$$\text{st}(g^* f) = g_0^*(\text{st } f). \quad (33.7)$$

This is the pullback (5.6) for the affine map g : its real part g_0 is a real Poincaré transformation and its correction has positive support, so [theorem 5.2](#) and [proposition 5.5](#) apply.

33.3 What remains valid on all of K^4 ?

Polynomial functions $K[x^0, x^1, x^2, x^3]$ have a globally defined full-Poincaré action. Their formal partial derivatives obey the ordinary chain rule at arbitrary K -valued points. Rational functions admit the same operations on their explicitly stated nonvanishing denominator domains. One can develop additional analytic classes on valuation balls, but their summability domains must be specified separately.

Consequently, the finite algebraic tensor theory has full K -Lorentz covariance, while the coherent smooth theory just constructed has real-spacetime and bounded-halo covariance. An infinite Lorentz transformation need not preserve the chosen coefficient function algebra. [Section 40](#) proves an explicit obstruction rather than hiding this boundary in notation.

33.4 Codifferentiation and the de Rham wave operator

For a p -form with $p \geq 1$, define the codifferential in Cartesian components by

$$(\delta\alpha)_{\mu_2 \dots \mu_p} = -\partial^\rho \alpha_{\rho \mu_2 \dots \mu_p}, \quad (33.8)$$

and put $\delta = 0$ on scalars. With our four-dimensional Lorentzian Hodge convention, this is $\delta = *d*$. The identity follows on each basis wedge from the metric signs and the complementary permutation. In particular $\delta^2 = 0$. Expanding the alternating derivative gives, in every degree,

$$(d\delta + \delta d)\alpha = -\square\alpha, \quad (33.9)$$

where the right side acts componentwise. The mixed second derivatives cancel and only $-\partial^\rho \partial_\rho$ on each component remains. The resulting de Rham operator is hyperbolic, rather than the positive elliptic operator associated with Euclidean Hodge theory. This is another place where the metric signature matters.

34 Electromagnetic fields over surreal coefficients

34.1 The two-form, sources, and the three-plus-one decomposition

A potential is a coherent one-form $A = A_\mu dx^\mu$. With $A_\mu = (\phi, -\mathbf{A})$, its field strength $F = dA$ is

$$(F_{\mu\nu}) = \begin{pmatrix} 0 & E_1 & E_2 & E_3 \\ -E_1 & 0 & -B_3 & B_2 \\ -E_2 & B_3 & 0 & -B_1 \\ -E_3 & -B_2 & B_1 & 0 \end{pmatrix}, \quad \mathbf{E} = -\text{grad } \phi - \partial_\tau \mathbf{A}, \quad \mathbf{B} = \text{curl } \mathbf{A}. \quad (34.1)$$

In particular $F^{0i} = -E_i$, $F^{i0} = E_i$, and $F^{ij} = F_{ij}$. With $J^\mu = (\rho, \mathbf{j})$, the equations

$$\partial_{[\lambda} F_{\mu\nu]} = 0, \quad \partial_\mu F^{\mu\nu} = J^\nu \quad (34.2)$$

are exactly

$$\operatorname{div} \mathbf{E} = \rho, \quad \operatorname{curl} \mathbf{B} - \partial_\tau \mathbf{E} = \mathbf{j}, \quad (34.3)$$

$$\operatorname{div} \mathbf{B} = 0, \quad \partial_\tau \mathbf{B} + \operatorname{curl} \mathbf{E} = 0. \quad (34.4)$$

Every operation here is a finite polynomial differential operation in the coherent algebra. The equations can therefore be interpreted either as equations of coherent fields on real spacetime or, for polynomial fields, as equations on all of K^4 .

The differential-form version, with the Hodge convention [equation \(31.6\)](#), is

$$dF = 0, \quad d * F = - * J^\flat, \quad J^\flat = J_\mu dx^\mu. \quad (34.5)$$

The minus sign is convention dependent and is intentional. For example the spatial 123 component of $d * F$ is $-\operatorname{div} \mathbf{E}$, whereas the corresponding component of $*J^\flat$ is ρ . Equivalently, the excitation form $- * F$ has exterior derivative $*J^\flat$.

34.2 Gauge symmetry and constraints

For any coherent scalar χ ,

$$A \mapsto A + d\chi \quad (34.6)$$

leaves F unchanged. Conversely, on a star-shaped real domain, potentials for the same field differ by such a gauge term, by the coefficientwise Poincaré lemma. Antisymmetry and commuting derivatives imply current conservation:

$$\partial_\nu J^\nu = \partial_\nu \partial_\mu F^{\mu\nu} = 0. \quad (34.7)$$

The Lorenz condition, written with a z , is

$$\partial_\mu A^\mu = 0. \quad (34.8)$$

In this gauge the field equation reduces to $\square A^\nu = J^\nu$. Reaching the gauge requires solving $\square \chi = -\partial_\mu A^\mu$ with suitable initial or support conditions; it is not a purely algebraic consequence of having a field extension.

The constraints in [equations \(34.3\) and \(34.4\)](#) propagate under the evolution equations. Indeed,

$$\partial_\tau (\operatorname{div} \mathbf{E} - \rho) = -\operatorname{div} \mathbf{j} - \partial_\tau \rho = 0, \quad (34.9)$$

$$\partial_\tau \operatorname{div} \mathbf{B} = -\operatorname{div} \operatorname{curl} \mathbf{E} = 0. \quad (34.10)$$

Thus compatible coherent initial data and a conserved coherent source give coefficientwise constraint propagation. Existence and uniqueness are obtained by lifting the ordinary Maxwell Cauchy problem, or by an appropriately constrained wave equation as in [section 36](#).

34.3 Invariants, duality, and observer dependence

The Hodge dual acts on the three-vectors by

$$* : (\mathbf{E}, \mathbf{B}) \mapsto (-\mathbf{B}, \mathbf{E}). \quad (34.11)$$

Direct contraction gives the two Lorentz invariants

$$I_1 = F_{\mu\nu} F^{\mu\nu} = 2(\mathbf{B}^2 - \mathbf{E}^2), \quad I_2 = F_{\mu\nu} (*F)^{\mu\nu} = 4\mathbf{E} \cdot \mathbf{B}. \quad (34.12)$$

The sign of I_2 changes under a different orientation convention. It is a pseudoscalar under orientation-reversing transformations and a scalar under the proper group used here.

For a future unit observer u , the electric and magnetic parts can be defined covariantly as

$$E_u^\mu = F^{\mu\nu}u_\nu, \quad B_u^\mu = (*F)^{\mu\nu}u_\nu. \quad (34.13)$$

Both are orthogonal to u . In the rest frame of u , these are $(0, \mathbf{E})$ and $(0, -\mathbf{B})$ with our particular Hodge convention. One can instead put a minus sign in the definition of B_u to make its spatial entries equal to the displayed $\mathbf{B}$; no physical assertion depends on this naming choice. The field itself, rather than either observer decomposition, is the Lorentz tensor.

In vacuum, $F \mapsto aF + b * F$ preserves the equations for constant $a, b \in K$. It preserves the stress tensor when $a^2 + b^2 = 1$. This circle can be parameterized rationally over K without defining trigonometric functions of unlimited angles:

$$a = \frac{1 - s^2}{1 + s^2}, \quad b = \frac{2s}{1 + s^2}, \quad s \in K, \quad (34.14)$$

with the remaining point $(-1, 0)$ added separately. The unit circle and its angle group over the surreals are treated in `docs/surcomplex/trigonometry`. Infinitesimal duality angles can alternatively be treated by strongly summable power series.

34.4 An exact null wave at any amplitude scale

Let $a \in K$, let $f \in C^\infty(\mathbb{R}, \mathbb{R})$, and put $s = \tau - x^3$. Then

$$\mathbf{E} = af(s)(1, 0, 0), \quad \mathbf{B} = af(s)(0, 1, 0) \quad (34.15)$$

is a source-free coherent Maxwell field. For example, $\text{curl } \mathbf{B} = (af'(s), 0, 0) = \partial_\tau \mathbf{E}$, while $\text{curl } \mathbf{E} = (0, -af'(s), 0) = -\partial_\tau \mathbf{B}$; the divergences vanish. Both invariants in [equation \(34.12\)](#) are zero for every value of a , including infinitesimal and unlimited values.

The energy propagates in the $+x^3$ direction. With $k = (1, 0, 0, 1)$, the stress tensor derived below is

$$T^{\mu\nu} = a^2 f(s)^2 k^\mu k^\nu. \quad (34.16)$$

A plane wave independent of x^1, x^2 generally has infinite total energy on a spatial slice even for a real amplitude. It is an exact local or per-unit-area example, not an example satisfying the compact-support energy hypotheses below.

For a finite wave covector $k = k_0 + \kappa$, with real k_0 and infinitesimal components of κ , the surcomplex phase

$$e^{ik \cdot x} := e^{ik_0 \cdot x} \sum_{n \geq 0} \frac{(i\kappa \cdot x)^n}{n!} \quad (34.17)$$

belongs to the coherent complex coefficient algebra on real spacetime. With $k^2 = 0$ and transverse polarization, it gives the usual complex potential description of a vacuum wave. An unlimited covector requires a different function class; it is not covered by this definition.

35 Stress–energy, positivity, and leading scales

35.1 The electromagnetic stress tensor

Define

$$T^{\mu\nu} = -F^{\mu\alpha}F^\nu{}_\alpha + \frac{1}{4}\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}. \quad (35.1)$$

It is symmetric and traceless. Its Cartesian components are

$$T^{00} = \mathcal{E} = \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2), \quad (35.2)$$

$$T^{0i} = S_i = (\mathbf{E} \times \mathbf{B})_i, \quad (35.3)$$

$$T^{ij} = \mathcal{E}\delta_{ij} - E_iE_j - B_iB_j. \quad (35.4)$$

These follow by multiplying the finite matrices in [equation \(34.1\)](#). In particular positivity of T^{00} is positivity in the ordered coefficient field; no absolute value of an indefinite quadratic form has been substituted for energy.

Proposition 35.1 (Local energy–momentum balance). *If F satisfies Maxwell’s equations, then*

$$\partial_\mu T^{\mu\nu} = -F^{\nu\lambda}J_\lambda. \quad (35.5)$$

The time component is Poynting’s identity,

$$\partial_\tau \mathcal{E} + \operatorname{div} \mathbf{S} = -\mathbf{j} \cdot \mathbf{E}. \quad (35.6)$$

Proof. Differentiate [equation \(35.1\)](#). The term differentiating the first factor of $F^{\mu\alpha}F^\nu{}_\alpha$ is $-J^\alpha F^\nu{}_\alpha$. Contracting the Bianchi identity with $F^{\mu\alpha}$ gives

$$F^{\mu\alpha}\partial_\mu F^\nu{}_\alpha = \frac{1}{4}\partial^\nu(F_{\alpha\beta}F^{\alpha\beta}).$$

This cancels the differentiated scalar term. The displayed balance law follows, and its $\nu = 0$ component has the stated sign because $F^{0i} = -E_i$ and $J_i = -j_i$. $\square$

35.2 Dominant energy over a real-closed field

The Euclidean identity $|\mathbf{E} \times \mathbf{B}|^2 = \mathbf{E}^2\mathbf{B}^2 - (\mathbf{E} \cdot \mathbf{B})^2$ gives

$$\mathcal{E}^2 - |\mathbf{S}|^2 = \frac{1}{4}(\mathbf{E}^2 - \mathbf{B}^2)^2 + (\mathbf{E} \cdot \mathbf{B})^2 =: \chi^2 \geq 0. \quad (35.7)$$

All terms on the right are squares in K .

Theorem 35.2 (Pointwise dominant energy). *For any future unit timelike u , the vector $j_u^\mu = T^{\mu\nu}u_\nu$ is future causal or zero. If $F \neq 0$, then $T^{\mu\nu}u_\mu u_\nu > 0$. These conclusions hold for arbitrary, not merely finite, components in K .*

Proof. Use an algebraic rest boost for u . In that frame the current is $(\mathcal{E}, \mathbf{S})$. Equation [equation \(35.7\)](#) gives its causal character and the sum-of-squares expression gives its nonnegative time component. If $F \neq 0$, at least one entry of $\mathbf{E}$ or $\mathbf{B}$ is nonzero, so $\mathcal{E} > 0$. Transforming back preserves the future cone and the scalar energy measured by u . $\square$

35.3 The Rainich identity and the null case

As an endomorphism, let $\mathcal{T}^\mu{}_\nu = T^{\mu\alpha}\eta_{\alpha\nu}$. Then

$$\mathcal{T}^2 = \chi^2 I, \quad \chi^2 = \frac{I_1^2 + I_2^2}{16}. \quad (35.8)$$

Here is a finite algebraic proof. Write $C = \mathcal{E}I_3 - \mathbf{E}\mathbf{E}^\top - \mathbf{B}\mathbf{B}^\top$. The mixed tensor has blocks

$$\mathcal{T} = \begin{pmatrix} \mathcal{E} & -\mathbf{S}^\top \\ \mathbf{S} & -C \end{pmatrix}.$$

Orthogonality of $\mathbf{S}$ to both $\mathbf{E}$ and $\mathbf{B}$ gives $C\mathbf{S} = \mathcal{E}\mathbf{S}$. The identity $C^2 - \mathbf{S}\mathbf{S}^\top = (\mathcal{E}^2 - |\mathbf{S}|^2)I_3$ follows by expanding the two rank-one products and using the cross-product identity. The block square is therefore [equation \(35.8\)](#).

If $\chi > 0$, the minimal polynomial divides $(X - \chi)(X + \chi)$, so $\mathcal{T}$ is diagonalizable over K . Tracelessness makes the multiplicities of $+\chi$ and $-\chi$ both two. This diagonalizability uses the special electromagnetic identity; it is not a general spectral theorem for Lorentz-self-adjoint tensors.

If $F \neq 0$ and $\chi = 0$, ordered-field positivity forces $\mathbf{E}^2 = \mathbf{B}^2$ and $\mathbf{E} \cdot \mathbf{B} = 0$. Thus $|\mathbf{S}| = \mathcal{E} > 0$. Set $\mathbf{n} = \mathbf{S}/\mathcal{E}$ and $k = (1, \mathbf{n})$. A spatial orthonormal basis adapted to $\mathbf{E}, \mathbf{B}, \mathbf{n}$ gives

$$T^{\mu\nu} = \mathcal{E}k^\mu k^\nu, \quad q_\eta(k) = 0, \quad \mathcal{T}^2 = 0, \quad \mathcal{T} \neq 0. \quad (35.9)$$

The stress is rank one, whereas its mixed endomorphism is nonzero nilpotent. Its scalar polynomial invariants can all vanish without its energy vanishing.

35.4 Energy detects the leading electromagnetic scale

For a two-form at an event, let $v_V(F)$ be the least valuation of its six independent entries. For coherent fields use the analogous least exponent of a nonzero component function.

Theorem 35.3 (Exact doubling of the energy scale). *For every nonzero electromagnetic field at a K -valued event,*

$$v_V(T) = v(T^{00}) = 2v_V(F). \quad (35.10)$$

For a nonzero coherent field on a real domain,

$$v_H(T) = v_H(T^{00}) = 2v_H(F). \quad (35.11)$$

These statements remain true for null fields with $I_1 = I_2 = 0$.

Proof. Put $\alpha = v_V(F)$ and write each electric and magnetic entry as t^α times a finite Hahn number. At least one normalized entry has nonzero real residue. The coefficient at 2α in T^{00} is half the sum of the six real squared residues, and is strictly positive. Every component of T is quadratic in the entries, so no component has valuation less than 2α . This proves the pointwise statement.

For coherent fields the leading entries are real functions. Their squared sum is a nonnegative smooth function and is not identically zero, because at least one leading entry is not identically zero. The same conclusion follows for global exponent order, despite the presence of zero divisors in the coefficient algebra. $\square$

Together with [theorem 32.2](#), this gives a bounded-frame-invariant scale detector. It avoids the false criterion that a small or zero scalar invariant must imply a small tensor.

35.5 Integrated energy and conservation at every scale

Assume each electric and magnetic coefficient on a spatial slice is compactly supported or Schwartz, and that the corresponding spacetime coefficients have the regularity and decay needed for integration by parts. Define

$$\mathcal{E}(\tau) = \int_{\mathbb{R}^3} T^{00}(\tau, \mathbf{x}) d^3x \quad (35.12)$$

coefficientwise. A finite convolution of such coefficient functions is integrable. If α is the leading exponent of the nonzero initial electromagnetic data, the coefficient at 2α is

$$\frac{1}{2} \int_{\mathbb{R}^3} (|\mathbf{E}_\alpha|^2 + |\mathbf{B}_\alpha|^2) d^3x > 0. \quad (35.13)$$

Consequently $v(\mathcal{E}) = 2\alpha$. In vacuum, Poynting's identity and the absence of a boundary flux imply that $\mathcal{E}$ is constant, coefficient by coefficient. The leading field scale cannot disappear during this evolution. With sources, the corresponding work term is retained at every exponent.

This positivity argument is stronger than just reducing to the real shadow: an infinitesimal wave has zero real field and zero real energy shadow but still has a strictly positive exact Hahn energy at its first nonzero scale.

36 Causal linear equations and retarded solution operators

36.1 The imported real theorem and the lifted data class

The ordinary input is the smooth Cauchy theorem for a real normally hyperbolic operator on a globally hyperbolic spacetime: compatible smooth compactly supported initial data and source have a unique smooth solution, with finite propagation speed. We use its standard Minkowski specialization and the associated Green operators [25, 26]. This is an imported real theorem, not a consequence of real closedness.

Let P_0 be such an operator on a finite-rank real vector bundle, trivialized over the real domain under consideration. Basic examples are $\square$ and $\square + m_0^2$ with $m_0 \in \mathbb{R}$. On the Cauchy slice $\Sigma = \{\tau = 0\}$, prescribe coherent data

$$f = \sum t^\gamma f_\gamma, \quad g = \sum t^\gamma g_\gamma, \quad h = \sum t^\gamma h_\gamma, \quad (36.1)$$

where h is the source. Suppose the union S of their exponent supports is well ordered and each real coefficient belongs to the ordinary Cauchy theorem's data class. A common compact spacetime support is useful for uniform geometric statements, but is not needed merely to form the coefficientwise solution.

Theorem 36.1 (Support-preserving Cauchy lifting). *There is a unique coherent solution of*

$$P_0 u = h, \quad u|_\Sigma = f, \quad \nabla_n u|_\Sigma = g. \quad (36.2)$$

Its exponent support is contained in S . Each real coefficient satisfies the ordinary finite-propagation estimate. If all coefficient data and sources have support in a common closed set C for which the classical causal support theorem applies, then every coefficient solution is supported in $J(C)$; when $J(C)$ is closed, the closure of their union is also contained there.

Proof. For each $\gamma \in S$, solve the real Cauchy problem with data $(f_\gamma, g_\gamma, h_\gamma)$, setting missing coefficients to zero. Call the solution u_γ . The series $u = \sum_{\gamma \in S} t^\gamma u_\gamma$ is coherent because its support is a subset of S . The equations follow by coefficient extraction. If two coherent solutions have the same data, every coefficient of their difference satisfies a real homogeneous zero-data Cauchy problem and is zero. The real finite-propagation theorem applies at each coefficient, proving the support assertions. $\square$

For a nonzero solution of a homogeneous equation, the least exponent of its initial data equals the least exponent of the global solution, by real uniqueness. On an individual time slice, a field component may vanish while its normal derivative does not, so the invariant data are the pair of Cauchy traces, not a single chosen component.

Remark 36.2 (What kind of well-posedness has been proved?). The theorem gives existence, uniqueness, support preservation, and real continuous dependence at each coefficient. It does not assert a uniform Sobolev bound across all exponents, convergence after setting t equal to a positive real number, or well-posedness in the fine topology of **No**. Such assertions need additional norms, growth conditions, and estimates.

36.2 The retarded massless kernel

For the sign convention $\square = \partial_\tau^2 - \Delta$ in three spatial dimensions, the real retarded inverse is

$$(G_{0,\text{ret}}h)(\tau, \mathbf{x}) = \frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{h(\tau - |\mathbf{x} - \mathbf{y}|, \mathbf{y})}{|\mathbf{x} - \mathbf{y}|} d^3y, \quad (36.3)$$

for smooth compactly supported sources. Its distribution kernel is $\delta(\tau - r)/(4\pi r)$ with future support, interpreted distributionally at the cone vertex. Applying [equation \(36.3\)](#) to every coefficient gives the retarded Hahn inverse with unchanged exponent support. The sign is opposite to conventions that define the wave operator as $\Delta - \partial_\tau^2$; compare the radiation notes [\[28\]](#).

For a conserved current with admissible retarded support, $A^\nu = G_{0,\text{ret}}J^\nu$ obeys the Lorenz condition. Indeed, real differentiation commutes with this translation-invariant Green operator on its support class, so $\partial_\nu A^\nu = G_{0,\text{ret}}(\partial_\nu J^\nu) = 0$. Then $F = dA$ solves the sourced Maxwell equations. This does not prescribe arbitrary independent initial radiation; a homogeneous solution must be added when such data are intended.

36.3 Positive-valuation perturbations of a hyperbolic operator

Consider

$$P = P_0 + Q, \quad Q = \sum_{\sigma \in S_Q} t^\sigma Q_\sigma, \quad S_Q \subset \Gamma_{>0} \text{ well ordered}, \quad (36.4)$$

where every Q_σ is a real smooth local differential operator of a fixed finite order. Work in a retarded class in which $G_{0,\text{ret}}$ is defined for the intermediate real sources. One convenient class consists of fields supported in $J^+(C)$ for a common compact set C : it is past compact in Minkowski space, the standard extension of the real retarded Green operator applies, and local differential operators preserve that support. Alternatively, use a fixed real time slab and a zero-initial-data solution operator throughout. The retarded extension to sources supported in $J^+(C)$ can be constructed locally: for a compact target region D , the intersection $J^-(D) \cap J^+(C)$ lies in a compact causal diamond. Cut off the source outside a neighborhood of that diamond and apply the compact-source

Green operator. Finite propagation makes the result on D independent of the cutoff, and uniqueness glues these local results.

Theorem 36.3 (Exact causal inverse for a positive-order perturbation). *For coherent sources in the specified retarded class, the operator*

$$G_{P,\text{ret}} = \sum_{n \geq 0} (-G_{0,\text{ret}}Q)^n G_{0,\text{ret}} \quad (36.5)$$

is well defined by strong summation and gives the unique retarded coherent solution of $Pu = h$. If h has exponent support S , its solution has exponent support contained in

$$S + \langle S_Q \rangle. \quad (36.6)$$

Every coefficient remains supported in the ordinary causal future prescribed by the real background operator.

Proof. In a term of length n , choose n exponents $\sigma_1, \dots, \sigma_n$ from S_Q . A contribution to exponent γ has $\gamma = s + \sigma_1 + \dots + \sigma_n$ with $s \in S$. The positive-support lemma makes the union of all such supports well ordered and leaves only finitely many contributing words, over all lengths, at any fixed γ . Each word is a finite composition of real differential and Green operators. Thus every coefficient is a finite sum of smooth functions in the admitted support class.

The series solves $(I + G_{0,\text{ret}}Q)u = G_{0,\text{ret}}h$ by formal telescoping. Applying P_0 gives $(P_0 + Q)u = h$. For uniqueness, a nonzero retarded solution of the homogeneous equation would have a least exponent α . At that exponent Q contributes nothing from the difference, because all its exponents are positive. The leading coefficient would solve $P_0 u_\alpha = 0$ with zero retarded data, contradicting the real uniqueness theorem. Locality of Q_σ and causality of $G_{0,\text{ret}}$ prove the support claim for each word. $\square$

For example a finite mass-squared parameter $m^2 = m_0^2 + \mu$, with μ of positive order, is handled by $Q = \mu$. A positive-order perturbation of the principal coefficients is also admissible as a *formal coefficient problem*. The theorem's causal support is relative to the real body P_0 . It is not a statement that the exact characteristic cone of every K -valued principal symbol coincides with the real Minkowski cone, or that the formal inverse converges for small positive real substitutions of the scale.

36.4 Tensor wave systems

A real normally hyperbolic operator may act on any fixed finite tensor representation. The proof of [theorem 36.1](#) is unchanged because there are only finitely many components. Additional algebraic or differential constraints must be checked to propagate. For Maxwell theory this was done explicitly; for a proposed higher-rank gauge field it is a separate obligation. Scalar extension does not by itself remove gauge degeneracy or turn a constrained system into a normally hyperbolic one.

37 Actions, scalar fields, fluids, and conserved charges

37.1 A variational calculus with real test variations

Let ψ^a be finitely many coherent fields, and let $L(x, \psi, \partial\psi)$ be a polynomial in their entries and first derivatives with coherent coefficients. On a compact real spacetime region D , define

$$S_D[\psi] = \int_D L(x, \psi, \partial\psi) d^4x \quad (37.1)$$

coefficientwise. The first variation is the coefficient of s in $S_D[\psi + s\zeta]$ for an ordinary real formal parameter s . For compactly supported variations in the interior, coefficientwise integration by parts gives

$$\delta S_D[\psi; \zeta] = \int_D \sum_a \left(\frac{\partial L}{\partial \psi^a} - \partial_\mu \frac{\partial L}{\partial (\partial_\mu \psi^a)} \right) \zeta^a d^4x. \quad (37.2)$$

It is enough to test all ordinary real smooth compactly supported ζ : extracting any exponent and applying the real fundamental lemma of the calculus of variations proves the Euler–Lagrange equations at that exponent. Conversely, those equations make [equation \(37.2\)](#) zero for every coherent test variation for which the products and integral are defined.

This is a theory of stationarity, with the fundamental-lemma argument of [section 17](#). It does not infer the existence of action minimizers from the order completeness of K ; that completeness is not available. It also does not require a measure on all of K^4 .

37.2 Scalar fields and their stress tensor

For a real K -valued scalar ϕ , let

$$L_0 = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi), \quad L = L_0 + J\phi, \quad (37.3)$$

where V is a polynomial with constant coefficients in K . With the sign convention of [theorem 17.1](#) for V , the equation of motion is

$$\square\phi + V'(\phi) = J. \quad (37.4)$$

The symmetric scalar stress tensor is

$$T_\phi^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi - \eta^{\mu\nu} L_0. \quad (37.5)$$

Differentiating and cancelling the mixed-derivative terms gives

$$\partial_\mu T_\phi^{\mu\nu} = (\square\phi + V'(\phi)) \partial^\nu \phi = J \partial^\nu \phi. \quad (37.6)$$

For $V(\phi) = \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4}\phi^4$, with $m^2, \lambda \geq 0$, the rest-frame energy density

$$T_\phi^{00} = \frac{1}{2}(\partial_\tau \phi)^2 + \frac{1}{2}|\text{grad } \phi|^2 + V(\phi) \quad (37.7)$$

is nonnegative in K . Negative potentials or other matter Lagrangians require their own energy analysis; positivity is not automatic for every tensor field.

For electromagnetism the Lagrangian

$$L_{\text{EM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - J^\mu A_\mu \quad (37.8)$$

produces $\partial_\mu F^{\mu\nu} = J^\nu$ with the conventions already fixed. Its symmetric gauge-invariant field stress is [equation \(35.1\)](#). The sign of the source term differs from the scalar source convention because the fields and their defining kinetic terms differ; each equation follows from its displayed Lagrangian.

37.3 Translations, Lorentz transformations, and charges

For a translation-invariant Lagrangian the canonical current is

$$\Theta^\mu{}_\nu = \sum_a \frac{\partial L}{\partial(\partial_\mu \psi^a)} \partial_\nu \psi^a - \delta^\mu{}_\nu L. \quad (37.9)$$

A direct application of the Euler–Lagrange equations proves its conservation when there is no explicit coordinate dependence. For a symmetric conserved stress tensor,

$$\mathcal{M}^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu}, \quad \partial_\lambda \mathcal{M}^{\lambda\mu\nu} = 0. \quad (37.10)$$

Indeed the differentiated coordinates produce $T^{\mu\nu} - T^{\nu\mu} = 0$, and the remaining terms contain the stress divergence.

On a real Cauchy slice, with coefficientwise integrability and vanishing boundary flux, the quantities

$$P^\nu = \int T^{0\nu} d^3x, \quad J^{\mu\nu} = \int \mathcal{M}^{0\mu\nu} d^3x \quad (37.11)$$

are conserved Hahn-valued charges. More invariantly, one integrates $T^{\mu\nu} d\Sigma_\mu$ over a real spacelike slice. Stokes' theorem gives independence of the slice under the stated boundary conditions. Ordinary real Lorentz covariance follows coefficientwise. Integration over a genuinely surreal hypersurface would be an additional construction; the real-chain definition does not silently supply it.

For finite fields satisfying polynomial differential equations with finite constant coefficients, [equation \(13.1\)](#) gives the corresponding real equations for their shadows. With appropriate integrability the same is true of charges. Thus shadow reduction is compatible with the local dynamics in this regime, although it discards positive-order energy and momentum.

37.4 A perfect-fluid tensor

Let u be a future unit coherent vector field and let ρ, p be coherent scalar fields. Algebraically,

$$T^{\mu\nu} = (\rho + p)u^\mu u^\nu - p\eta^{\mu\nu} \quad (37.12)$$

is a well-defined perfect-fluid stress. Put $D_u = u^\mu \partial_\mu$, $\vartheta = \partial_\mu u^\mu$, and $a^\nu = D_u u^\nu$. Contracting $\partial_\mu T^{\mu\nu} = 0$ first with u_ν and then with the spatial projector gives

$$D_u \rho + (\rho + p)\vartheta = 0, \quad (\rho + p)a^\nu = h^{\nu\mu} \partial_\mu p, \quad h^{\nu\mu} = \eta^{\nu\mu} - u^\nu u^\mu. \quad (37.13)$$

In an instantaneous rest frame the second equation has the usual negative spatial pressure gradient. The identity $u \cdot a = 0$ follows by differentiating $u \cdot u = 1$.

If $\rho \geq |p|$, the tensor satisfies the dominant energy condition. In its rest frame the mixed tensor is $\text{diag}(\rho, -p, -p, -p)$, so the energy current of a future causal vector $(v^0, \mathbf{v})$ is $(\rho v^0, -p\mathbf{v})$, whose squared length is at least $\rho^2((v^0)^2 - |\mathbf{v}|^2) \geq 0$ and whose time component is nonnegative. This is an ordered-field proof. An equation of state, regularity, shock theory, and hyperbolic well-posedness are separate dynamical inputs, not consequences of [equation \(37.12\)](#).

38 A nonlinear wave theorem by transfinite coefficient recursion

38.1 Fixing the real background

Let $I \subset \mathbb{R}$ be an open time interval containing zero and let $U = I \times \mathbb{R}^3$. Fix $m_0^2, \lambda \in \mathbb{R}$ and a smooth real solution ϕ_0 on U of

$$(\square + m_0^2)\phi_0 + \lambda\phi_0^3 = f_0. \quad (38.1)$$

No claim is made that every real datum has such a global background. Choose positive-order coherent perturbations of the source and initial data. Assume their nonzero exponents have well-ordered union $S_r \subset \Gamma_{>0}$ and that the real coefficient data are smooth, with compact support or other hypotheses sufficient for the real linear Cauchy problems used below. Let $\mathfrak{M} = \langle S_r \rangle$.

The linearized real operator

$$L_0 = \square + m_0^2 + 3\lambda\phi_0^2 \quad (38.2)$$

is normally hyperbolic. It therefore has a smooth real linear Cauchy theory on the fixed background slab. When a recursive source is not compact in time, the needed smooth Cauchy solution is obtained by causal localization on compact diamonds and uniqueness; compact initial supports and finite propagation are sufficient for the recursive products, and an arbitrary smooth source may likewise be localized. Smooth coefficients can grow toward the ends of I ; the construction asserts smoothness on U , not a uniform bound at an endpoint outside I .

Theorem 38.1 (Positive-order cubic-wave deformation). *There is a unique coherent solution $\phi = \phi_0 + H$, with H of strictly positive exponent support, having the prescribed perturbed initial data and satisfying*

$$(\square + m_0^2)\phi + \lambda\phi^3 = f_0 + r. \quad (38.3)$$

The support of H is contained in $\mathfrak{M} \setminus \{0\}$. Each coefficient is obtained by solving one real linear Cauchy problem for L_0 with a source formed from finitely many lower-exponent coefficients.

Proof. Subtract the background equation and expand the cubic:

$$L_0 H = r - \lambda(3\phi_0 H^2 + H^3). \quad (38.4)$$

The positive-support lemma makes $\mathfrak{M}$ well ordered and gives only finitely many decompositions into words from S_r at a fixed exponent. Suppose the coefficients H_β have been constructed at all positive $\beta < \gamma$ in $\mathfrak{M}$. The equation for the coefficient at γ is

$$\begin{aligned} L_0 H_\gamma &= r_\gamma - 3\lambda\phi_0 \sum_{\substack{\beta+\zeta=\gamma \\ \beta,\zeta>0}} H_\beta H_\zeta \\ &\quad - \lambda \sum_{\substack{\beta+\zeta+\kappa=\gamma \\ \beta,\zeta,\kappa>0}} H_\beta H_\zeta H_\kappa. \end{aligned} \quad (38.5)$$

Every exponent on the right is smaller than γ , and each sum is finite. The prescribed initial coefficients at γ and the real Cauchy theorem determine H_γ uniquely. Transfinite recursion along $\mathfrak{M} \setminus \{0\}$ constructs the family; at limit positions there is no additional topological limit, only the previously constructed lower coefficients.

The family has well-ordered support by construction, and coefficientwise substitution into [equation \(38.4\)](#) verifies the equation. To prove uniqueness among all coherent positive-order

deformations, not just those preassigned to $\mathfrak{M}$, let a nonzero difference have least exponent γ . All terms in the nonlinear difference have that difference multiplied by at least one positive-order perturbation, so they cannot contribute at γ . The coefficient difference solves a homogeneous zero-data Cauchy problem for L_0 and must be zero, a contradiction. $\square$

The result is an exact formal existence and uniqueness theorem over a Hahn coefficient algebra. It does not prove convergence of the deformation at a small real parameter, nor prevent a different real solution from developing a singularity outside the background domain. It is stronger than writing an unevaluated perturbation ansatz: every coefficient is determined, all convolution sums are certified finite, and the support class is closed.

The theorem is the Lorentzian instance of the recursion of [theorem 16.1](#), with a real Cauchy problem in place of a bijective real linearization.

38.2 A worked spatially homogeneous example

On a real time interval, consider

$$y'' + y^3 = 0, \quad y(0) = t, \quad y'(0) = 0. \quad (38.6)$$

This is the spatially homogeneous, massless, unit-coupling special case. It is used as an ODE example rather than as compactly supported spatial data. The recursion yields

$$y(\tau) = t - \frac{\tau^2}{2}t^3 + \frac{\tau^4}{8}t^5 - \frac{3\tau^6}{80}t^7 + \frac{7\tau^8}{640}t^9 + \cdots. \quad (38.7)$$

For example $y_1 = 1$, $y_3'' = -1$, and $y_5'' = -3y_3 = 3\tau^2/2$, with zero higher-order initial data. At exponent seven the source is $-(3y_5 + 3y_3^2) = -9\tau^4/8$, giving $y_7 = -3\tau^6/80$. At exponent nine it is $-(3y_7 + 6y_3y_5 + y_3^3) = 49\tau^6/80$, giving $y_9 = 7\tau^8/640$.

The exact conserved energy is

$$\frac{1}{2}(y')^2 + \frac{1}{4}y^4 = \frac{1}{4}t^4. \quad (38.8)$$

Its derivative is $y'(y'' + y^3)$, so it vanishes coefficientwise and the initial value determines it. The accompanying symbolic program verifies the displayed truncation through exponent nine; the theorem, not the finite test, establishes the all-order solution.

38.3 Extension to other systems

The proof uses three ingredients: a specified real background, a well-posed real linearized Cauchy problem on a common domain, and nonlinear terms that raise positive exponent order. Thus the same architecture applies to finite polynomial systems when those ingredients are separately verified. A nonpolynomial nonlinearity can be treated by a support-admissible formal Taylor expansion around the body, but its domain and coefficient regularity must be stated.

For nonabelian gauge fields, the finite formulas $F_A = dA + A \wedge A$ and $d_AF_A = 0$ extend over coherent matrix coefficients, the latter by associativity and the graded Leibniz rule. A full evolution theorem additionally requires a gauge choice, constraint propagation, and a real linearized hyperbolic theory. Those extra steps are not replaced by the scalar recursion above.

39 Worldlines, local scales, and derivative choices

39.1 Parameterized surreal-valued trajectories

A coherent trajectory on an ordinary real interval is $X \in \mathcal{A}_\Gamma(I)^4$. Its velocity and acceleration are defined coefficientwise. If its parameter s satisfies $q_\eta(\dot{X}) = 1$ and $\dot{X}^0 > 0$, it is proper-time normalized within the coefficient theory. Differentiating gives

$$\eta(\dot{X}, \ddot{X}) = 0. \quad (39.1)$$

Such a trajectory is a real-parameter family of K -valued events, not an ordinary path in $\mathbb{R}^4$ unless all coefficients outside exponent zero vanish. It also need not be continuous for the fine topology on the surreal target.

A proper-time lapse can be constructed locally under a nondegenerate leading-term hypothesis. Suppose

$$q_\eta(\dot{X}) = t^{2\alpha}(r_0 + R), \quad r_0 \in C^\infty(I, \mathbb{R}_{>0}), \quad R \in \mathcal{A}_{>0}(I). \quad (39.2)$$

Then

$$\sqrt{q_\eta(\dot{X})} = t^\alpha \sqrt{r_0} \sum_{n \geq 0} \binom{1/2}{n} (R/r_0)^n \quad (39.3)$$

is a coherent positive lapse. If the leading real coefficient vanishes at some parameter value, a single such expansion need not extend through it. One must reanalyze the scale or restrict the domain. The existence of pointwise square roots in K alone does not give one coherent smooth square-root field across every zero.

39.2 The Lorentz force and admissible composition

For nonzero constant mass $m \in K$, charge $e \in K$, and proper velocity U , the Lorentz force equation is

$$m\dot{U}^\mu = eF^{\mu\nu}(X)U_\nu, \quad \dot{X} = U. \quad (39.4)$$

The expression $F(X)$ must be admitted by the chosen function class: for finite trajectories with real body in the domain one uses the halo construction; for polynomial fields one may evaluate at arbitrary K -valued trajectories. Antisymmetry gives

$$\frac{d}{ds} q_\eta(U) = \frac{2e}{m} U_\mu F^{\mu\nu}(X) U_\nu = 0. \quad (39.5)$$

Thus any solution with unit initial velocity remains unit. Existence of solutions, especially for unlimited eF/m , is not implied by this conserved algebraic identity.

For constant coefficients of finite body plus positive-order perturbation, the corresponding linear system has a coefficientwise perturbative fundamental matrix built around its ordinary real fundamental matrix. Conversely, an unlimited oscillatory generator can fall outside the coherent class. Unlimited coefficients are not universally forbidden: a nilpotent constant matrix has a finite exponential polynomial, so its evolution can remain coherent even at unlimited scale. It is the support of the actual operation, not a blanket finiteness slogan, that decides admissibility.

39.3 Inner coordinates and scale-dependent differential operators

Fix a center $a \in K^4$ and a scale $t^b > 0$. In an inner chart put

$$x = a + t^b X. \quad (39.6)$$

For an admissible scalar expression, the chain rule yields

$$\partial_{x^\mu} = t^{-b} \partial_{X^\mu}, \quad \square_x = t^{-2b} \square_X. \quad (39.7)$$

The metric written in the X coordinates is $t^{2b} \eta$, and the volume form is scaled by t^{4b} . The inverse metric supplies the inverse factor in $\square$. Thus coordinate derivative scales cannot be inferred just from the amplitude of the field.

This inner chart is an algebraic or halo parameterization of a region of K^4 . When $b \neq 0$, it is not an ordinary diffeomorphism between two open subsets of real spacetime. Inner and outer coherent coefficient expansions require an explicit overlap and matching rule. Such scale bookkeeping is useful for boundary layers, concentrated sources, and asymptotic regimes without declaring a divergent quantity regular.

40 Covariance, singular sources and interpretation

40.1 An infinite boost leaves the coherent class

[Theorem 19.1](#) explains a genuine covariance boundary. A real null plane wave transformed by a suitable infinite boost acquires an unlimited frequency in an ordinary real coordinate direction. Its would-be transformed phase would satisfy an equation of the form [equation \(19.1\)](#); it therefore cannot remain a nonzero element of the same coherent function algebra. Full K -Lorentz covariance and this particular globally coherent smooth coefficient class cannot both be asserted without enlarging the class or changing the base.

40.2 Singular sources and punctured domains

The electrostatic Coulomb field

$$\mathbf{E}(\mathbf{x}) = \frac{e}{4\pi} \frac{\mathbf{x}}{r^3}, \quad r = \sqrt{(x^1)^2 + (x^2)^2 + (x^3)^2}, \quad r \neq 0, \quad (40.1)$$

has an algebraic interpretation at every nonzero K -valued spatial point. At an infinitesimal radius its components may be unlimited. The denominator is still zero at the origin, so scalar extension has not defined a field there. A distributional source at the origin belongs to the coefficientwise linear distribution theory, but products needed for nonlinear self-energy require additional regularization and cannot be inferred from that linear statement.

This is the flat-space analogue of the repository's broader warning that scalar extension can preserve a singular expression's exact asymptotic content without regularizing its domain [\[6\]](#). No claim about black-hole interiors or quantum gravity follows from the Minkowski construction.

40.3 What standard part does and does not preserve

For coherent fields with nonnegative support, standard part preserves polynomial differential expressions. It preserves rational expressions only when the denominators have nonvanishing real

body on the chosen domain, so that their inverses have coherent finite expansions. For example the finite nonzero scalar t has an unlimited inverse, and reduction cannot turn the identity $\varepsilon t^{-1} = 1$ into multiplication of two real shadows.

Even when every displayed field and tensor is finite, shadow reduction need not preserve a strict causal classification: the near-null example of [section 32](#) proves otherwise. It also fails to commute with unbounded frame changes, as [theorem 32.3](#) shows. Accordingly, a real readout map defined by standard part has a natural bounded-frame covariance domain, not a universal full-Lorentz equivariance property.

40.4 Mathematics versus a physical interpretation

The construction provides exact relativistic tensor identities, coherent multiscale fields, and controlled formal evolution. It does not by itself select the physically realized exponent group, attach operational units to an arbitrary surreal scale, prescribe detector response, or produce experimentally different predictions. A physical model would need those additional choices together with a correspondence to measured real quantities.

In particular, an unlimited field value is an exact mathematical value in an ordered extension, not a proof that an ordinary apparatus can measure an infinite real energy. An infinitesimal correction has no nonzero standard-part readout without a specified amplification or scaling procedure. The infinite-boost example is an algebraic illustration of amplification; it is not a claim that an observer of unlimited Lorentz factor can be physically prepared.

These interpretive limits are source 02's own. Together with the firewall of [section 1.2](#) they govern the whole of Part III.

Part IV

Synthesis

41 What the theory establishes, and what it leaves as choices

The resulting theory has three genuine strengths. It accommodates every finite tensor type in every finite dimension with exact infinite and infinitesimal scalar values. It provides a coordinate-compatible calculus with certified infinite operations, including actual intrinsic derivatives on a large class of surreal-coordinate domains. And it converts specified real linear solvability into exact nonlinear multiscale solutions without requiring ordinary convergence of perturbation series.

Its boundaries are equally concrete. A bare map into $\mathbf{No}^n$ is not a smooth field. Pointwise nondegeneracy is not a certificate that an inverse belongs to a particular function algebra. The intrinsic topology of a fixed workspace is not the topology inherited from the full proper class. Coefficientwise integration is not an unrestricted measure theory on all surreal subsets. Standard part preserves regular geometry but can destroy rank in a multiscale metric. A field derivation on scalars is not a spatial partial derivative. None of these distinctions disappears when n is specialized to three or four; Parts II and III show them at work there.

Source 03's three-dimensional development has a definite core. Vectors and finite tensors over surreal scalars obey the expected algebraic identities; compact-chain integration obeys exact Stokes identities; cohomology keeps the ordinary topological degrees of freedom while periods become

surreal; Hodge decomposition survives both scalar extension and positive-order metric perturbation; and invertible real operators and quadratic nonlinearities admit exact support-controlled lifts. Its boundary is equally definite: a common-support field is more restrictive than a collection of pointwise surreal values, pointwise invertibility does not give a field inverse, singular scale operators can force solutions out of the calculus, and standard part can lose rank, flux, energy and curvature and can fail to commute with raising indices for a degenerate leading metric.

Source 02's Lorentzian package is a compatible collection of algebraic, coherent and halo theories with explicit overlaps, not a single unrestricted analytic theory on all of $\mathbf{No}^4$. The finite algebra has full K -Lorentz covariance; the coherent calculus has real-spacetime and bounded-halo covariance; Maxwell theory has exact gauge symmetry, local conservation, dominant energy, null radiation and a leading-scale energy detector; and real hyperbolic solution operators lift without enlarging support, while positive-order perturbations enlarge it only by explicitly generated monoids. The infinite-frequency theorem and the infinite-time example show that omitting these boundaries would produce false results.

41.1 Open directions and their status

Each source ends with proposed directions. They are merged below; where another source answers part of a direction, the answered part and its source are recorded and the rest stays open. None of these is asserted to be unsolved in every existing formalism.

1. *Larger intrinsic function classes* admitting coordinate gluing and integration (source 01), in particular an amplitude algebra enlarged by explicitly controlled oscillatory phases and boundary-layer charts with precise transition rules (sources 02 and 03), so that a larger class of infinite boosts can act without violating [theorem 19.1](#). Open.
2. *Hodge and elliptic results for genuinely surreal leading metrics* (source 01), and a Hodge theory for anisotropic metrics whose real reduction is singular, tracking scale-dependent cohomological and coercivity data (source 03). *Partly answered:* for a regular Riemannian Hahn metric, a nondegenerate positive-definite real leading part plus a positive-support perturbation, source 03 proves the orthogonal decomposition, harmonic representatives and Betti dimension ([theorem 12.4](#)). *Open:* regular metrics of indefinite signature, and every metric whose standard part is degenerate, such as $\text{diag}(1, t^2, t^4)$.
3. *Scale charts around rank-changing real limits* (source 01). Open; [section 14.2](#) and the softened monopole of [section 28](#) are worked examples, not a classification.
4. *Support certificates for specific gauge-fixed nonlinear systems* (source 01), and a variable Lorentzian metric with nondegenerate real body (source 02). *Partly answered:* the inverse, Levi-Civita connection, curvature, Bianchi identities, first-order response and standard-part reduction of a regular metric of any signature are proved in [sections 8, 9 and 13](#) (source 01); specific nonlinear instances are proved for stationary incompressible flow on the three-torus ([theorem 26.1](#), source 03) and for a cubic wave equation around a real background ([theorem 38.1](#), source 02). *Open:* gauge fixing, causal propagation and the nonlinear gravitational equations, and any global Einstein, Yang-Mills or Navier-Stokes evolution statement.
5. *A locally coherent sheaf theory with precise global support criteria for integration and evolution* (source 02). *Partly answered:* the sheaf $\mathcal{A}_\Gamma^{\text{loc}}$, its local calculus, and finite localization over compact sets and compact chains for integration ([section 4](#), source 01). *Open:* global support criteria for evolution on noncompact domains.

6. *Computation and formalization*: a computational treatment of support certificates that reliably detects which truncations are finite (source 03), and formalization of the prolongation and unit-admissibility interfaces (source 01), following the layer orders of [section 20](#). Open.

The constructions already proved give a working foundation for these projects, while retaining a sharp distinction between exact scale representation and analytic or physical regularization.

A Notation and hypothesis ledger

Symbol or phrase	Meaning and scope
No	The proper class of Conway surreal numbers; not a set-sized scalar type without a universe qualification.
Γ	A nonzero set-sized divisible ordered additive subgroup of No .
t^γ	Conway monomial $\omega^{-\gamma}$, not an implicit use of surreal exponentiation.
K	The real closed Hahn workspace $\mathbb{R}((t^\Gamma))$, with its intrinsic topology when derivatives are taken.
$\mathcal{A}_\Gamma(U)$	Smooth real coefficient functions on one common real domain, with one well-ordered outer Hahn support.
$\mathcal{A}_\Gamma^{\text{loc}}$	The sheaf of local such presentations; globally uniform support is not assumed on noncompact bases.
v, v_H	Minimum exponent of a scalar; minimum exponent of a nonzero coefficient function in a local field presentation.
U^h	$\text{st}^{-1}(U) \subset K^n$; a nearstandard open set, not all of K^n .
D_i	Coefficientwise spatial differentiation, fixing every constant in K .
∂_{y^i}	Intrinsic ordered-field derivative of a realized function; agrees with $\text{Ev } D_i$ only where the bridge applies.
∂_{BM}	The optional Berarducci–Mantova scalar derivation, not a spatial derivative.
Admissible metric	Symmetric metric whose inverse lies in the chosen coefficient algebra; its density is also required when used.
Regular metric	$g = g_0 + h$ with real nondegenerate g_0 and positive-support h .
Δ_g, Δ_H	Divergence–gradient scalar Laplacian; $d\delta + \delta d$. On functions $\Delta_H = -\Delta_g$.
$\int^H, \int^{\text{alg}}$	Coefficientwise integration over real chains; polynomial moment integration on affine K -simplices.
Strong sum	Well-ordered support union and finitely many nonzero contributions at every exponent; no ordinary convergence assertion.
ε	A chosen positive infinitesimal t^η ; source 02’s ε is t here.
t, τ	$t = t^1 = \omega^{-1}$ when $1 \in \Gamma$ (Parts II and III); τ is ordinary time.
∂_i, ∂_μ	In Parts II and III, the coefficientwise derivatives D_i .
$\mathcal{D}_\chi, \mathcal{D}_i$	Character scale derivation and scale connection of section 18 ; not spatial derivatives.
v_V	Least component valuation of a finite tensor in a stated frame (source 02’s ν).
Coherent field	An element of $\mathcal{A}_\Gamma(U)$: one well-ordered support on all of U .

Symbol or phrase	Meaning and scope
Regular Riemannian metric	Regular metric with g_0 positive definite (source 03's regular Hahn metric).
$\epsilon_{ijk}, \epsilon_{\mu\nu\rho\sigma}$	Alternating symbols.
$\eta_{\mu\nu}, q_\eta$	Minkowski metric $\text{diag}(1, -1, -1, -1)$ and its quadratic form (Part III only).
$\mathcal{G}, \mathcal{G}_{OK}, \mathcal{G}_1$	$\text{SO}^+(1, 3; K)$, its entrywise finite subgroup, and the kernel of reduction.
$\square$	$\partial_\tau^2 - \Delta$.
S_Q, S_r	Exponent sets of a positive-order operator perturbation and of cubic-wave data (source 02's Δ).
$\kappa_E, \kappa_B; \lambda_L, \mu_L$	Permittivity and permeability; Lamé constants (source 03's $\epsilon_{\text{el}}, \mu$ and λ, μ).
$\text{in}(F)$	Leading normalized observable $t^{v_H(F)} F_{v_H(F)}$.

B Lorentzian sign conventions

The conventions of Part III (source 02).

The following table collects signs that must be changed together when importing a different convention.

Object	Convention in Part III
Metric and wave operator	$\eta = \text{diag}(1, -1, -1, -1); \square = \partial_\tau^2 - \Delta$
Alternating tensor	$\epsilon_{0123} = +1, \epsilon^{0123} = -1$
Potential components	$A_\mu = (\phi, -\mathbf{A}); F = dA$
Electric and magnetic entries	$F_{0i} = E_i, F_{ij} = -\epsilon_{ijk} B_k$
Hodge action on fields	$(\mathbf{E}, \mathbf{B}) \mapsto (-\mathbf{B}, \mathbf{E})$
Hodge square by degree 0, 1, 2, 3, 4	$-1, +1, -1, +1, -1$
Inhomogeneous Maxwell equation	$\partial_\mu F^{\mu\nu} = J^\nu; d * F = - * J^\flat$
Stress divergence	$\partial_\mu T^{\mu\nu} = -F^{\nu\lambda} J_\lambda$
Wave kernel	$G_{\text{ret}} = \delta(\tau - r)/(4\pi r)$ as a distribution
Small Hahn monomial	$t^\gamma \mapsto \omega^{-\gamma}, \gamma > 0$
Spacetime differentiation	$\partial_\mu c = 0$ for all constant $c \in K$

Three small diagnostics prevent large errors. A pure electric field in the x^1 direction has $F \wedge *F = -E_1^2 \text{vol}$; a null plane wave has a nonzero positive T^{00} even though $I_1 = I_2 = 0$; and a source-free field has $\partial_\tau \mathcal{E} + \text{div } \mathbf{S} = 0$. These check, respectively, the signature/Hodge pairing, the distinction between invariants and energy, and the source sign in the stress balance.

C Sources, provenance and repository audit

C.1 The three manuscripts

Source	Delivered as	Checks	Placed in this report
01	n -dimensional article, 31 pages; README, Makefile, <code>verify.py</code> , recorded output	7 groups, 5,533 exact assertions	Base text of Part I, the synthesis, the ledger and this audit's first part
02	Minkowski article, 37 pages; README, source audit, Makefile, requirements, <code>check_identities.py</code> , log and JSON record	42 named exact checks	Part III; section 19 , theorem 12.1 (with 03), remarks in Part I, and section B
03	Three-dimensional article, 33 pages; README, <code>verify_examples.py</code> , recorded output	11 check groups	Part II; section 4.2 , theorems 5.1, 12.1, 12.4, 16.2 and 18.1 and remarks in Part I

All three state the repository pin `d22a5b35d5b3040e870c3cfd6d1c8f7259e094b0` and the date September 22, 2026. The manuscripts themselves are not shipped; their code and recorded data are, byte-identical, under `code/` and `data/`, and source 02's audit is `02-minkowski-SOURCE_AUDIT.md`. The page counts are those stated in the delivered READMEs. The report was merged in the collection after placement commit `5fe7f8d`.

C.2 Statements printed once

The following results are proved by more than one source and printed once; the sources are named at the place of printing.

- The workspace $K = \mathbb{R}((t^\Gamma))$, its real closedness, the localization of set-sized data, valuation, $\mathcal{O}_K$, $\mathfrak{m}_K$ and standard part; strong summability, the Neumann support lemma, and the examples showing that strong sums are not limits: all three sources, [section 2](#).
- Finite tensor algebra over K and the invariance of component valuations under bounded frames: all three sources, [section 3](#); source 02's bounded-Lorentz version is kept as [theorem 32.2](#) because it adds the equivariance of standard part and the infinite-boost counterexample.
- The common-domain differential algebra, the failure of the sheaf property for global supports, and the pointwise-invertible field $x^2 + \varepsilon^2$: all three sources or sources 01 and 03, [section 4](#).
- Taylor–Hahn prolongation and the flat-function remark: all three sources, [theorem 5.2](#).
- Lie brackets, forms, Cartan calculus, Lie derivatives and covariant derivatives of tensors: all three sources, [sections 7 and 8](#).
- Inverse, density, signature, Levi–Civita connection, curvature and Bianchi identities of regular metrics: sources 01 and 03, [sections 8 and 9](#).
- The Hodge star, its square and the codifferential: all three sources, [section 10](#); source 02's four-dimensional formulas are kept in [section 31](#) as the case $n = 4$, $q = 3$.
- Coefficientwise integration and Stokes, the radial Poincaré lemma and polynomial-simplex integration: all three sources or sources 01 and 03, [section 11](#).

- The cohomology of the Hahn-extended complex: sources 02 and 03, [theorem 12.1](#); compact base change with products is source 01’s [theorem 12.2](#). The Hodge decomposition for a real metric: sources 01 and 03, [corollary 12.3](#).
- Standard-part reduction: all three sources, [theorem 13.1](#) and the remark after it.
- The infinite-frequency obstruction: sources 02 and 03, [theorem 19.1](#); distributional coefficients: sources 02 and 03, [section 19.3](#).
- The distinction between spatial and scalar derivations, the Euler scale derivation and the curvature of a scale connection: all three sources or sources 02 and 03, [section 18](#).
- The radial flux computation is [section 14.3](#) (source 01); its case $n = 3$ is kept in [section 25.2](#) for the vector-potential obstruction.

Kept twice, because the hypotheses or proofs differ: nonlinear lifting ([theorem 16.1](#), source 01) and quadratic lifting by trees ([theorem 16.2](#), source 03); the Poynting balance with material constants (source 03, [section 28](#)) and the covariant stress balance in vacuum units (source 02, [theorem 35.1](#)); the general variational calculus with a scalar action (source 01) and with a polynomial first-order Lagrangian (source 02).

C.3 Repository audits of the sources

Source 01. The repository was inspected read-only at commit

d22a5b35d5b3040e870c3cfd6d1c8f7259e094b0.

The targeted review used the root `README.md`, `docs/README.md`, the opening scope of `docs/foundations-and-computation/foundations/article.tex`, and the normal-form, workspace, function-class, derivative and support conventions in `docs/surcomplex/analysis/article.tex`. The documentation catalogue was used to identify the nearby contours, differential-equations, spectral and physics reports. This was not a line-by-line audit of every manuscript or a local rebuild of the Lean project. The catalogue’s report count is not used as a claim that the entire repository, including newly added material, was exhaustively surveyed.

The points carried forward are the use of increasing Hahn exponents, set-sized divisible workspaces, common domains, strong summability, the separation of scalar and coordinate derivations, and explicit statements of formalization scope. Source 01’s tensor and PDE developments are proved independently above. The repository itself describes its manuscripts as research drafts with theorem-specific, rather than blanket, Lean coverage.

The external literature check included the original Berarducci–Mantova paper, the general-base-field calculus of Bertram–Glöckner–Neeb, the van den Dries–Ehrlich paper, Poonen’s discussion of Hahn units, and classical smooth and de Rham geometry sources. Two sources relevant to the cutoff date are Poonen’s January 2026 revision on units and Mantova’s January 2026 Taylor-approximation preprint. Their roles are identified in the bibliography; neither is represented as a theorem about all tensor fields on all of $\mathbf{No}^n$.

Source 02. The repository was inspected at the same pin through a connected GitHub reader: the recursive tree and a large excerpt of the root `README` (both responses truncated), `docs/README.md` lines 1–160, the physics report’s `README` lines 1–170, `docs/NORMAL_FORM_BRIDGE.md` lines 1–95, directory listings, and a repository search for “Minkowski tensor”, which returned no matches and

is not evidence that every topic is absent. The full physics article, the other manuscripts and the Lean sources were not read line by line, and no Lean build was performed. The external sources checked were the Mantova–Matusinski survey, Berarducci–Mantova, Bär and Bär–Ginoux–Pfäffle, Tong’s notes (whose PDF pages were viewed to fix conventions and the sign of the retarded kernel), and Madarász–Stannett–Székely; Neumann’s paper was identified bibliographically, and Conway and Gonshor are cited as foundational monographs cross-checked against the survey. The delivered audit is shipped as `02-minkowski-SOURCE_AUDIT.md`.

Source 03. The repository was inspected at the same pin: the root README, the documentation catalogue and the contours-and-stokes README. The spectral, physics and surquaternion reports are cited as described in the catalogue. External sources are Gonshor, Berarducci–Mantova (both papers), Costin–Ehrlich, Lee and Warner.

Checked for this merge. Every repository statement of the three sources was checked against the tree at the pin and against the current tree. The cited paths exist at both: `docs/foundations-and-computation/foundations/article.tex`, `docs/surcomplex/analysis/article.tex`, `docs/surcomplex/contours-and-stokes`, `docs/surcomplex/spectral-theory`, `docs/physics/surreal-scalars-and-spacetime`, `docs/surquaternions/surquaternions` and `docs/NORMAL_FORM_BRIDGE.md`. The foundations report proves that small subsets and small-index nets are discrete (`found:thm:discrete`). The contours-and-stokes report contains the coefficientwise Stokes theorem, the support-preserving Poincaré lemma, polynomial-chain Stokes and the finite-partition obstruction, as source 03 states. The physics README distinguishes scalar extension from regularization and records bounded-frame valuation invariance, as source 02 states. The opening assessment of `docs/NORMAL_FORM_BRIDGE.md`, which records the infinite normal-form ordered-field bridge, is the same at the pin and now. The root README’s statement that the Hahn layer proves the full Neumann support lemma is present at the pin, supporting source 03’s remark.

C.4 Stale or incomplete statements corrected

1. Source 01 described “the repository’s current documentation”. The statement is now pinned: it held at `d22a5b3` and still holds.
2. Source 01 developed Hahn Stokes, the Poincaré lemma, de Rham base change and polynomial Stokes without naming the report `docs/surcomplex/contours-and-stokes`, which at the pin already contained their complex-coefficient versions. The relation is now stated ([section 1.4](#)), and those results are not claimed as new.
3. Source 01’s remark that a Hodge decomposition for any metric other than a real one “requires a separate argument” is updated: source 03 supplies the argument for regular Riemannian Hahn metrics ([theorem 12.4](#)); the rest stays open ([section 41.1](#)).
4. Source 02’s future direction on a variable Lorentzian metric with nondegenerate real body is partly answered inside this report by source 01, and source 02’s direction on a locally coherent sheaf theory partly by source 01’s sheaf $\mathcal{A}_\Gamma^{\text{loc}}$; both are re-scoped in [section 41.1](#).
5. Source 02’s audit recorded that a repository search for “Minkowski tensor” returned no matches. That held at the pin; the collection now contains this report. The audit file is shipped as delivered.

6. Source 03’s remark that parametrizing $\mathrm{SO}(3, K)$ by large angles needs a separate trigonometric convention now points to the report `docs/surreal/euclidean-three-space`, placed with this one, and to the spin description in the surquaternion report.
7. The three sources name their programs by their delivered paths (`code/verify.py`, `verification/check_identities.py`, `verify_examples.py`) and their records correspondingly. The prefixed names under which they are shipped are used throughout (section 20.3). The delivered Makefiles of sources 01 and 02 are shipped under `code/` and still refer to the delivered layout.

C.5 Non-claims, source by source

Every limitation stated by a source is kept in the text; this ledger collects them.

Source 01 (18). (1) No completed Lean formalization and no Lean build; no repository file modified. (2) No priority claim for every result, no resolution of a named open problem, no exhaustive novelty search. (3) The finite checks do not prove the infinite-support and existence theorems, and typesetting has no bearing on correctness. (4) The framework represents exact hierarchies of scales; it is not an automatic regularization of singularities or a transfer of all real analytic existence theorems. (5) A bare map into $\mathbf{No}^n$ is not a smooth field, and pointwise nondegeneracy does not certify an admissible inverse. (6) The K -valued norm carries no completeness theorem. (7) Localization of scalar data localizes neither regularity claims nor closure under scalar derivations. (8) The intrinsic topology of a workspace is not the topology inherited from the full class, and prolongation is not an identity theorem for analytic functions nor unique outside the chosen class. (9) An arbitrary element of $K[[Z]]$ is not promised to evaluate on the whole halo; a formal inverse function theorem is not a convergence theorem. (10) The algebraic integration is not a countably additive Lebesgue theory on subsets of K^n . (11) The positivity statement concerns smooth common-support densities, not arbitrary Hahn-valued measures. (12) The Hodge corollary is not an unrestricted Hodge theorem for every surreal metric, every intrinsic K -manifold or a completed normed space. (13) The Hahn cohomology is not singular cohomology of the halo. (14) The parametric sphere is not a fine-continuous map, and the singularity at $r = 0$ is not assigned a value. (15) Flows: neither completeness of arbitrary real fields nor all-real-time flows for infinite leading coefficients. (16) Lifting does not establish global Einstein evolution, Navier–Stokes regularity or singularity removal; gauge kernels and compatibility conditions must be handled first. (17) Variational and gauge equations come without existence theorems. (18) The repository review was targeted, not a line-by-line audit, and the mixed scalar-derivation connection is not forced to be a physical direction.

Source 02 (16). (1) A mathematical framework, not a new experimentally validated theory; it does not select a physical exponent group, units or detector response. (2) No complete formal verification, no Lean build, and no claim that its theorems have checked Lean proofs. (3) No exhaustive novelty or priority claim; classical identities are not new because the coefficient field changed. (4) The 42 checks do not establish Neumann summability, real closedness, transfinite recursion, PDE existence, halo functoriality or a Lean theorem. (5) Cauchy lifting gives no uniform Sobolev bound across exponents, no convergence at a positive real scale, and no fine-topology well-posedness. (6) Retarded perturbation: causal support is relative to the real body; no statement about the characteristic cone of a K -valued principal symbol, and no convergence at small real scales. (7) Cubic wave: no convergence at a small real parameter, no control of singularities of other real solutions outside the background domain, and no claim that every real datum has a global

background. (8) Infinite boosts are algebraic facts, not statements about attainable observers. (9) An unlimited field value is not a measurable infinite energy, and an infinitesimal correction has no real readout without a specified amplification. (10) No claim about black-hole interiors or quantum gravity; scalar extension does not define the Coulomb field at the origin; products of distributions remain undefined. (11) The infinite-frequency theorem is a limitation of the chosen algebra, not an impossibility of every oscillatory extension. (12) Conservation of $q_\eta(U)$ under the Lorentz force does not give existence of solutions. (13) Stationarity only: no minimizers from order completeness, no measure on K^4 . (14) The fluid tensor carries no equation of state, regularity, shock theory or well-posedness. (15) The halo prolongation is a specified extension, not a uniqueness theorem for functions on K^4 nor smoothness in the full fine topology. (16) The source audit was targeted, and its search for “Minkowski tensor” is not evidence of absence.

Source 03 (17). (1) No unrestricted integration theory on $\mathbf{No}^3$ and no theorem about all functions on $\mathbf{No}^3$; replacing full-class language by a workspace does not preserve an unmentioned topology or function class. (2) No physical singularity resolution and no new experimentally validated field theory. (3) No Lean source, no Lean-checked proof, and a formalization section that is an architecture proposal; the repository was not modified. (4) No priority claim; Hahn summability, coefficientwise Stokes, the radial Poincaré operator and the real-closed spectral theorem are not presented as new. (5) The finite checks do not prove support lemmas, arbitrary-rank summability, the operator theorems, the imported Hodge theorem or the infinite-dimensional consequences. (6) The stationary-flow corollary is periodic, stationary and restricted to infinitesimal data around zero; nothing is asserted about time-dependent three-dimensional Navier–Stokes regularity. (7) Quadratic lifting is not a claim that a real nonlinear problem is solvable for arbitrary real forcing. (8) The regular-metric Hodge theorem does not cover metrics with singular standard part; pointwise positive definiteness is no substitute. (9) Reduction is a consistency result, not a converse existence theorem. (10) The spectral theorem is pointwise; no globally smooth eigenvector fields across repeated eigenvalues. (11) Taylor prolongation is not a canonical extension and does not solve the extension problems of Costin and Ehrlich. (12) The rotation groups are algebraic; no parametrization by large angles is provided. (13) The fast-time obstruction is a limitation of the chosen calculus, not a contradiction with real oscillators or formal time series. (14) Algebraic positivity of the elastic energy gives no analytic coercivity. (15) Products of distributions remain undefined, and the jet distribution is not an ordinary distribution at a nonreal point. (16) A Lie derivative does not give a global flow. (17) Its open directions are proposals, not statements that the problems are unsolved in every formalism.

D A proof-dependency summary

The support-sensitive claims have short, explicit dependency chains. The field and size setup uses normal forms and the classical Hahn-field closure theorem. Differential algebra uses the finite-pair support lemma. Prolongation adds positive-support substitution and a valuation remainder estimate. Metrics add only a certified geometric inverse and a certified square root. Curvature identities are finite differential-algebra identities. Stokes and Poincaré use fixed real coefficient operators. Compact de Rham base change uses finite real cohomology and support-preserving linear splittings; the Hodge refinement imports the real Hodge theorem. Nonlinear lifting uses a specified real linear inverse plus positive-support recursion. No circular reliance on the desired surreal existence theorem occurs in these constructions.

Parts II and III add the following chains. The finite vector identities use sums of squares in an ordered field; the vector differential identities are finite differential-algebra identities extended by [theorem 4.2](#). The Helmholtz decomposition, the elliptic example and the stationary-flow corollary use real Fourier multipliers on the torus, the Leray projection, and [theorems 4.3](#) and [16.2](#). The Lorentz algebra uses real closedness and sums of squares; the bounded reduction uses the infinitesimal exponential and logarithm ([2.3](#)); the energy scale uses positivity of sums of real squares. Cauchy and retarded lifting import the real Cauchy and Green theorems [[25](#), [26](#)]; the cubic-wave theorem adds positive-support recursion. The regular-metric Hodge theorem imports the real Hodge theorem and uses the positive-support inverse.

References

- [1] Vladimir Reshetnikov, *Surreal*, repository documentation, inspected by all three sources at commit d22a5b35d5b3040e870c3cfd6d1c8f7259e094b0, September 22, 2026. <https://github.com/VladimirReshetnikov/Surreal>. The root README and docs/README.md give the implementation and manuscript scope; these are project sources, not a blanket verification claim.
- [2] *Foundations for Surreal and Surcomplex Mathematics*, Surreal project, docs/foundations-and-computation/foundations/article.tex, same pinned revision as [[1](#)]. Targeted use: sizes, workspaces and the fine-topology warning.
- [3] *Surcomplex Analysis*, Surreal project, docs/surcomplex/analysis/article.tex, same pinned revision as [[1](#)]. Targeted use: monomials, coherent coefficient fields, summability and the distinction between derivative operators.
- [4] *Contours and Stokes: Jordan separation, Cauchy theory, and Stokes calculus on the surcomplex plane*, Surreal project, docs/surcomplex/contours-and-stokes, same pinned revision as [[1](#)].
- [5] *Spectral theory*, Surreal project, docs/surcomplex/spectral-theory, same pinned revision as [[1](#)].
- [6] *Surreal Scalars in Theoretical Physics: Black-Hole Singularities, Asymptotic Structure, and the Limits of Replacing the Scalar Field*, Surreal project, docs/physics/surreal-scalars-and-spacetime, same pinned revision as [[1](#)].
- [7] *Surquaternions*, Surreal project, docs/surquaternions/surquaternions, same pinned revision as [[1](#)].
- [8] *Next steps for the infinite normal-form bridge*, implementation note, Surreal project, docs/NORMAL_FORM_BRIDGE.md, assessment dated September 22, 2026; same pinned revision as [[1](#)].
- [9] J. H. Conway, *On Numbers and Games*, Academic Press, 1976; second edition, A K Peters, 2001.
- [10] Harry Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986. <https://www.cambridge.org/core/books/an-introduction-to-the-theory-of-surreal-numbers/312AE504A3E88E804054BFB390446374>.
- [11] Lou van den Dries and Philip Ehrlich, “Fields of surreal numbers and exponentiation,” *Fundamenta Mathematicae* 167 (2001), 173–188. DOI: [10.4064/fm167-2-3](https://doi.org/10.4064/fm167-2-3). See also their published erratum, *Fundamenta Mathematicae* 168 (2001). Used as background, not as an unrestricted transfer theorem for this calculus.
- [12] Bjorn Poonen, “Maximally complete fields,” *L’Enseignement Mathématique* (2) 39 (1993), 87–106. <https://math.mit.edu/~poonen/papers/amsval.pdf>. Used for the algebraic-closure theorem for Hahn fields with algebraically closed residue field and divisible value group.

- [13] B. H. Neumann, “On ordered division rings,” *Transactions of the American Mathematical Society* 66 (1949), 202–252. DOI: [10.1090/S0002-9947-1949-0032593-5](https://doi.org/10.1090/S0002-9947-1949-0032593-5). Classical source for the well-ordered support and positive-word finiteness arguments.
- [14] Bjorn Poonen, “Units in Hahn–Mal’cev–Neumann rings,” author manuscript, January 14, 2026. <https://math.mit.edu/~poonen/papers/malcev.pdf>. Theorems 1.1 and 4.1 give the positive-support unit theorem and its explicit geometric-series form.
- [15] V. Mantova and M. Matusinski, “Surreal numbers with derivation, Hardy fields and transseries: a survey,” in *Ordered Algebraic Structures and Related Topics*, Contemporary Mathematics 697, American Mathematical Society, 2017, 265–290. Preprint: <https://arxiv.org/abs/1608.03413>, version 2 (2016).
- [16] Alessandro Berarducci and Vincenzo Mantova, “Surreal numbers, derivations and transseries,” *Journal of the European Mathematical Society* 20 (2018), 339–390. DOI: [10.4171/JEMS/769](https://doi.org/10.4171/JEMS/769). <https://arxiv.org/abs/1503.00315>.
- [17] Alessandro Berarducci and Vincenzo Mantova, “Transseries as germs of surreal functions,” arXiv:1703.01995 (2017). <https://arxiv.org/abs/1703.01995>. Constructs composition on the omega-series and distinguishes it from the situation on all surreals; not a claim about arbitrary surreal-coordinate tensor fields.
- [18] Vincenzo Mantova, “Monotonicity and a Taylor approximation theorem for transseries,” preprint, January 12, 2026, arXiv:2601.07747. <https://arxiv.org/abs/2601.07747>. Recent context for omega-series composition and its Taylor theory; not used in the proof of the smooth finite-halo prolongation theorem.
- [19] O. Costin and P. Ehrlich, “Integration on the Surreals,” arXiv:2208.14331 (2022). <https://arxiv.org/abs/2208.14331>. Cited for its specified function-extension and integration program and its limitations, not as a premise that all smooth functions have a global canonical extension.
- [20] Wolfgang Bertram, Helge Glöckner and Karl-Hermann Neeb, “Differential calculus, manifolds and Lie groups over arbitrary infinite fields,” preprint, 2003, arXiv:math/0303300. <https://arxiv.org/abs/math/0303300>.
- [21] J. X. Madarász, M. Stannett and G. Székely, “Groups of worldview transformations implied by Einstein’s special principle of relativity over arbitrary ordered fields,” *The Review of Symbolic Logic* 15 (2022), 334–361. DOI: [10.1017/S1755020321000149](https://doi.org/10.1017/S1755020321000149).
- [22] John M. Lee, *Introduction to Smooth Manifolds*, second edition, Graduate Texts in Mathematics 218, Springer, 2012 (copyright 2013). DOI: [10.1007/978-1-4419-9982-5](https://doi.org/10.1007/978-1-4419-9982-5).
- [23] Peter Petersen, *Manifolds, Transversality, and de Rham Cohomology*, author lecture notes, accessed September 22, 2026, especially Chapters 6–7. <https://www.math.ucla.edu/~petersen/manifolds.pdf>.
- [24] Frank W. Warner, *Foundations of Differentiable Manifolds and Lie Groups*, Graduate Texts in Mathematics 94, Springer, 1983, Chapter 6. DOI: [10.1007/978-1-4757-1799-0](https://doi.org/10.1007/978-1-4757-1799-0). The real compact-manifold Hodge theorem is the imported analytic input to [corollary 12.3](#) and [theorem 12.4](#).
- [25] C. Bär, “Linear wave equations on Lorentzian manifolds,” 2010. <https://arxiv.org/abs/1006.2354>. The real Cauchy and finite-propagation theorems are used with their stated hypotheses.
- [26] C. Bär, N. Ginoux and F. Pfäffle, *Wave Equations on Lorentzian Manifolds and Quantization*, ESI Lectures in Mathematics and Physics, European Mathematical Society, 2007. DOI: [10.4171/037](https://doi.org/10.4171/037). <https://arxiv.org/abs/0806.1036>.

-
- [27] D. Tong, *Electromagnetism*, lecture notes, chapter “Electromagnetism and Relativity,” accessed September 22, 2026. <https://davidtong.org/pdfs/teaching/electromagnetism/electro4.pdf>.
- [28] D. Tong, *Electromagnetism*, lecture notes, chapter “Electromagnetic Radiation,” accessed September 22, 2026. <https://davidtong.org/pdfs/teaching/electromagnetism/electro5.pdf>.